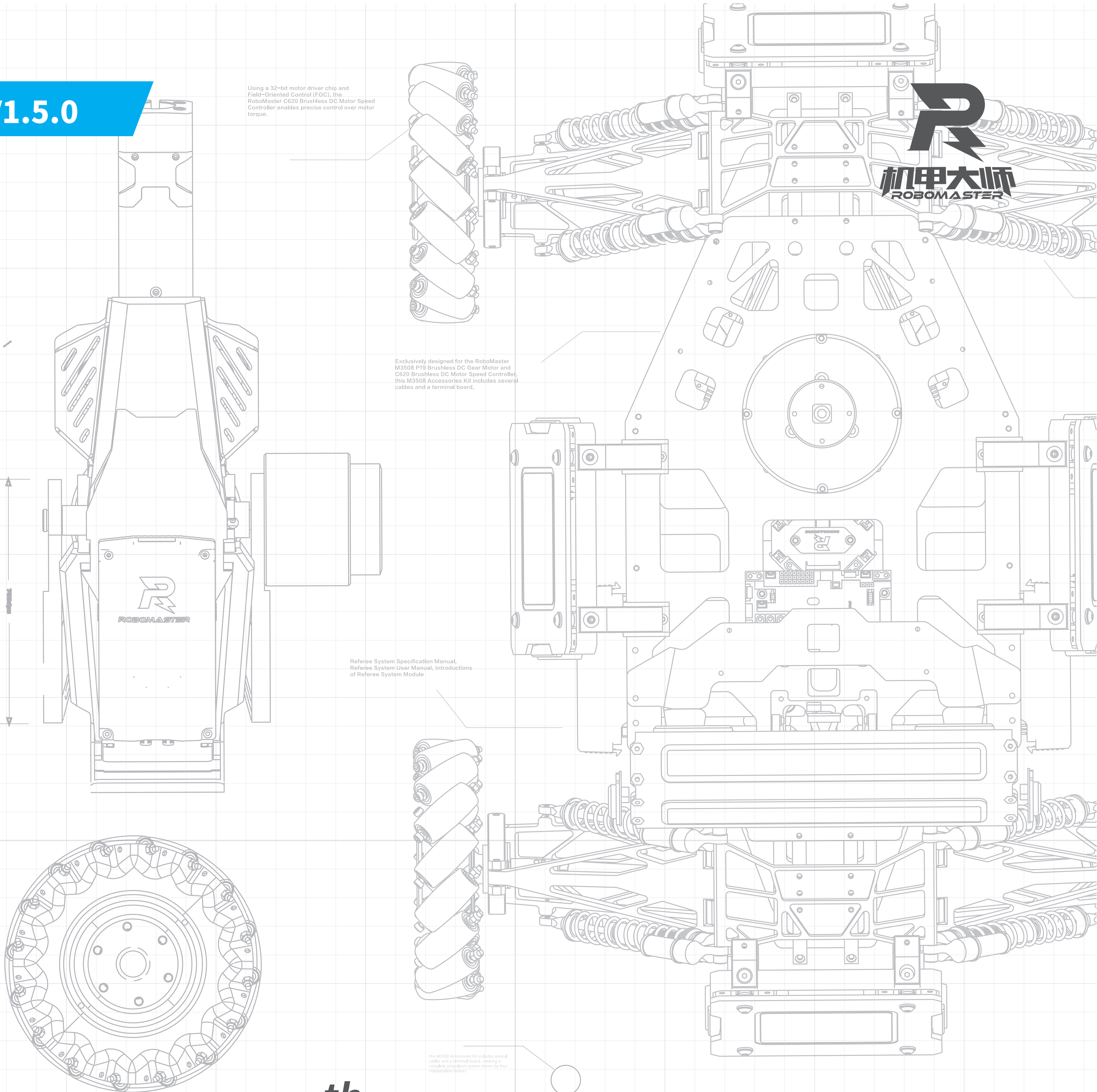
V1.5.0



The 25th China University Robot Competition

ROBOMASTER 2026

University Championship

Rule Manual

Prepared by RoboMaster Organizing Committee

Released in May, 2026

Intellectual Property Statement

The RoboMaster Organizing Committee (RMOC) encourages and promotes technical innovation and open-source technology and respects the intellectual property rights of the Participating Teams. All rights related to the intellectual property developed during the competition are owned by individual teams. The RMOC will not participate in the handling of intellectual property disputes within the team. The Participating Teams must properly handle all aspects of intellectual property rights among internal school members, company members, and other members of the team.

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https://bbs.robomaster.com/article/4820

Using this Manual

The RoboMaster competition is a robotics event aimed at cultivating youth engineers. Documents related to the competition may contain sensitive terms, names, operations, and technical content. These do not reflect any existing technologies or actions in reality, nor do they intend to involve any specific country, organization, or individual. This manual is intended solely to explain the rules of the RoboMaster competition and must not be used for any other purposes. It should not be interpreted as being associated with any real events or entities.

Version Number Description

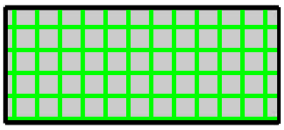
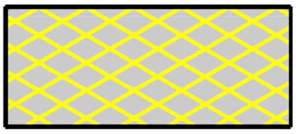
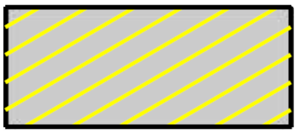
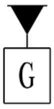
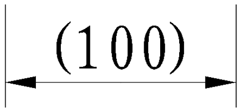
The Rules Manual adopts semantic versioning in the format of X.Y.Z, the meanings of which are as follows:

- X: Updates for different competition stages, which are indicated as follows: "1" for the Regional Competition and "2" for Final Tournament. The specific year is displayed in the file name and on the cover.
- Y: Updates to major content such as the Competition Mechanism and Competition Area, for example, changes to the Competition Mechanism or to the properties of robots.
- Z: Corrections of errors without altering the original Competition Mechanism or Competition Area, such as fixing a typo, clarifying ambiguous wording, or replacing an incorrect figure.

Example:

- **The first version of the manual for a season (for the Regional Competition) is numbered V1.0.0.**
- **Some competition mechanisms were adjusted based on the first version (for the Regional Competition), and the updated manual is numbered V1.1.0.**
- **Three typos were corrected based on the second version (for the Regional Competition), and the updated manual is numbered V1.1.1.**
- **Some competition mechanisms were adjusted based on the third version (for the Final Tournament), and the updated manual is numbered V2.0.0.**

Legend for Battlefield Drawings

		
Buff Point for both sides	Buff Point for one side	Penalty Zone for both sides
		
Penalty Zone for one side	Battlefield is located on the lowest plane	Dimensions are for reference only

Release Notes

Date	Version	Revisions
May 19, 2026	V1.5.0	Update Base defense mechanisms
April 29, 2026	V1.4.2	● Revised the handling procedure for the Tech Core getting stuck. ● Added the description requiring robots to pass through slowly during tunnel Inspection. ● Added penalties for the repeated use of Energy Units. ● Supplemented partial dimensions of the central rectangular area on the Base and Outpost Armor Modules, refer to "Figure 5-16 Rectangular Area". ● Added and revised multiple descriptions.
April 17, 2026	V1.4.1	● Add description of motion resistance in Energy Unit assembly step 2. ● Added safety lock and motion resistance torque description in

Date	Version	Revisions
		Energy Unit assembly step 5. ● Update Radar Foundation schematic diagram. ● Update Forfeiture processing.
March 23, 2026	V1.4.0	● Modify Radar Analysis of Information Waves parsing mechanism, refer to “5.6.6 Special Mechanism of Radar“. ● Supplement Radar Analysis of Information Waves spectrum distribution, air interface frame structure, time-domain characteristics and other information. For details, see “Appendix 1: Radar Information Wave Characteristics“.
February 13, 2026	V1.3.0	● Added several parameters in the Radar electromagnetic wave analysis Mechanism. ● Improved the description of the Energy Unit assembly process. ● Modified the Radar Counting Drone mechanism. ● Revised the relative Location relationship between the two tech cores in the level 4 tech core assembly difficulty section. ● Added requirements for the time intervals between each step of Level-4 assembly of the Tech Core. ● Added and revised multiple descriptions.
January 7, 2026	V1.2.0	● Adjusted the personnel scope of Operators. ● Added information about Battlefield PVC Flooring and Battlefield inclination. ● Adjusted the range of Assembly Zone, Resupply Zone, and Terrain Crossing Buff Points. ● Added information about the Resource Station and Assembly Zone, as well as the Energy Unit Mechanism for the Tech Core. ● Adjusted the shapes and dimensions of Base and Outpost Armor Modules. ● Adjusted penalty mechanisms for interface blockage caused by exceeding the Barrel Heat Limit, and added detailed rules for interface blockage under different mechanisms. ● Adjusted the Attack Damage Mechanism for the Dart Detection Module and Base Armor Module. ● Added a handling method when the Laser Detection Module goes disconnected. ● Adjusted the Respawn Mechanism. ● Adjusted the exchange limit of the 17 mm Projectile Allowance.

Date	Version	Revisions
		● Improved light effect descriptions for the Large Power Rune. ● Optimized the Terrain Crossing Buff Point (Tunnel) and Fortress Buff Point mechanisms. ● Specified the multiplication logic of the Sentry pose mechanism. ● Specified the quantity limit for ancillary equipment. ● Adjusted the quantity limit for Pit Crew members. ● Added penalties related to Captains' management responsibilities. ● Supplemented specific dimensions for the Assembly Area. ● Corrected errors in the description and diagrams for Step 5 of the Tech Core - Energy Unit mechanism. ● Added mechanisms regarding the Drone being countered by the Radar ● Added mechanisms regarding the Radar decoding electromagnetic waves. ● Adjusted penalties regarding the improper installation of Referee System onboard modules. ● Added schematic diagrams for Battlefield Component Training. ● Fixed several known issues and optimized the wording of certain content.
October 21, 2025	V1.0.0	First release

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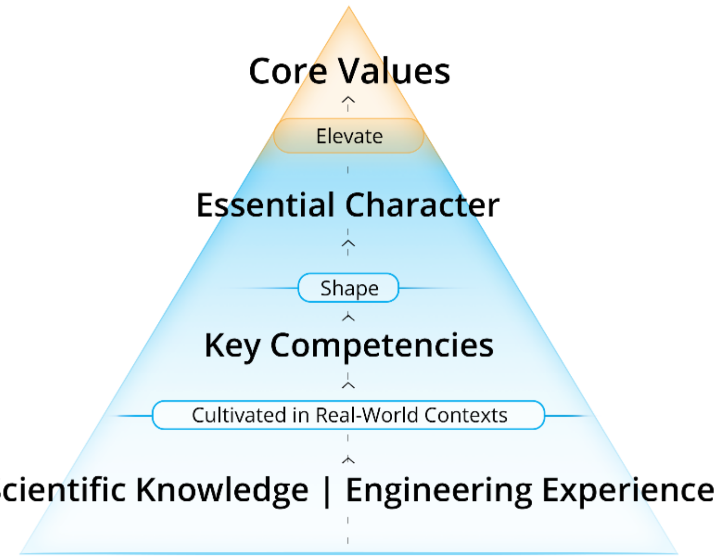
1 Foreword

1.1 About the Competition

The RoboMaster University Championship ("RMUC") under the China University Robot Competition is a shooting sport-style robotics competition. The Participating Teams are required to design, develop, and create a fleet of robots in compliance with specifications. During a seven-minute round, each team strives to destroy the opponent's Base through tactical combat to win the match.

RMUC is dedicated to cultivate excellent youth engineers. Participants are encouraged to put what they have learned into practice and achieve continued growth during the process.



- **Core values:**
  Empirical Inquiry, Pursuit of Excellence, Self-Reflection
  - Cultivate deep thinking and root-cause analysis
  - Set ambitious goals, pursue them relentlessly, and maintain a proactive, positive mindset
  - Be self-aware and regularly assess your abilities
- **Essential Character:**
  Proactivity, Positivity, Factuality, Objectivity
- **Key Competencies:**
  Critical Thinking, Communication, Problem Solving, Self-directed Learning, Positive Design

Figure 1-1 Talent Development Philosophy

1.2 About the Competition Rules Documents

The competition rules documents include the Rules Manual, Participant Manual, Robot Building Specifications Manual, Communication Protocol, and supplements to these documents. The competition rules documents are applicable to all the Participating Teams, referees, competition staff, and other partners.

- During the Preparation Period, the RMOC reserves the right to update the competition rules documents as necessary.
- During the competition, to allow the Participating Teams to fully showcase their technical achievements, the RMOC may adjust the competition rules in the following phases, but such adjustments will not involve direct structural changes to the robots.
  - During Regional Competitions: after the end of the competition for a single division.
  - During the Wild Card Competition and Final Tournament: after the end of a competition phase (for example, Group Stage or 8/16 Knockout Stage).

The RMOC reserves the right of final interpretation over the competition rules documents. During the event, only the Chief Referee can answer queries on the competition rules documents on behalf of the RMOC. Any questions related to the competition rules documents can only be directed to the Chief Referee for consultation.

For more reference materials on RMU, please go to the Rules/Resources Hub, a site built and maintained by the RMOC that provides all the official materials required for participation to help teams efficiently access information. Materials on this site include:

- Competition rules: Rules Manual, Robot Building Specifications Manual, Participant Manual, and Communication Protocol.
- Event products: Referee System (including Onboard Module and RoboMaster Champion), robot kits & accessories, product details for RM Assistant, documentation downloads, and purchase & after-sales policies.
- Technical assessment: Technical Assessment criteria and important notes about all sections.
- Brand promotion: brand promotion asset packages.
- Sponsorship guidelines: Sponsors Manual and sponsorship guides.
- Training system: participant guides, course salons, and open-source materials.

1.3 Q&A

Any participating team or other relevant personnel who have questions about the Competition Rules documents can submit them through our official channel. The RMOC will reply to them through the following Q&A process.

- To submit questions about the Competition Rules documents, please complete a questionnaire available at: https://qingflow.com/f/c5rf6rkkbs02
- The RMOC will respond at: https://qingflow.com/appView/c5rf6rkkbs02/shareView/c5rf6slgbs02

Notes about timeliness:

- Questions submitted during the Competition Period will be answered by the RMOC within one calendar day.
- Questions submitted during the Preparation Period will be answered by the RMOC within five business days.

- Competition Period: From the check-in day of the event until the end of the final competition day.
- Preparation Period: Any time outside of the Competition Period.

Note: The Preparation Period and Competition Period are calculated separately for different events and stages.

The Rules Q&A is considered as authoritative as the Competition Rules documents. In the case of any discrepancy between the Q&A and Competition Rules documents, the latest document will prevail. The Q&A for each season applies only to the current season.

2 Key Terms

In this chapter, we will provide a list of commonly used terms in the competition rules. For details on each term, please refer to the relevant section using associated keywords.

Table 2-1 Overview of Key Terms

Term	Definition
	Robot
Robot’s Main States	● **Alive:** The Referee System Main Control Module is normally connected to the Referee System server and the robot's HP is not 0. ➢ **Out of Combat:** An alive robot has not launched a Projectile or suffered any HP loss for six consecutive seconds. By default, robots are out of combat at the start of the competition. ● **Defeated:** The robot's HP drops to 0 (Reasons include: Armor Module suffering damage from attacks or collision, Referee System Going Offline, being ejected, and HP losses caused by a Dart). ➢ **Ejected:** The robot is issued a Red Card. ➢ **Temporarily Activated:** The robot's Chassis and Gimbal are powered on temporarily after the robot has been defeated. The Launching Mechanism of the robot is powered off in this period. ● **Irregular Disconnection:** The Referee System Main Control Module is disconnected from the Referee System server during the competition, due to a power outage on the robot or other reasons. Note: After a robot is defeated, the Referee System will cut off the power supply to the robot (except for the Mini PC).
Ground Robot	An umbrella name for Hero, Engineer, Infantry, and Sentry.
Referee System	An electronic penalty system integrating computation, communication, and control features. It includes Onboard Module, as well as RoboMaster Champion (Server, Referee's Client, and Player's Client) installed on the computer. It can monitor robots' power, Projectile launches, and damage, and automatically determine the outcome according to competition rules.
Inter-Robot Communication	An interaction method for robots to communicate with one another using the Referee System Serial Port.

Term	Definition
Ancillary Equipment	An umbrella name for Custom Controller, Custom Client, and wireless charging devices (Transmitters).
Robot Chassis	A mechanism that carries the robot propulsion system and its accessories, and supports the body of a robot.
Chassis Power	The power of the system that enables robots to move horizontally and rotate. For details, refer to the definition of Chassis Power in the "Referee System Mounting Specifications" section of the RoboMaster 2026 University Series Robot Building Specifications Manual.
Launching Mechanism	A mechanism capable of launching Projectiles from a robot on a fixed trajectory and at a certain initial speed.
Launch Mechanism Locked	If the Launching Mechanism is locked, it will be in a power-off state.
Launch Mechanism Unlocked	If the Launching Mechanism is unlocked, it will be powered on or off depending on its Projectile Allowance.
Launch	A robot's action of releasing Projectiles or Darts. "Launch" mentioned in the following "Competition Mechanism" section is considered valid when a launch action is detected by the Referee System.
Initial Launching Speed	The initial speed measured by the relevant modules of the Referee System after Projectiles or Darts has completed its acceleration using the Launching Mechanism.
Barrel Heat	Barrel heat accumulates as a robot launches Projectiles. It is used to limit continuous Projectile firing.
Projectile Allowance	The quantity of Projectiles each robot is allowed to launch currently.
Initial HP	The HP value set by the Referee System for a robot at the start of the competition. Note: The Referee System will round off the HP loss to the nearest integer when calculating the HP.
Current HP	A robot's real-time HP.
Maximum HP	The maximum value to which a robot's HP can be restored.

Term	Definition
Experience Point	The accumulated points required for a robot to level up, which can be obtained in various ways.
Invincible	A state in which a robot is immune to Projectiles or Darts attack or collision.
Attack	A robot's action of launching Projectiles or Darts.
Destroy	A robot attacks the Armor Module of the opponent's robots, Base, or Outpost until the latter's HP drops to 0.
Occupy	An alive robot has reached a Buff Point, its RFID Interaction Module has detected the RFID Interaction Module Card in the area or has met other conditions, and it has obtained certain special effects.
Entanglement	Mechanisms of robots are entangled with one another during the competition, i.e., one robot remains connected to the other robot and is pulled with said robot whichever direction it moves.
Collision	An active act of collision by a robot during the competition.
Battlefield	
Buff Point	A zone that, once occupied by a robot during the competition, will generate a special effect.
Penalty Zone	An area into which a robot's entry is forbidden.
Battlefield Component	Composite elements of the Battlefield, including but not limited to: Base, Outpost, and Power Rune.
Personnel	
Arbitration Commission	A body consisting of the Chief Referee and other members of the RMOC, responsible for handling appeals.
Referee	Personnel responsible for maintaining the order of the competition and enforcing its rules.
Chief Referee	The person with the right of final interpretation over the competition rules documents during the competition.
Head Referee	The lead referee responsible for maintaining the order of the competition and enforcing its rules.

Term	Definition
Head Inspector	The referee responsible for leading and assigning Pre-Match Inspection tasks, with the right of final interpretation over the inspection standards.
Participating Team	A team that has registered and been recorded in the registration system for the current competition season.
Participant	An individual who has registered and been recorded in the registration system for the current competition season.
Pit Crew	Regular Members, Reserve Members, Team Coaches, and Coaches who have registered and been recorded in the registration system for the current competition season, and can enter the Staging Area and Competition Area.
Operator	The Pit Crew responsible for controlling robots during the competition, including Ground Robot Operators, Gimbal Operators, and Pilots.
Offending Team	A participating team that violates the competition rules.
Offender	A participant that violates the competition rules.
Competition Process	
Competition Time	“X minutes and Y seconds” in this manual refers to a countdown of (6 – X) : (59 – Y). “X minutes and Y seconds to M minutes and N seconds” refers to a countdown of (6 – X) : (59 – Y) to (7 – M) : (60 – N). **Example:** ● **Five minutes into the round, the countdown is 1:59.** ● **Two to five minutes into the round, the countdown is 4:59–2:00.**
Round	A complete competition that includes the Setup Period, the Referee System Initialization Period, a Five-Second Countdown Period, and the competition round.
Match	A match may contain several rounds, depending on the match format.
Official Technical Timeout	A technical timeout initiated by the Head Referee during the Setup Period or Referee System Initialization Period.
Team Technical Timeout	A technical timeout requested by a participating team during the Setup Period.

Term	Definition
Regular Battle Damage	During a Three-Minute Setup Period or a Seven-Minute Round of a match, a Referee System Onboard Module experiences a fault.
Victory Determination	
Attack Damage	HP lost when a robot, Base, or Outpost is struck by Projectiles or Darts. Note: HP loss caused by the imposition of penalty for the violation of one side's robots, Base, or Outpost will be counted as a part of the opponent's Attack Damage.
Non-attack Damage	HP lost as a result of collision on its Armor Module or Referee System being disconnected.
Net Base HP	It is equal to the remaining HP of one side's Base minus the remaining HP of the opponent's Base at the end of each round.
Total Remaining HP	The total remaining HP of one side's alive robots at the end of each round.

3 Robot and Operator

RoboMaster requires robots to fight together as a team with good coordination and teamwork. For robot building specifications, please refer to the RoboMaster 2026 University Series Robot Building Specifications Manual.

The required Robot Line-up is as follows:

Table 3-1 Robot Line-up

Type	Number	Full Line-up Quantity (sets)	Competition Stage
Hero	1	1	Regional Competition, Wild Card, and Final Tournament
Engineer	2	1	
Infantry	3/4	2	
Drone	6	1	
Sentry	7	1	
Dart System	8	1	
Radar	9	1	

Minimum Line-up for the first round of each match: at least four robots, excluding Radar and Dart System.

The Operator Line-up is as follows:

Table 3-2 Operator Line-up

Type	Robot Operated	Full Line-up Size
Ground Robot Operator	Hero	1
	Engineer	1
	Infantry	2
	Sentry	0
Gimbal Operator	Drone (optional) and Dart System (optional)	1
Pilot	Drone	1

- The Operator must be a Pit Crew member of the Participating Team for the current season, except for the Coach.
- After the end of each round, the Operator can be replaced by a Pit Crew member for the current match, except for the Coach.
- Whether the Drone and Dart System are fielded or not, the Gimbal Operator can control the Referee System Player's Client for the Drone and Dart System and wear the corresponding headset, but they must remain at the designated area after the Three-Minute Setup Period.
- Pilot must obtain pilot certification and wear corresponding labels in order to operate Drone during the match. For details, please refer to the "Technical Assessment" section of the RoboMaster 2026 University Championship Participant Manual.

3.1 Hero

Only Hero can launch 42 mm Projectiles on the Battlefield.

Table3-3 Key Features of Heroes

Key Feature	Description
Initial Zone	Starting Zone
Operating Mode	No restrictions, up to one Remote Controller and one Custom Controller
Inter-robot Communication	Allowed
Launching Mechanism	One Launching Mechanism (42mm Projectile)
Initial Launching Speed Limit (m/s)	12m/s; increases to 16.5m/s when the Performance System is set to prioritize long-range attacks and enters Deployment Mode
HP Loss and Penalty for Exceeding Limit	• Exceeding Initial Launching Speed Limit: applicable • Exceeding Barrel Heat Limit and Cooling: applicable • Exceeding Chassis Power Consumption Limit: applicable • Attack Damage or Collision: applicable • Referee System Going Offline: applicable • Penalty Damage: applicable For details, see "5.1 HP Loss and Penalty for Exceeding Limit".

Key Feature	Description
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone or use Remote Exchange for HP. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn. For details, see "5.2 HP Recovery and Respawn Mechanism".
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 42mm and 17mm Projectiles can be pre-loaded. The Initial Projectile Allowance is 0. ● During the Seven-Minute Round, you can either exchange for Projectile Allowance in the Resupply Zone, Base Buff Point, and Outpost Buff Point, or perform a remote exchange. For details, see "5.3 Economy System".
Experience and Performance Systems	Applicable Note: Parameters such as Chassis Power Limit, Initial HP, Maximum HP, Barrel Heat Limit, and Barrel Heat Cooling Rate are related to the Level, Chassis Type, and Launching Mechanism Type. For details, see "5.4 Experience and Performance Systems".
Buff Point Occupation	● Base Buff Point ● Central Elevated Ground Buff Point ● Trapezoid-Shaped Elevated Ground Buff Point ● Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel) ● Outpost Buff Point ● Resupply Zone Buff Point For details, see "5.5.3 Battlefield Buff Mechanism".

3.2 Engineer

Engineers can be equipped with Energy Units to provide a variety of team buffs.

Table3-4 Key Features of Engineers

Key Feature	Description
Initial Zone	Starting Zone
Operating Mode	No restrictions, up to one Remote Controller and one Custom Controller.
Inter-robot Communication	Allowed
Launching Mechanism	Not applicable
Initial Launching Speed Limit (m/s)	Not applicable
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: not applicable ● Exceeding Barrel Heat Limit and Cooling: not applicable ● Exceeding Chassis Power Consumption Limit: applicable ● Attack Damage or Collision: applicable ● Referee System On-board Module Disconnection: applicable ● Penalty Damage: applicable For details, see “5.1 HP Loss and Penalty for Exceeding Limit“.
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn. For details, see “5.2 HP Recovery and Respawn Mechanism“.
Projectile Loading and Reloading	During the Three-Minute Setup Period, 42mm and 17mm Projectiles can be pre-loaded. For details, see “5.3 Economy System“.
Experience and Performance Systems	The Initial HP or Maximum HP is 250, and the Chassis Power Limit is 120W. Apart from this, the Experience and Performance Systems do not apply.

Key Feature	Description
Buff Point Occupation	● Base Buff Point ● Trapezoid-Shaped Elevated Ground Buff Point ● Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel) ● Outpost Buff Point ● Assembly Zone Buff Point ● Resupply Zone Buff Point For details, see "5.5.3 Battlefield Buff Mechanism".

3.3 Infantry

Infantries can launch 17 mm Projectiles.

Table3-5 Key Features of Infantries

Key Feature	Description
Initial Zone	Starting Zone
Operating Mode	No restrictions. Up to one Remote Controller and one Custom Controller.
Inter-robot Communication	Allowed
Launching Mechanism	One Launching Mechanism (17 mm Projectile)
Initial Launching Speed Limit (m/s)	25
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: applicable ● Exceeding Barrel Heat Limit and Cooling: applicable ● Exceeding Chassis Power Consumption Limit: applicable ● Attack Damage or Collision: applicable ● Referee System On-board Module Disconnection: applicable ● Penalty Damage: applicable For details, see "5.1 HP Loss and Penalty for Exceeding Limit".

Key Feature	Description
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone or use Remote Exchange for HP. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn. For details, see "5.2 HP Recovery and Respawn Mechanism".
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 42mm and 17mm Projectiles can be pre-loaded. The Initial Projectile Allowance is 0. ● During the Seven-Minute Round, you can either exchange for Projectile Allowance in the Resupply Zone, Base Buff Point, and Outpost Buff Point, or perform a remote exchange. For details, see "5.3 Economy System".
Experience and Performance Systems	Applicable Note: Parameters such as Chassis Power Limit, Initial HP, Maximum HP, Barrel Heat Limit, and Barrel Heat Cooling Rate are related to the Level and Chassis Type. For details, see "5.4 Experience and Performance Systems".
Buff Point Occupation	● Base Buff Point ● Central Elevated Ground Buff Point ● Trapezoid-Shaped Elevated Ground Buff Point ● Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel) ● Outpost Buff Point ● Resupply Zone Buff Point ● Fortress Buff Point For details, see "5.5.3 Battlefield Buff Mechanism".

3.4 Drone

Drones can initiate Air Support, have a First-Person View (FPV), and fire 17mm Projectiles from the air.

For the purpose of this manual, a Drone is considered alive while providing Air Support.

Table3-6 Key Features of Drones

Key Feature	Description
Initial Zone	Landing Pad
Operating Mode	No restrictions, up to two Remote Controllers and one Custom Controller.
Inter-robot Communication	Allowed
Launching Mechanism	One Launching Mechanism (17mm Projectile)
Initial Launching Speed Limit (m/s)	25
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: applicable ● Exceeding Barrel Heat Limit and Cooling: applicable ● Exceeding Chassis Power Consumption Limit: not applicable ● Attack Damage or Collision: not applicable ● Referee System On-board Module Disconnection: applicable ● Penalty Damage: not applicable For details, see "5.1 HP Loss and Penalty for Exceeding Limit".
HP Recovery and Respawn Mechanism	Not applicable
Projectile Loading and Reloading	● During the Three-Minute Setup Period, 17mm Projectiles can be pre-loaded. The Initial Projectile Allowance is 750. ● During the Seven-Minute Round, a Drone is not allowed to exchange for additional Projectile Allowance. For details, see "5.3 Economy System".

Key Feature	Description
Experience and Performance Systems	Applicable Note: Parameters such as Barrel Heat Limit and Barrel Heat Cooling Rate are related to the Level. For details, see “5.4 Experience and Performance Systems“.
Buff Point Occupation	Not applicable

3.5 Sentry

Sentries can operate in fully automatic or semi-automatic mode and can launch 17 mm Projectiles.

Table3-7 Key Features of Sentries

Key Feature	Description
Initial Zone	Starting Zone
Operating Mode	Up to one Remote Controller for commissioning
Inter-robot Communication	A Sentry can send data to all other robots on its own side, but can only receive data from the Radar.
Launching Mechanism	One Launching Mechanism (17mm Projectile)
Initial Launching Speed Limit (m/s)	25
HP Loss and Penalty for Exceeding Limit	● Exceeding Initial Launching Speed Limit: applicable ● Exceeding Barrel Heat Limit and Cooling: applicable ● Exceeding Chassis Power Consumption Limit: applicable ● Attack Damage or Collision: applicable ● Referee System On-board Module Disconnection: applicable ● Penalty Damage: applicable For details, see “5.1 HP Loss and Penalty for Exceeding Limit“.
HP Recovery and Respawn Mechanism	● HP Recovery Method: Occupy the Resupply Zone or use Remote Exchange for HP. ● Respawn Method: Use Progressive Respawn or exchange for Instant Respawn.

<table>
<tr><th>Key Feature</th><th>Description</th></tr>
<tr><td></td><td>For details, see “5.2 HP Recovery and Respawn Mechanism“.</td></tr>
<tr><td>Projectile Loading and Reloading</td><td>During the Three-Minute Setup Period, 17mm and 42mm Projectiles can be pre-loaded. The Initial Projectile Allowance is 300.
During the Seven-Minute Round, you can exchange for Projectile Allowance in the Resupply Zone, Base Buff Point, and Outpost Buff Point, obtain it directly, or perform a remote exchange.
For details, see “5.3 Economy System“.</td></tr>
<tr><td>Experience and Performance Systems</td><td>The Experience and Performance Systems do not apply. A robot's performance depends on how it is operated, as detailed below:
Full-automatic Sentry
The Initial HP or Maximum HP is 400.
The Barrel Heat Limit is 260.
The Barrel Heat Cooling Rate is 30/second.
The Chassis Power Limit is 100W.
Semi-automatic Sentry
The Initial HP or Maximum HP is 200.
The Barrel Heat Limit is 100.
The Barrel Heat Cooling Rate is 10/second.
The Chassis Power Limit is 60W.
For details, see “5.6.4 Special Mechanism of Sentry”.</td></tr>
<tr><td>Buff Point Occupation</td><td>Base Buff Point
Central Elevated Ground Buff Point
Trapezoid-Shaped Elevated Ground Buff Point
Terrain Crossing Buff Point (Road, Elevated Ground, Launch Ramp, and Tunnel)
Outpost Buff Point
Resupply Zone Buff Point
Fortress Buff Point
For details, see “5.5.3 Battlefield Buff Mechanism“.</td></tr>
</table>

3.6 Dart System

The Dart System can attack the opponent's Outpost and Base by launching Darts.

Table3-8 Key Features of the Dart System

Key Feature	Description
Initial Zone	Dart Launching Station
Operating Mode	No restrictions, up to one Remote Controller.
Inter-robot Communication	Allowed

3.7 Radar

A Radar can acquire Battlefield information autonomously and send it to robots on its own side or the Referee System Player's Client via Inter-robot Communication. A Radar can also intercept the opponent's information or counter Air Support from opponent Drones.

Table3-9 Key Features of Radars

Key Feature	Description
Initial Zone	Radar Foundation
Operating Mode	Autonomous, up to one Remote Controller for commissioning.
Inter-robot Communication	A Radar can send data to all robots on its own side, but can only receive data from Sentries.

4 Competition Area

4.1 Battlefield Overview

- Unless otherwise specified, all dimensions described in this document are subject to certain tolerances. Dimensional tolerances for structures smaller than 100mm are within ±5mm. Dimensional tolerances for structures of 100mm or larger are within ±5%. Angular tolerances of Battlefield structures are within ±3°. Angular tolerances of Battlefield Components are within ±1°. All dimension parameters in Battlefield Instruction drawings are indicated in mm. To ensure the stability and safety of the Battlefield, adjustments may be made to specific dimensions and workmanship during actual construction. For Battlefield elements or dimensions not explicitly specified in the drawings, please refer to the actual Battlefield.
- The Battlefield has a centrally symmetrical layout. All descriptions and illustrations of Battlefield modules in this document will be based on the Red Team, with the same applying to the Blue Team. The Battlefield has a complex environment that may feature non-uniform magnetic fields, air conditioner winds, and other environmental conditions that are not fully symmetrical.
- The Renderings are for illustrative purposes only. For the specific Battlefield dimensions, see the corresponding drawings. Any matters not covered are subject to the actual conditions during the Practice Match.
- Multiple Buff Points and Penalty Zones are set in the Battlefield. To keep this document concise and clear, this chapter will not go into their details. Their specific definitions will be explained in a unified way in “5.5.3.1 Battlefield Buff Points“ and “7.2.3.2 Interaction between Robots and Battlefield Components“ respectively.

The core Competition Area of the RMUC is called Battlefield. The Battlefield is 28m long and 15m wide, with a wooden inner structure. Its surface is covered with PVC Flooring that is 2mm thick. The lower areas of Terrain Crossing Buff Points (Elevated Ground and Road) and Bumpy Roads are covered with wire-embedded PVC Flooring that is 2mm thick. Some structures have metal protective layers. It mainly includes a Base Zone, an Elevated Zone, a Road Zone, and a Flight Zone. On the periphery of the Battlefield is a black steel Perimeter Wall with a height of 2.4m from its upper edge to the Battlefield ground.

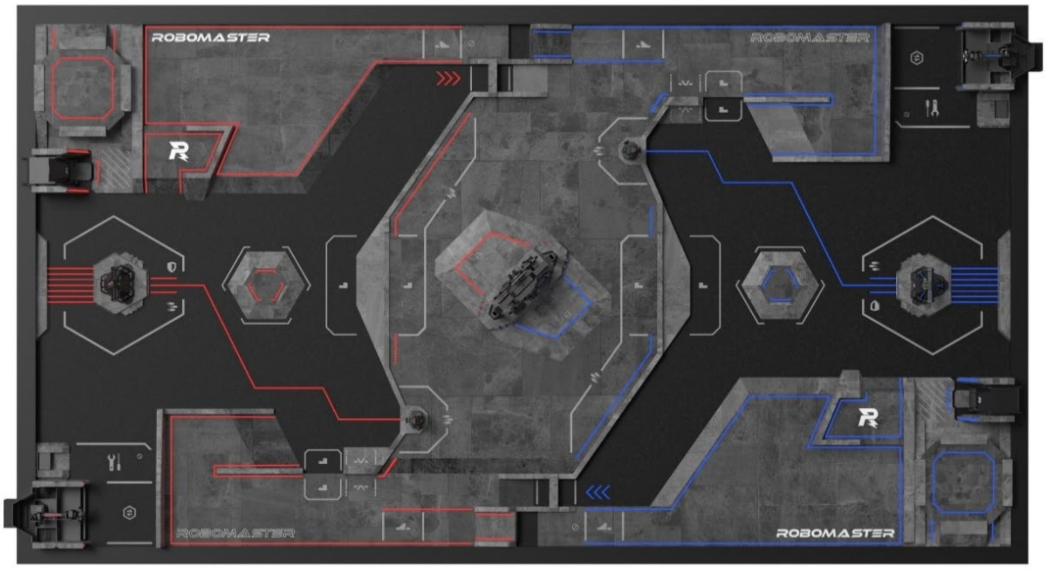


Figure 4-1 Top-view Rendering of the Battlefield

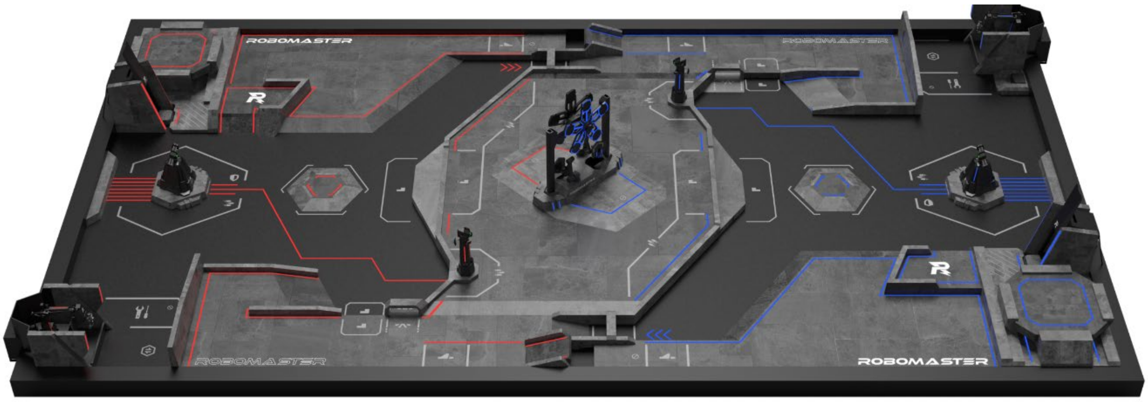
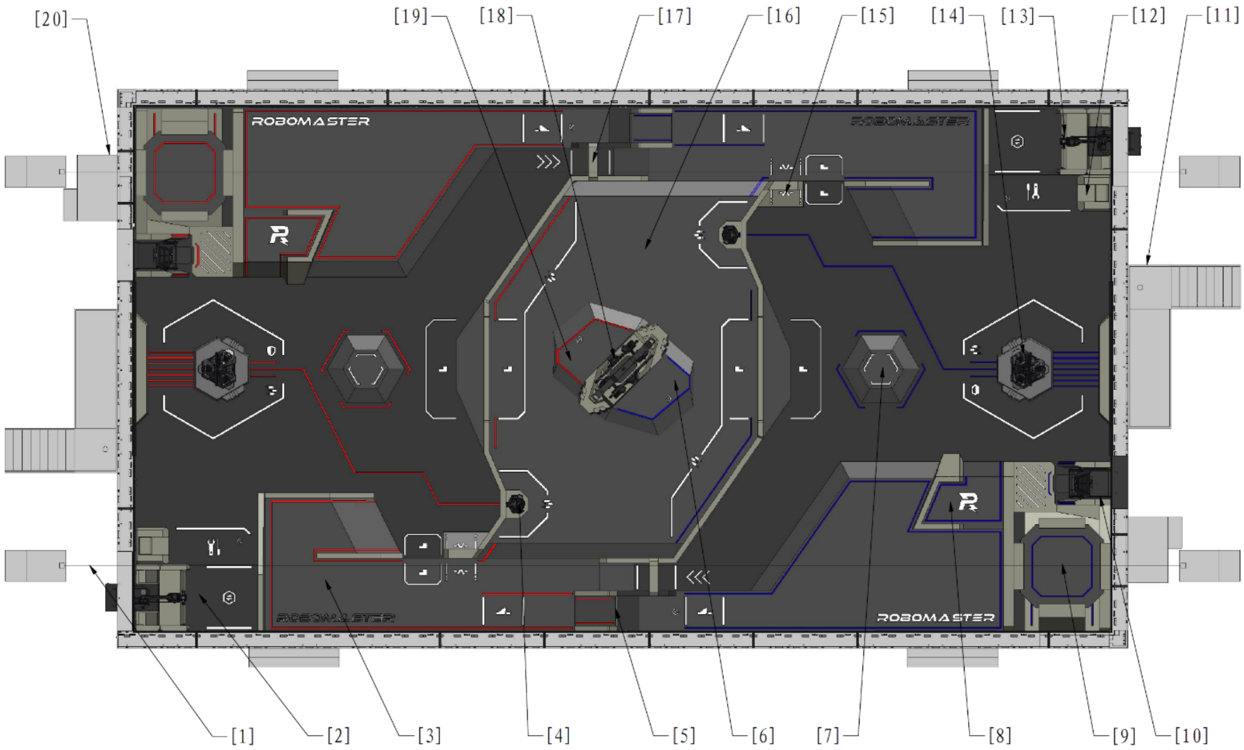


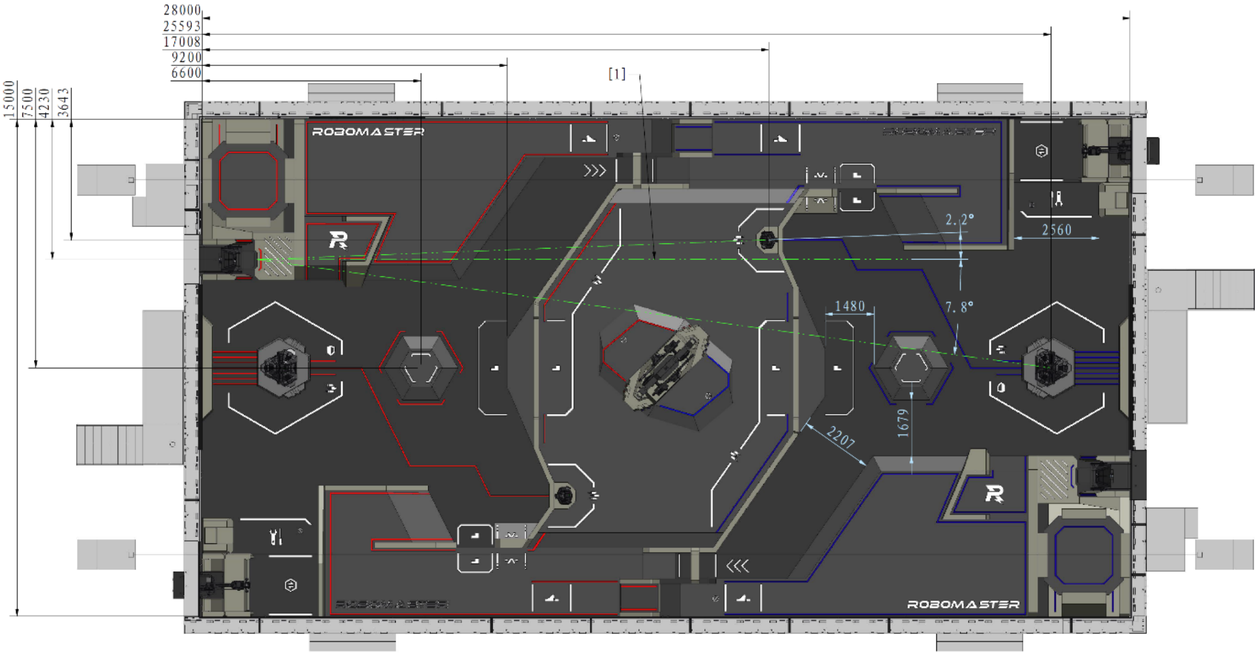
Figure 4-2 Oblique Rendering of the Battlefield

Figure 4-3 Axonometric Rendering of the Battlefield



[1]	Drone Safety Rope	[2]	Resupply Zone	[3]	Road Zone	[4]	Outpost
[5]	Launch Ramp	[6]	Blue Team's Assembly Zone	[7]	Fortress	[8]	Trapezoid-Shaped Elevated Ground
[9]	Landing Pad	[10]	Dart Launching Station	[11]	Radar Foundation	[12]	Wireless Charging Device Zone
[13]	Resource Station	[14]	Base	[15]	Road Tunnel	[16]	Central Elevated Ground
[17]	Road Tunnel	[18]	Power Rune	[19]	Red Team's Assembly Zone	[20]	Pilot Operation Room

Figure 4-4 Battlefield Modules

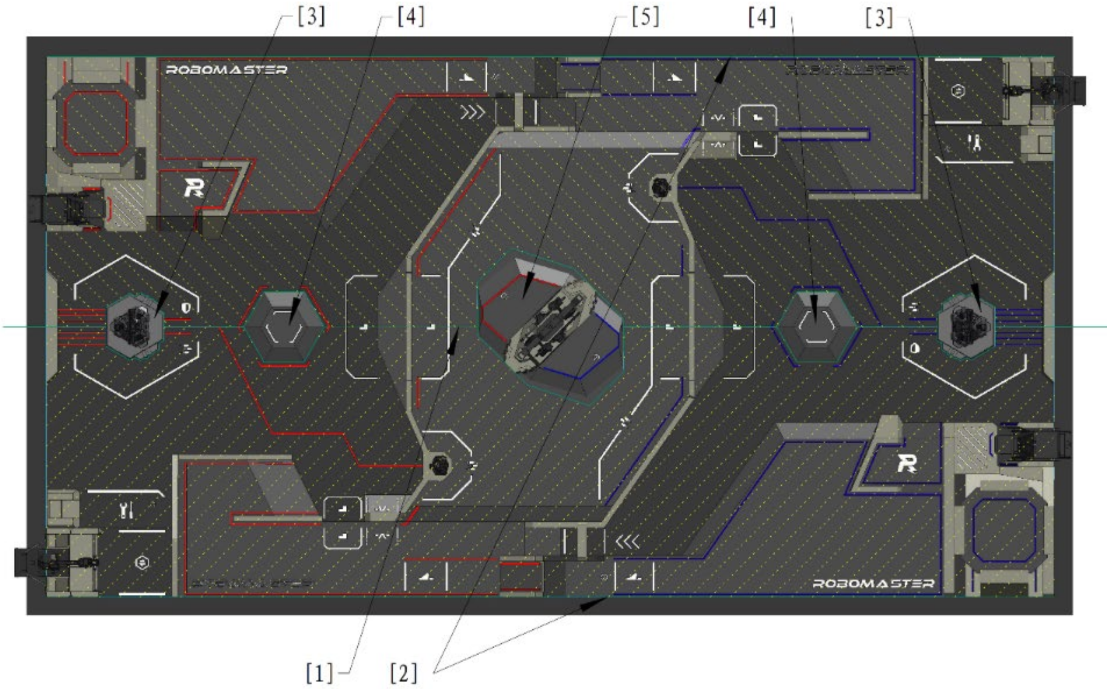


[1] Parallel with the long side of the Battlefield and the orientation of the Dart Launching Station

Figure 4-5 Battlefield Module Localization Dimensions

All site measurements are indicated based on the left and right inclined planes. In practice, the Battlefield floor is not level; it tilts 1° to 2° toward both long sides along the central axis of the Battlefield, as illustrated in the Figure below. Battlefield inclination will cause Projectiles to flow toward both long sides.

To ensure smooth Projectile recovery, there may be a groove approximately 50mm deep along the intersection between some wooden structures and the Battlefield surface.

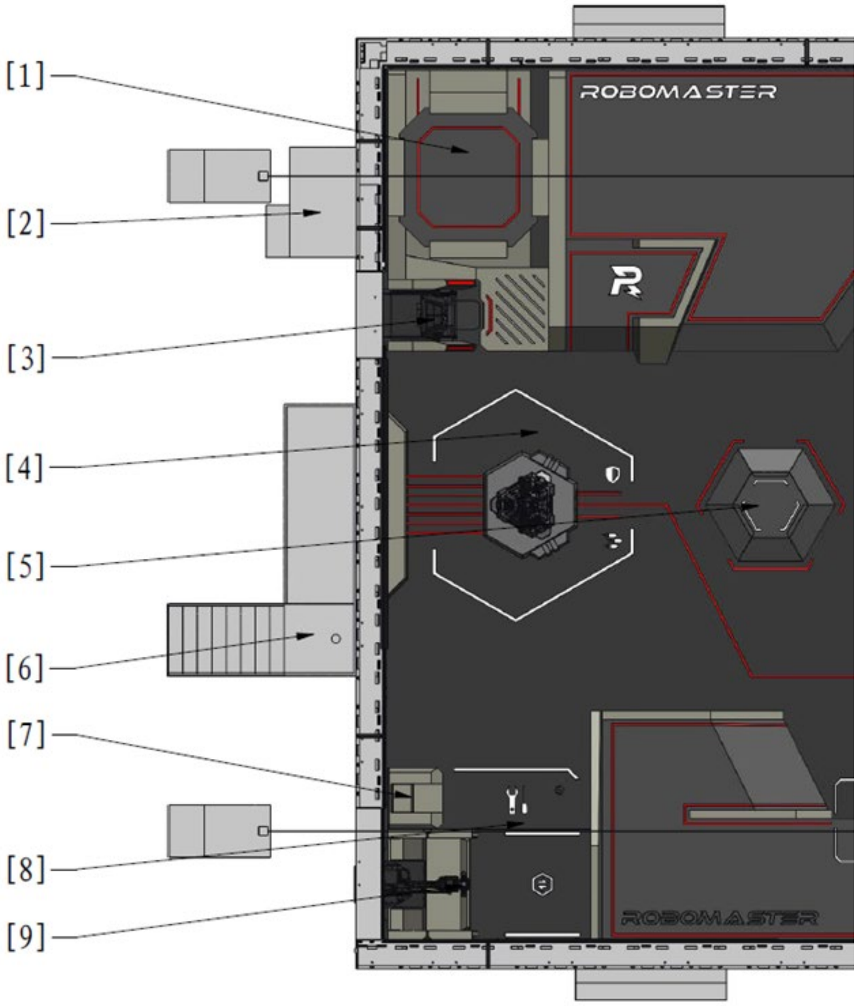


[1]	Higher Area	[2]	Lower Area	[3]	Base Foundation
[4]	Fortress Foundation	[5]	Assembly Zone Foundation		

Figure 4-6 Battlefield Inclination

4.2 Base Zone

The Base Zone consists of the Starting Zone, Base, Dart Launching Station, Landing Pad, Radar Foundation, Resupply Zone, and Fortress.



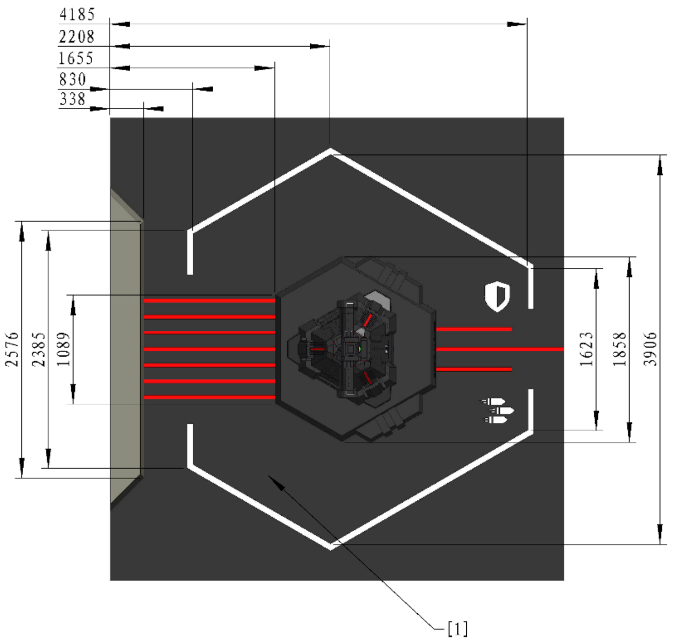
[1]	Landing Pad	[2]	Pilot Operation Room	[3]	Dart Launching Station
[4]	Starting Zone	[5]	Fortress	[6]	Radar Foundation
[7]	Wireless Charging Device Zone	[8]	Resupply Zone	[9]	Resource Station

Figure 4-7 Base Zone

4.2.1 Starting Zone

The Starting Zone is a hexagonal area around the Base, designated as the area for placing Ground Robots before the match begins.

In actual competitions, the steps on the side near the Perimeter Wall have been removed.



[1] Starting Zone

Figure 4-8 Robots Starting Zone

4.2.2 Base

The Base is at the core of offense and defense on both sides, and is placed on the Base Foundation at the center of each team's Starting Zone.

A Base consists of the main body, six Base Armor Modules, Dart Detection Module, Base Protective Armor, etc. The Base Protective Armor has two states: closed or expanded.

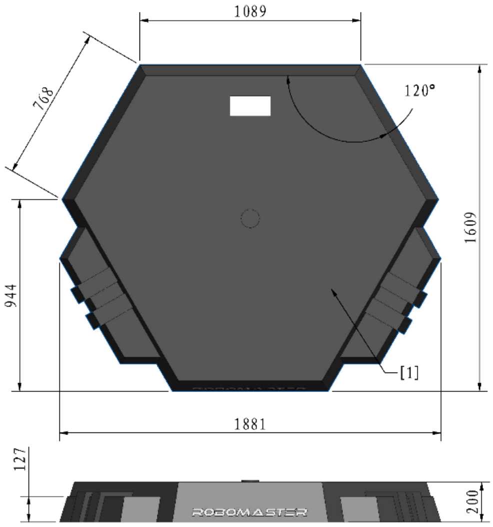
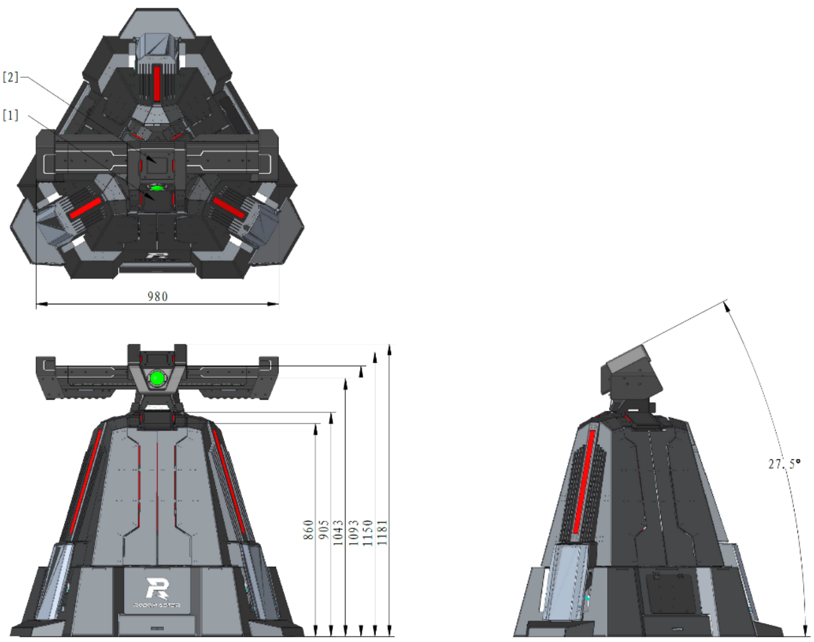


Figure 4-9 Base Foundation



[1] Base Armor Module [2] Dart Detection Module

Figure 4-10 Closed Base Protective Armor

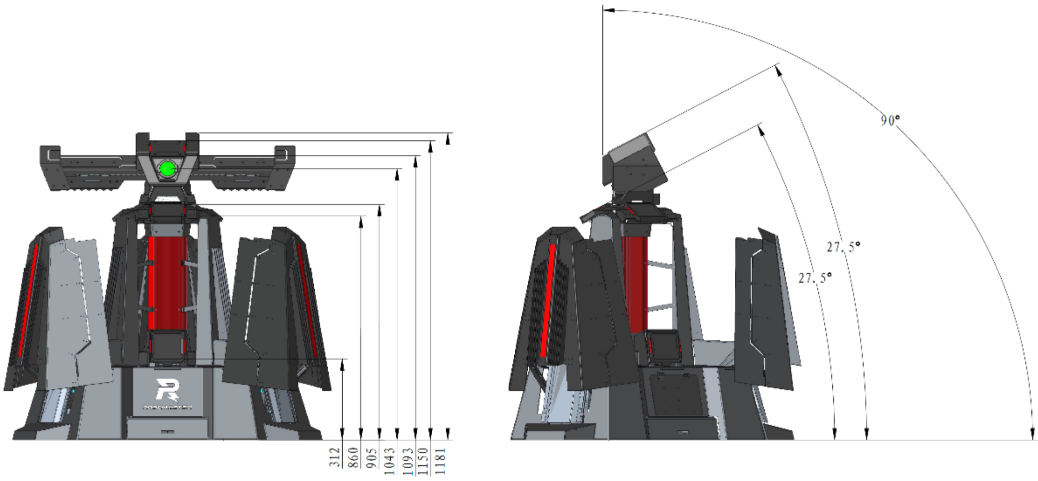
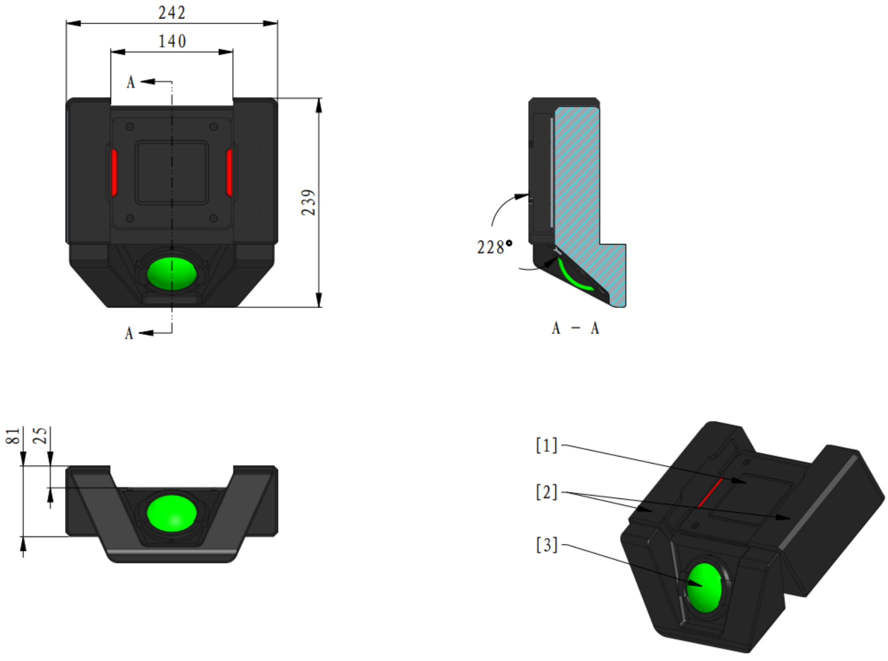


Figure 4-11 Expanded Base Protective Armor

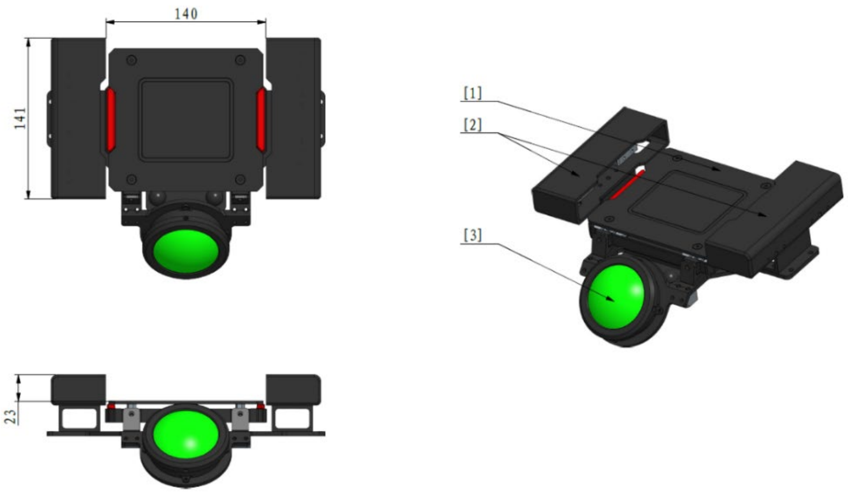
The Dart Detection Module is located on the top of both the Base and Outpost, consisting of a Small Armor Module, Dart Detection Sensor, and Dart Guiding Light.

The Small Armor Module of the Dart Detection Module can detect strikes from both Darts and 42mm Projectiles. The Dart Detection Sensor can detect infrared beams emitted by a Dart Trigger. When the projection of the Dart Trigger onto the Dart Detection Module falls entirely within the Small Armor Module, and the Dart Detection Module simultaneously detects an infrared beam and a strike, the system determines the module has been hit by a Dart. If only a strike is detected, the system considers the module to have been hit by a Projectile. The Dart Guiding Light emits green visible light with a wavelength of 520nm. When approximating a point light source, its luminous intensity is around 10cd, and the diameter of its light-emitting part is about 55mm. It is for guiding Darts toward their targets.

If the projection of the Dart Trigger onto the Dart Detection Module does not fall entirely within the Small Armor Module, there is still a certain chance that the Dart Detection Module will successfully recognize it.



Base Dart Detection Module



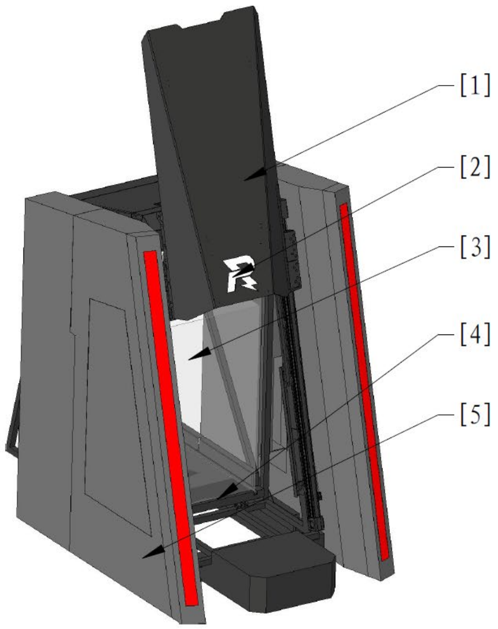
Outpost Dart Detection Module

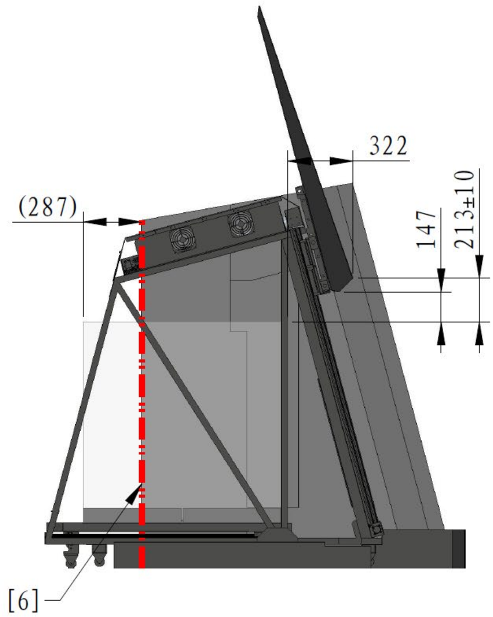
[1] Small Armor Module [2] Dart Detection Sensor [3] Dart Guiding Light

Figure 4-12 Dart Detection Module

4.2.3 Dart Launching Station

The Dart System can only be placed in the Dart Launching Station, which consists of the main body, Gliding Platform, and gate. A lighting device is installed inside the Dart Launching Station and cannot be turned off. During the Referee System Initialization Period, the gate of the Dart Launching Station opens for approximately five seconds. Throughout this time, the Dart Detection Module remains static at the fixed target position at the Base. During the Three-Minute Setup Period and the Fifteen-Second Referee System Initialization Period, the Dart Guiding Lights at the Outpost and Base are illuminated.





[1]	Gate	[2]	Status Indicator	[3]	Space for placing the Dart System
[4]	Gliding Platform	[5]	Main body of the Dart Launching Station	[6]	Battlefield edge

Figure 4-13 Dart Launching Station

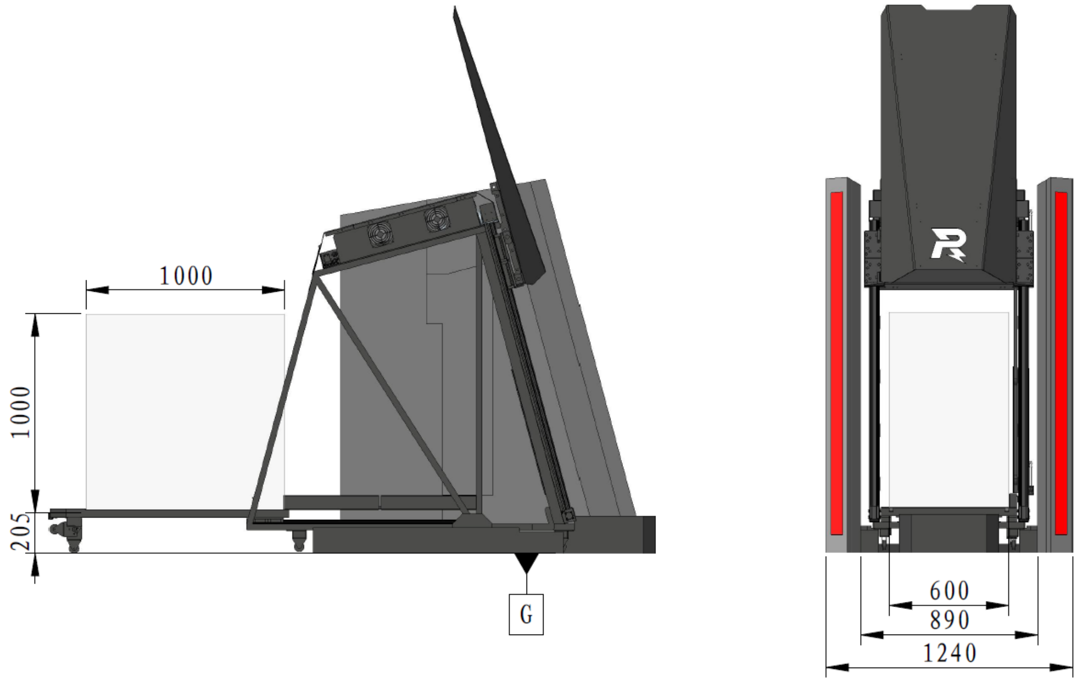
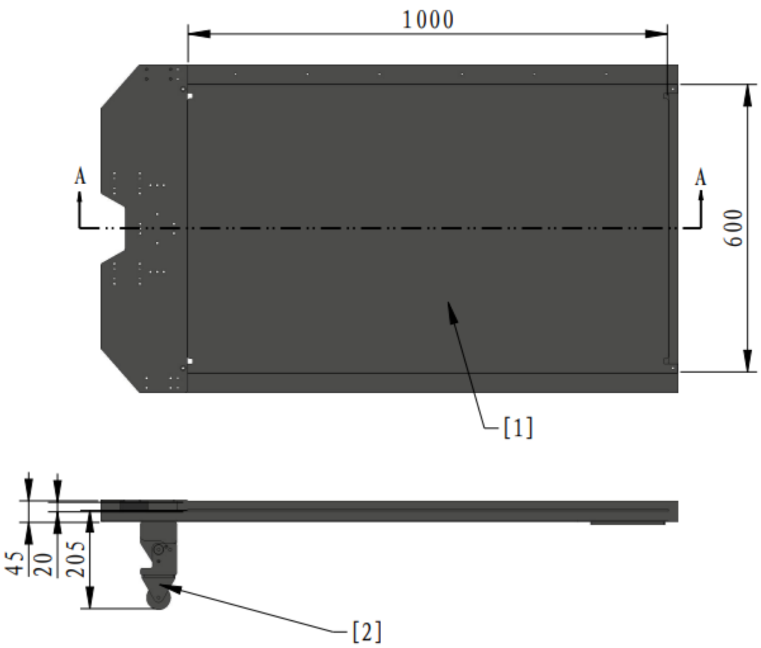


Figure 4-14 Gliding Platform Extension

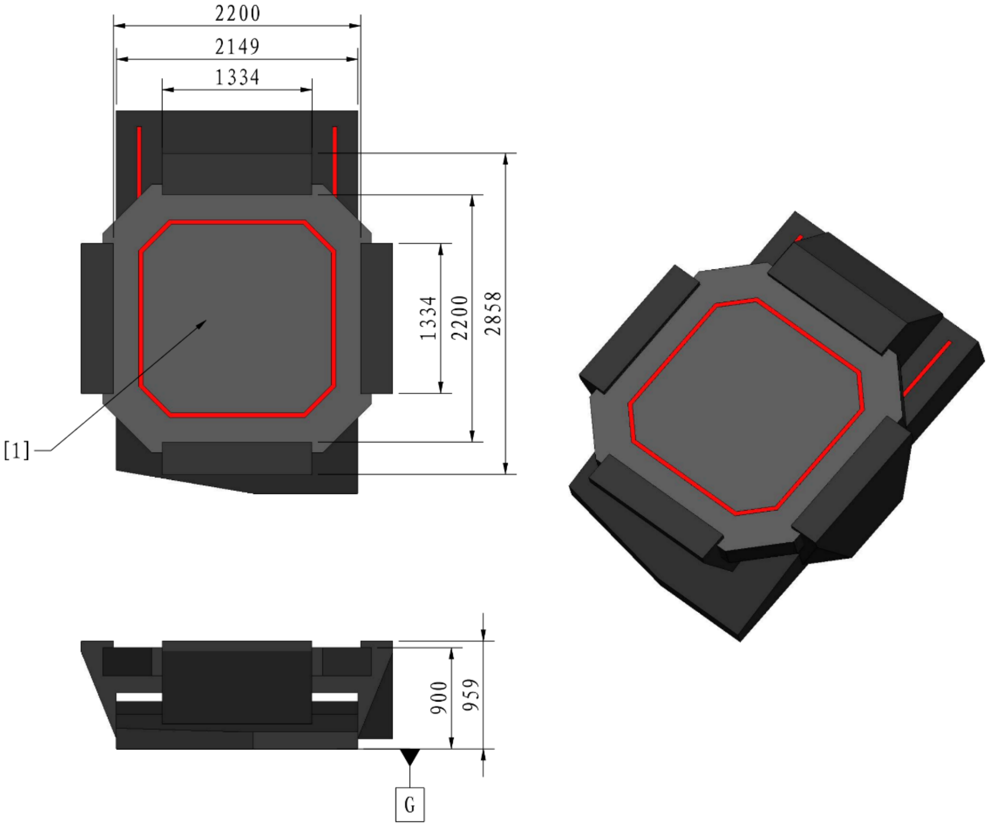


[1] Surface for placing the Dart System [2] Supporting Wheel

Figure 4-15 Gliding Platform Dimensions

4.2.4 Landing Pad

Prior to the start of the match, the Drone must be placed on a Landing Pad Platform, its projection must be within the boundaries of the Landing Pad Platform, and it must be connected to an Aerial Safety Rope in accordance with guidelines.



[1] Landing Pad Platform

Figure 4-16 Landing Pad

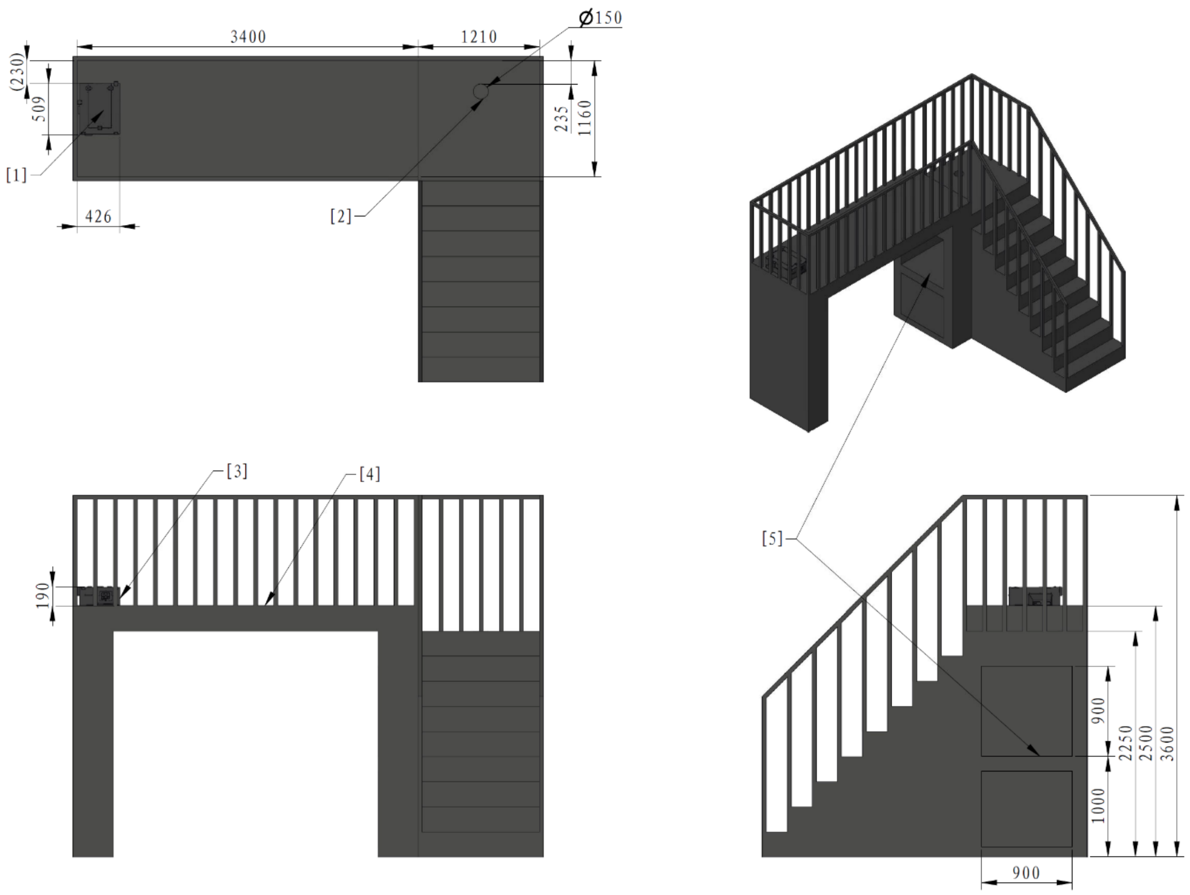
4.2.5 Radar Foundation

The Radar Foundation is used to place the Radar Sensor and laser launching device, featuring a top platform that measures 3.4m×1.16m. With the center aligned with the central axis of the Battlefield, the platform surface is approximately 2.5m above the Battlefield ground and surrounded by a fence that is 1.1m high. The platform is provided with a sensor data cable slot, which can be used as needed during the competition.

The Radar Computing Platform is powered by 220V mains supply. Its platform has the following:

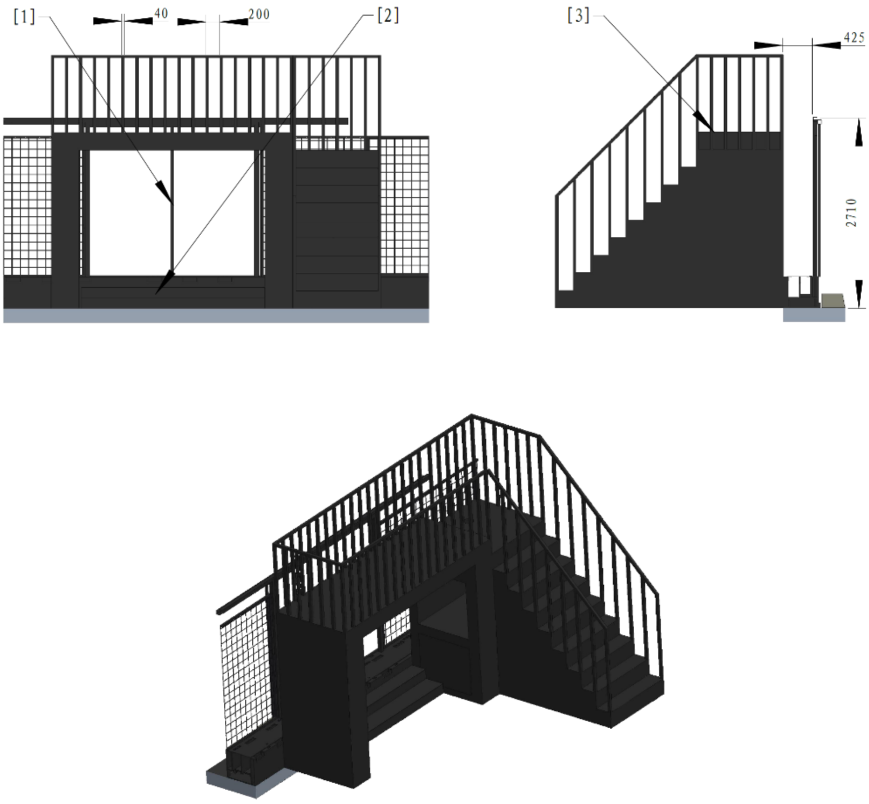
- One official display device that only supports HDMI signal input, with a resolution of 1920×1080. The Participating Teams may use it to confirm the operational status of the Radar.
- One HDMI cable for connecting the Radar to the official display device.

- One fixed 5-hole socket compliant with China's national standard GB/T 2099.7-2015, with both single-phase two-pin and single-phase three-pin outlets available for simultaneous use. This socket is designated for supplying power to the Radar.



[1]	Radar Information Wave	[2]	Sensor data cable slot hole	[3]	Radar Information Wave transmitting surface
[4]	Radar Sensor placement platform	[5]	Radar Computing Platform placement platform		

Figure 4-17 Radar Foundation

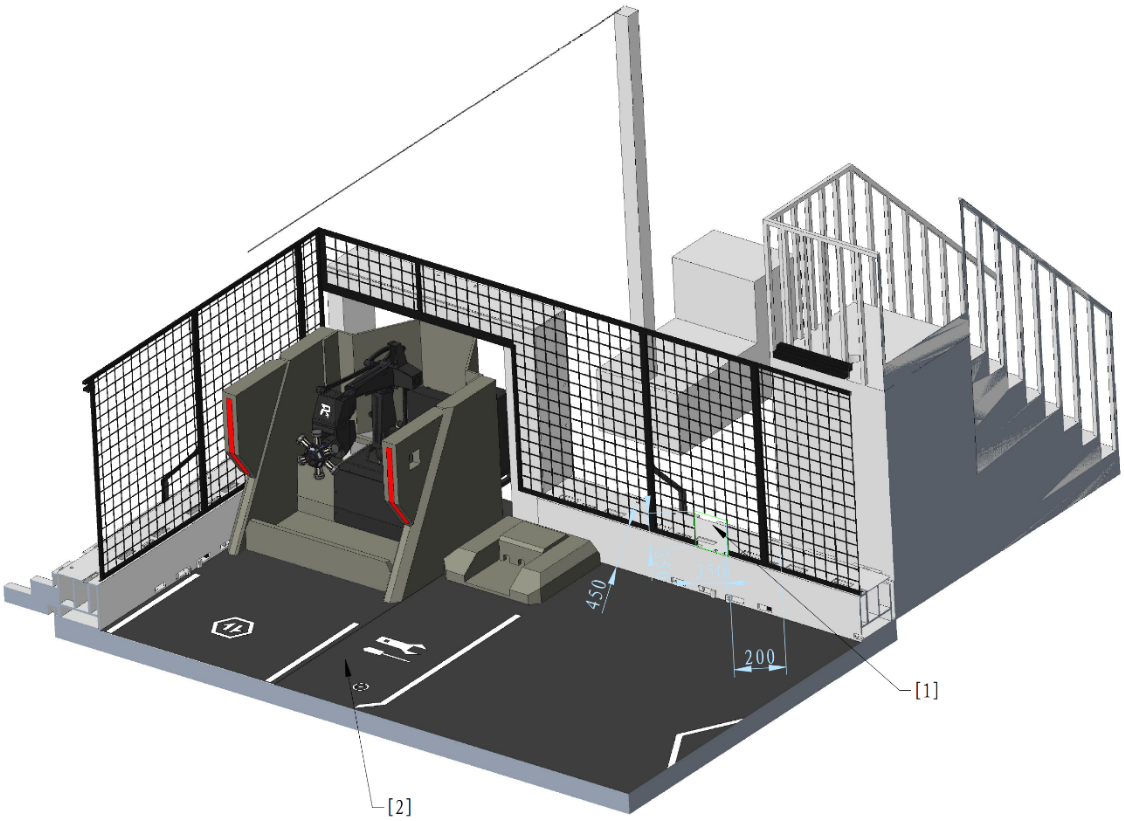


[1] Battlefield Entrance & Exit [2] Steps [3] Radar Foundation

Figure 4-18 Relative Location of the Radar Foundation

4.2.6 Resupply Zone

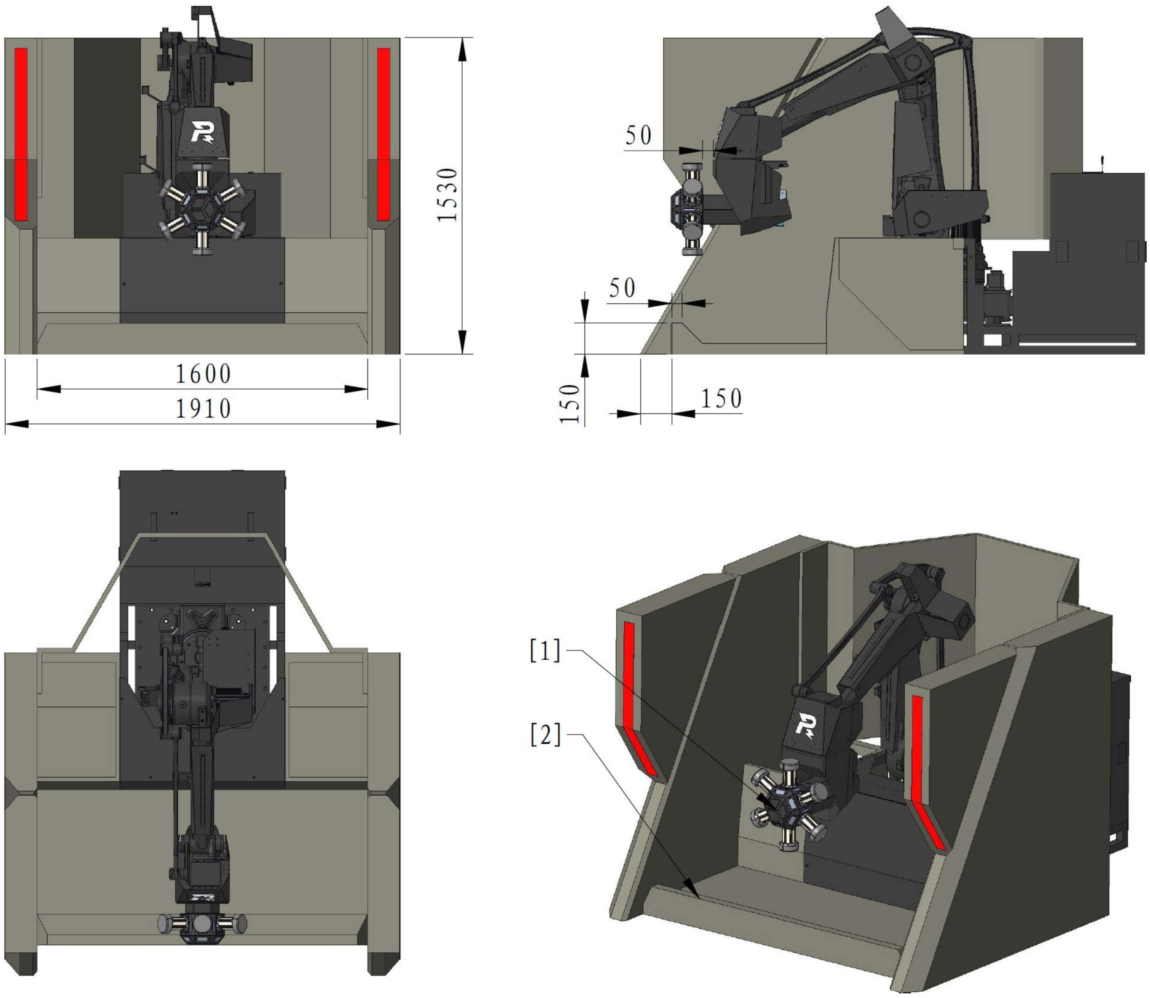
The Resupply Zone includes the Resource Station, Resupply Zone Buff Point, and Wireless Charging Device Zone. It is a key area for robots to perform wireless charging, restore HP, replenish their Projectile Allowance, and obtain Energy Units. A Dart Delivery Window is set in the Perimeter Wall at the edge of the Resupply Zone, allowing robots to pick up Darts from their own side and pass them through the Dart Delivery Window to team members outside the Battlefield for repairs, as shown in the following figure.



[1] Dart Delivery Window [2] Resupply Zone

Figure 4-19 Resupply Zone

The Resource Station is an area where Engineers obtain Energy Units. Six Energy Units are placed there, all held in place magnetically within the Energy Units Zone. Engineers need to remove Energy Units along their rotational symmetry axes. In addition to overcoming gravity, this process requires an additional force of approximately 10N to overcome magnetic attraction.



[1] Energy Units Zone [2] Guard

Figure 4-20 Resource Station

> The actual installed Location of the Energy Unit may deviate in angle around the axis of symmetry from what is shown in the Figure. The Figure is for illustration only. Please refer to the actual placement for the exact arrangement.

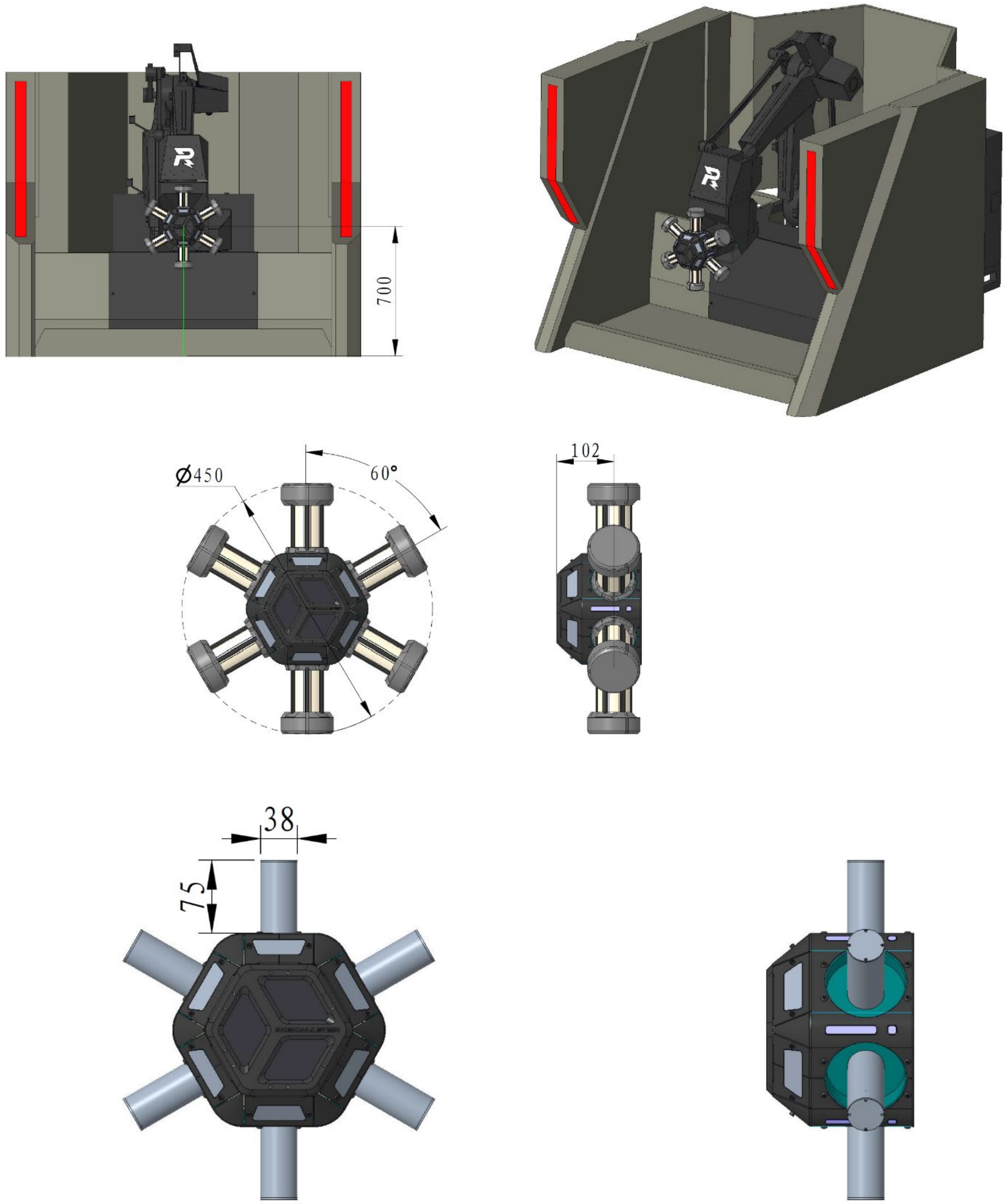
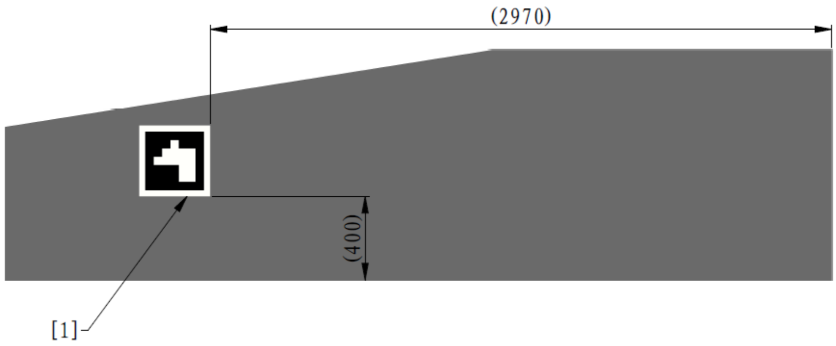
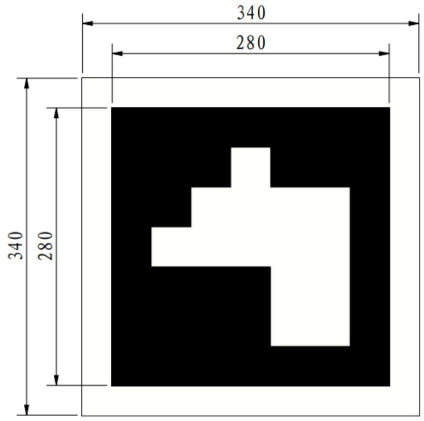


Figure 4-21 Energy Units Zone

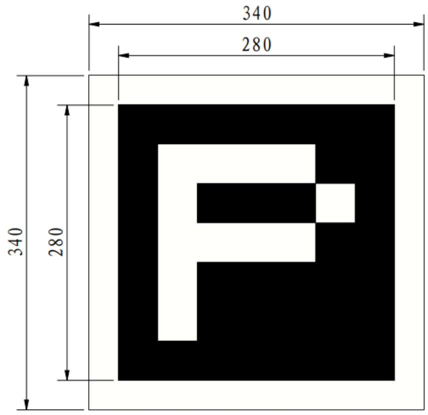
The wall of the Road Zone facing the Resupply Zone contains a metal Visual Marker, enabling the Video Transmitter Module to determine whether a robot is in the Resupply Zone, as shown below.



[1] Visual Marker



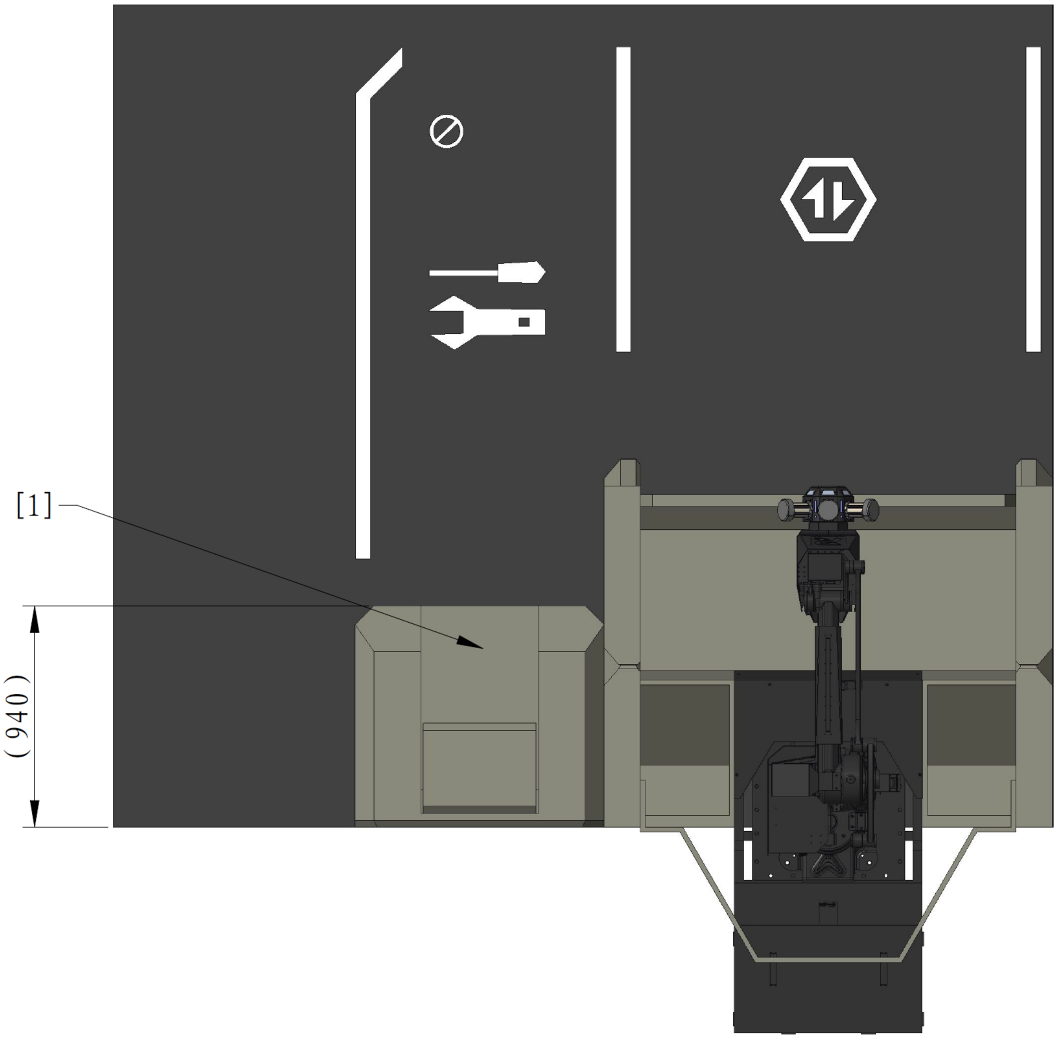
Red Team's Visual Marker



Blue Team's Visual Marker

Figure 4-22 Visual Marker

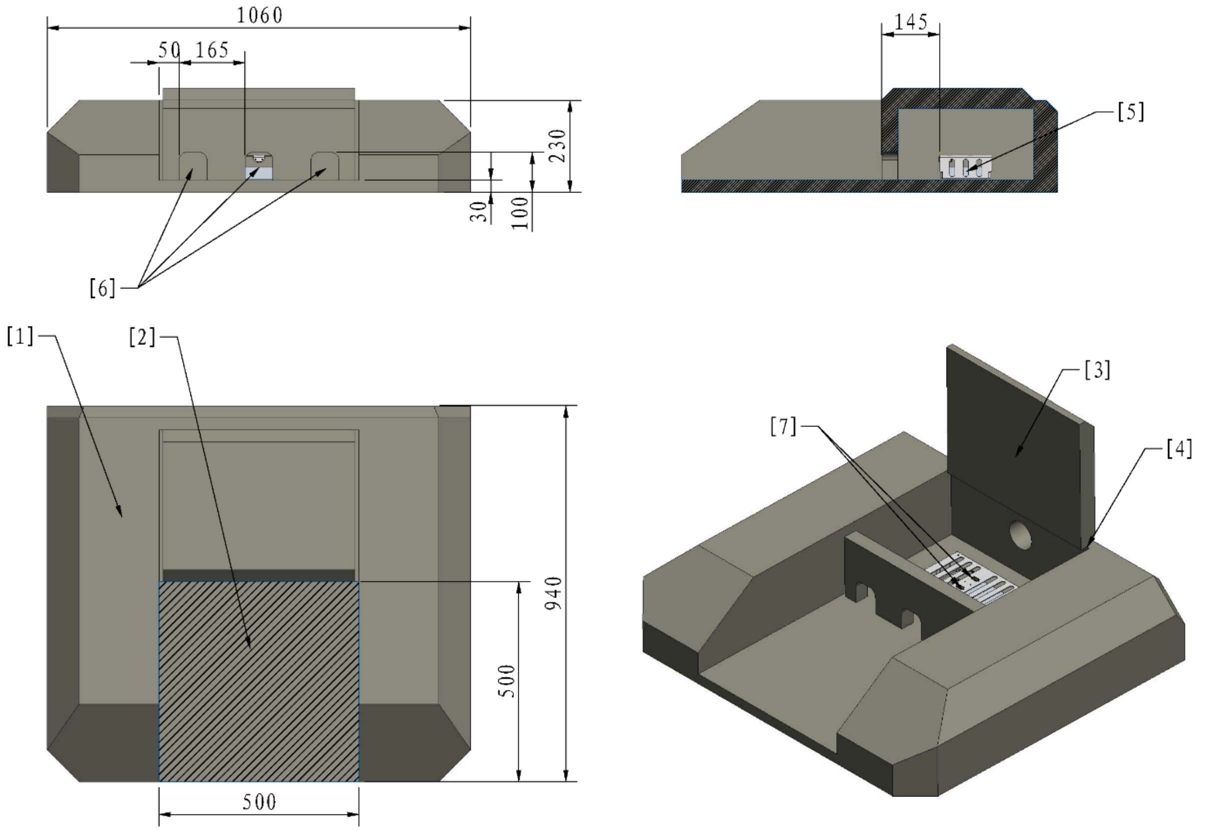
The wireless charging device is placed in the Resupply Zone. The Participating Teams can install their wireless charging devices (Transmitters) in the designated area during the Three-Minute Setup Period, but they must ensure the projection of the devices remains within the area. The specifics are shown in the figure below.



[1] Wireless Charging Device Zone

Figure 4-23 Wireless Charging Device Zone

The RMOC has provided two power supply interfaces with XT60 receptacles in the Wireless Charging Device Zone, offering 24V ± 1V DC power with a maximum total power of 200W.

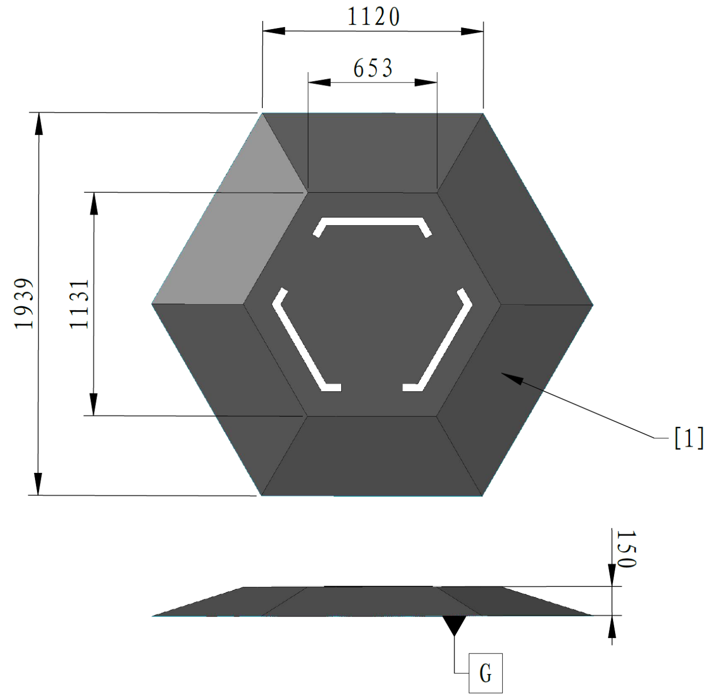


[1]	Perimeter Wall	[2]	Wireless Charging Device Zone	[3]	Cable management flip cover	[4]	Rotating shaft of flip cover
[5]	Power supply	[6]	Cable pass-through hole	[7]	Interfaces with XT60 receptacles		

Figure 4-24 Wireless Charging Device

4.2.7 Fortress

The Fortress is the key defensive area in the Base Zone.



[1] 20° slope

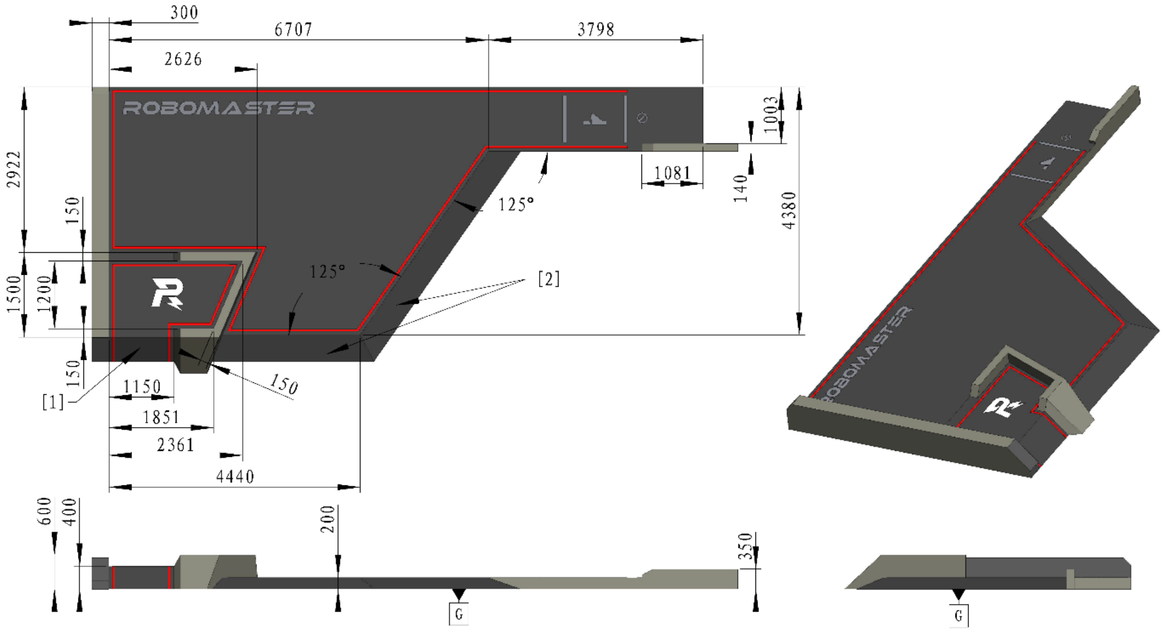
Figure 4-25 Fortress

4.3 Elevated Zone

The Elevated Zone refers to the elevated areas in the Battlefield that are higher than the Battlefield floor. On each half of the Battlefield, there is one Trapezoid-Shaped Elevated Ground designated for the red and blue teams, respectively. Additionally, there is a Central Elevated Ground located at the center of the Battlefield. These Elevated Grounds divide the Battlefield into different zones and create a three-dimensional space.

4.3.1 Trapezoid-Shaped Elevated Ground

The Trapezoid-Shaped Elevated Ground is located near the Landing Pad, with a height of 200 to 400 mm above the Battlefield ground. A Buff Point is located on the Trapezoid-Shaped Elevated Ground, as shown in the figure below.

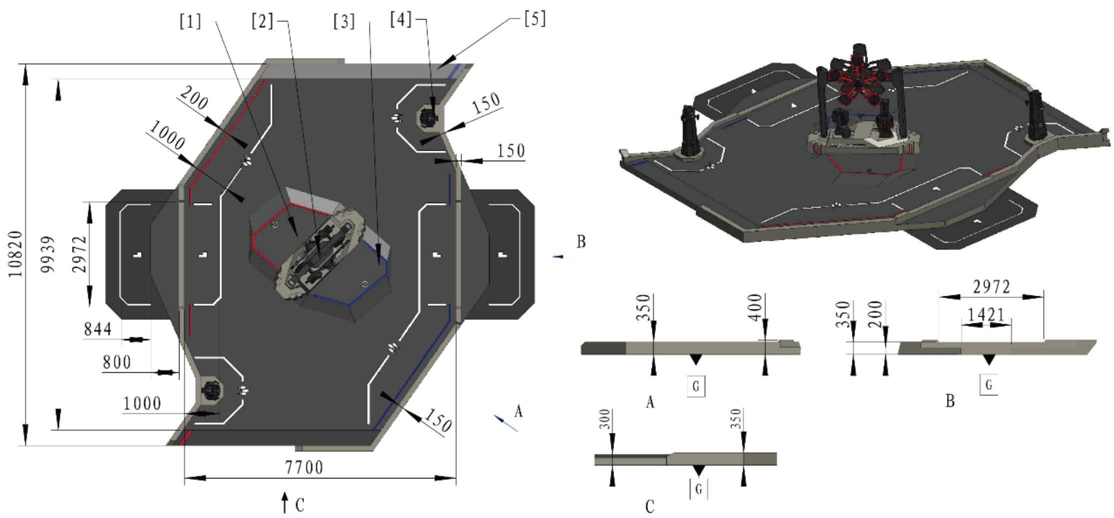


[1] 43° slope [2] 23° slope

Figure 4-26 Trapezoid-Shaped Elevated Ground

4.3.2 Central Elevated Ground

The Central Elevated Ground is located at the center of the Battlefield, with both ends connected to the Road Zone by slopes.



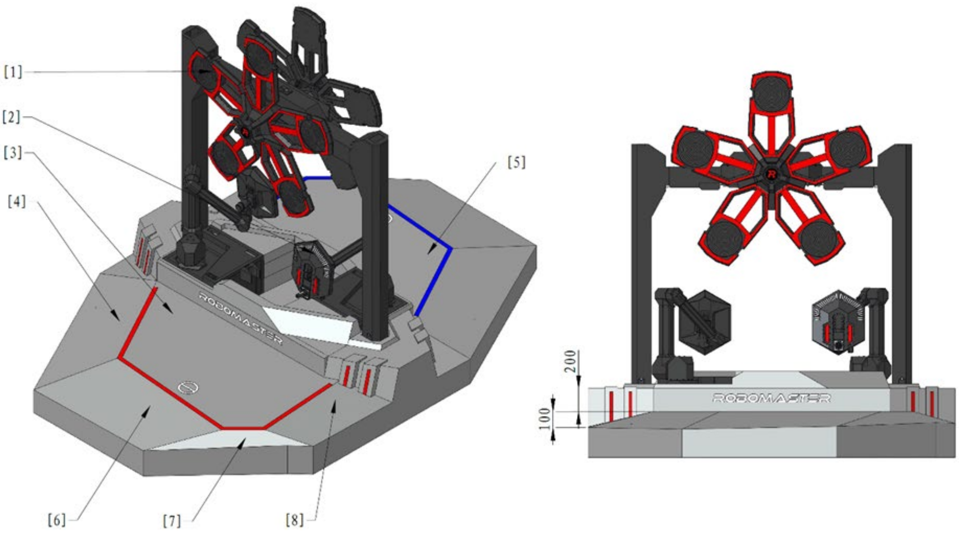
[1] Red Team's Assembly Zone [2] Power Rune [3] Blue Team's Assembly Zone

[4] Outpost [5] 10.5° slope

Figure 4-27 Central Elevated Ground

4.3.2.1 Assembly Zone

The Assembly Zone is located at the center of the Battlefield, directly beneath the Power Rune. The Assembly Zone is divided into red and blue zones. A Tech Core is positioned directly below the Power Rune. Engineers can assemble Energy Units onto the Tech Core. The specifics are shown in the figure below.



[1]	Power Rune	[2]	Tech Core	[3]	Red Team's Assembly Zone	[4]	14° slope
[5]	Blue Team's Assembly Zone	[6]	12° slope	[7]	45° slope	[8]	15° slope

Figure 4-28 Axonometric View of the Assembly Zone

The wooden structural base of the Tech Core throughout this document should follow the appearance shown in “Figure 4-28 Axonometric View of the Assembly Zone“.

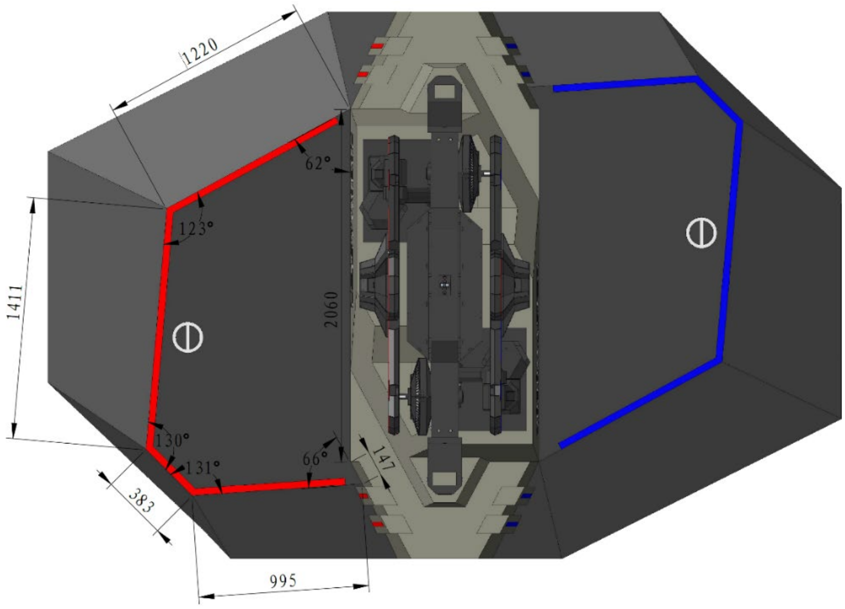


Figure 4-29 Assembly Zone Dimensions

4.3.2.2 Power Rune

The Power Rune is located directly above the Assembly Zone. The Power Rune is powered by the motor and rotates synchronously according to a specific pattern. The Power Rune is divided into red and blue sides, with one side being the Red Team's Power Rune and the other side the Blue Team's Power Rune.

- The Power Rune will have a slight dip in the middle due to its weight. The dip is 0–50mm.
- Due to the viewing angle and transmission gap, a team may see parts of the other team's Power Rune.

The Power Rune consists of five evenly distributed Light Arms. Their locations and dimensions are shown below.

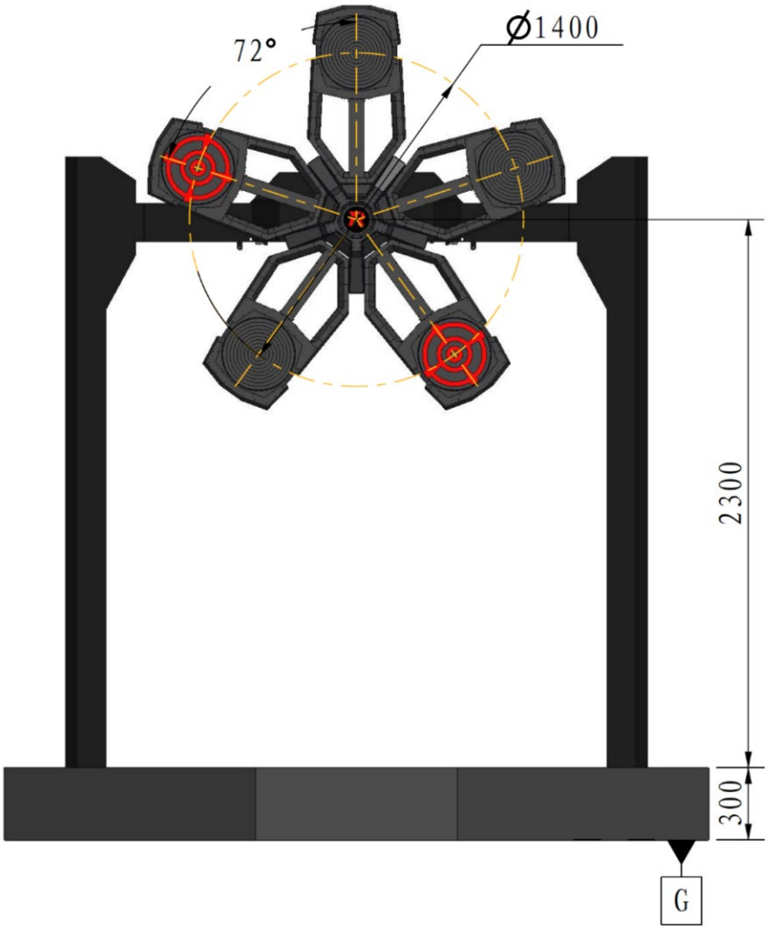


Figure 4-30 Power Rune

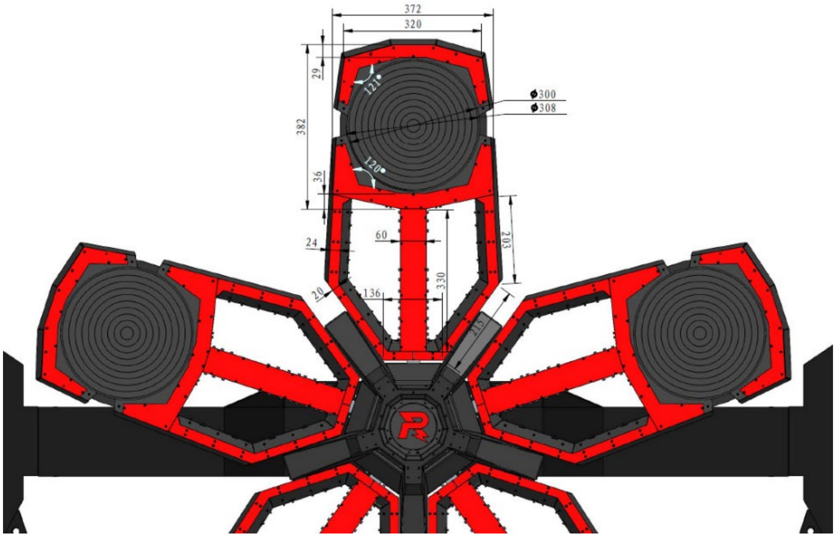


Figure 4-31 Power Rune Light Arm Dimensions

The actual maximum diameter of a Light Arm's round target is 308 mm, and its effective detection diameter is 300 mm.

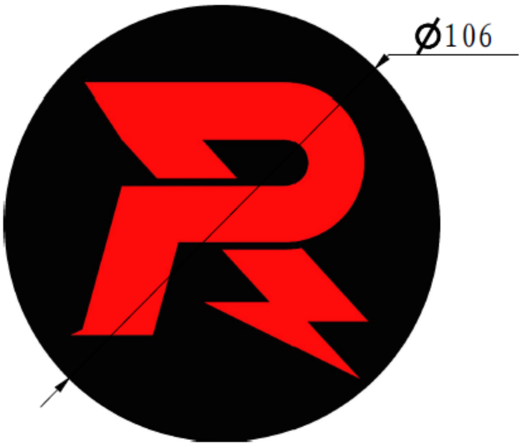
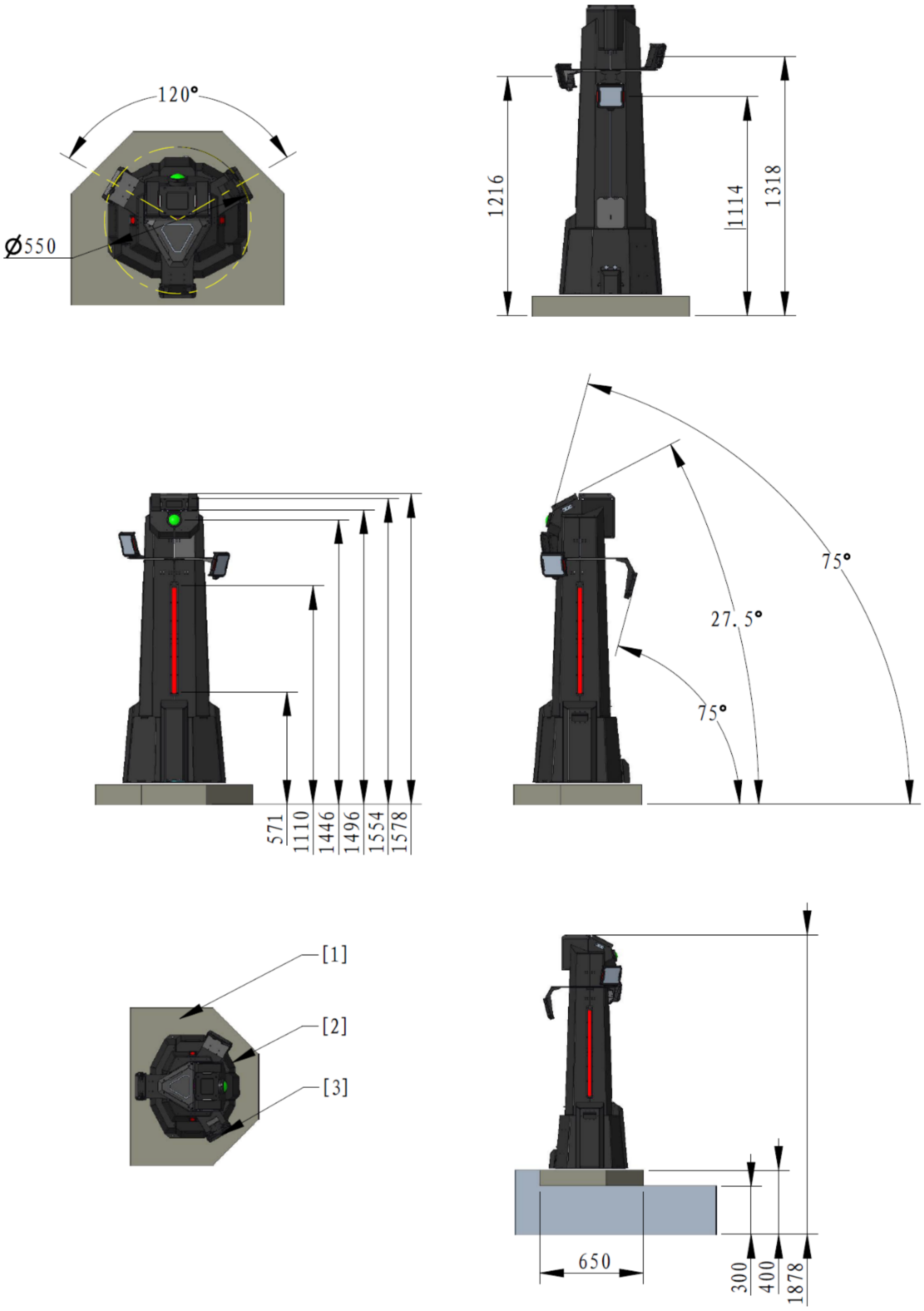


Figure 4-32 Power Rune Central Logo Dimensions

4.3.2.3 Outpost

The Outpost is located on the Outpost Foundation near the Road Launch Ramp and consists of its main body, three Outpost Armor Modules, and a Dart Detection Module. Please refer to the Dart Detection Module graph in "

Figure 4-12 Dart Detection Module". The Outpost Buff Point is located around the Outpost and its specific location is shown below.

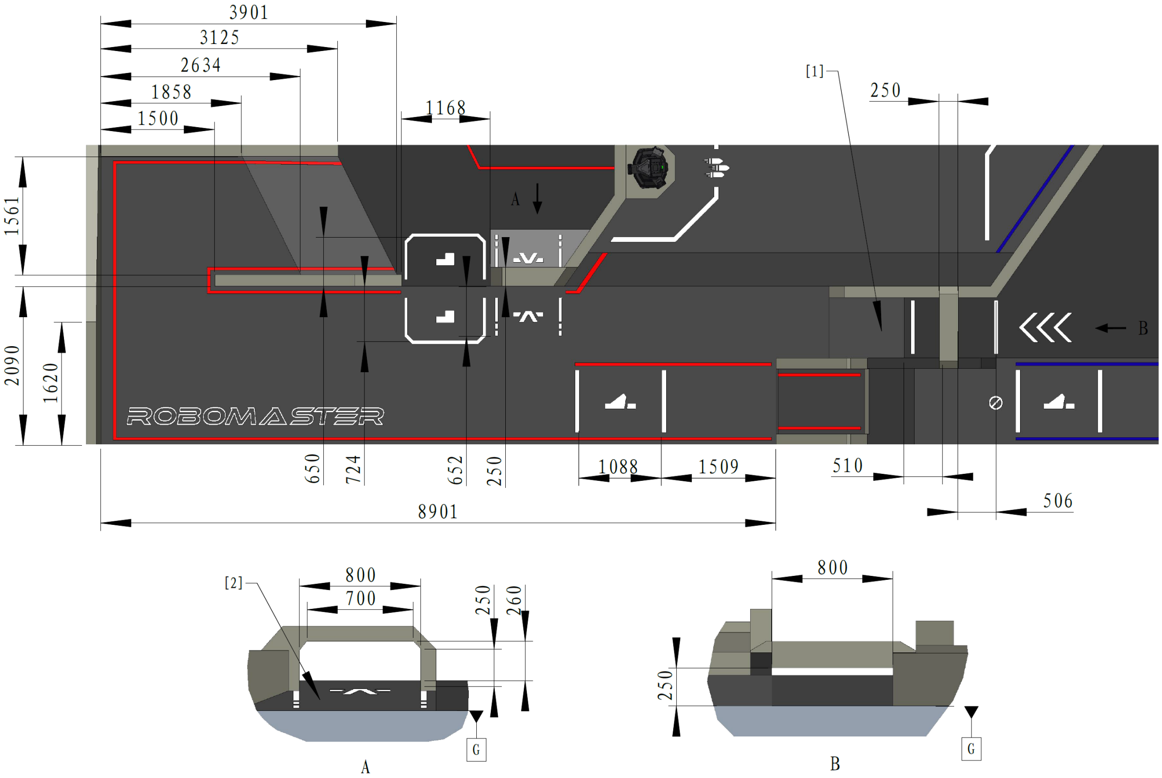


[1] Outpost Foundation [2] Outpost [3] Middle Armor

Figure 4-33 Outpost

4.4 Road Zone

The Road Zone includes the Road, Launch Ramp, Tunnel, and Bumpy Roads.

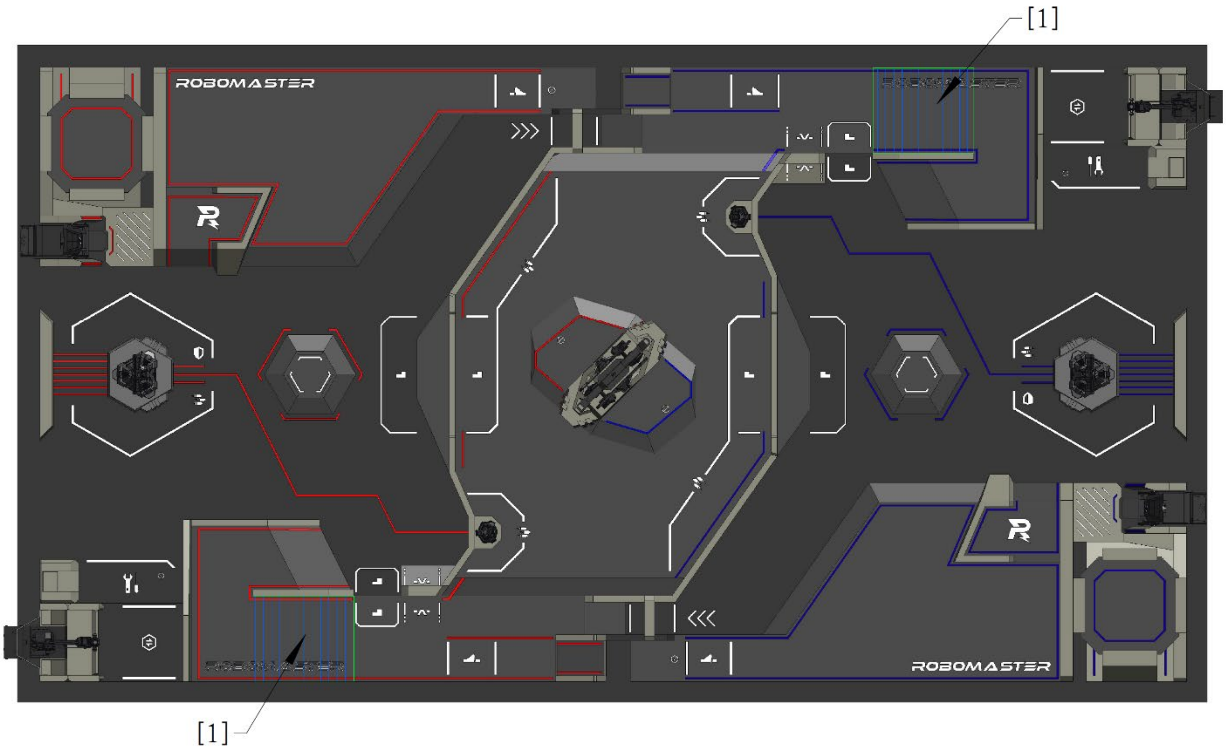


[1] 11° slope [2] 15° slope

Figure 4-34 Road Zone

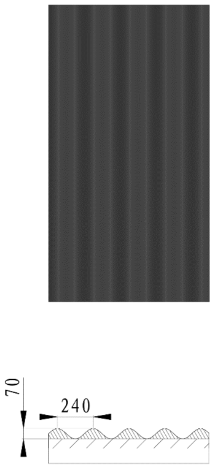
4.4.1 Bumpy Roads

Bumpy Roads are located in the Road Zone, with bumps arranged at regular intervals and in the direction shown in the figure below.



[1] Bumpy Roads

Figure 4-35 Bumpy Roads

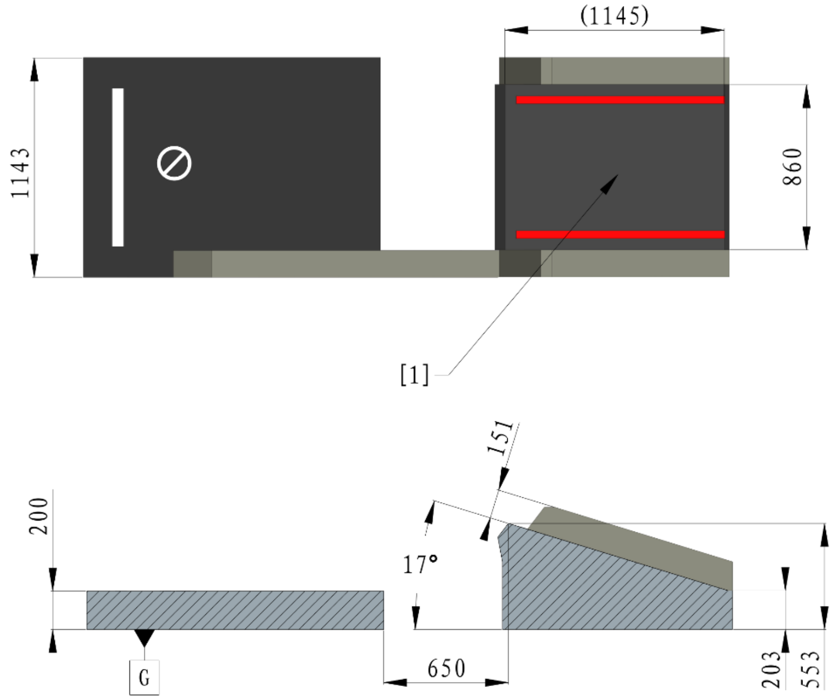


[1] Bump

Figure 4-36 Bump

4.4.2 Launch Ramp

The Launch Ramp is located in the Road Zone, with which robots can fly over the ravine and reach the territory of the other team quickly.



[1] 17° slope

Figure 4-37 Launch Ramp

4.5 Flight Zone

The Flight Zone is the flight area for Drones, which includes the Landing Pad and the air space above it, as well as the air space above the road connected to the Trapezoid-Shaped Elevated Ground.

To ensure its safety, a Drone must be attached to an Aerial Safety Rope during a match. The Aerial Safety Rope is 2.4 m long. The flight distance is restricted by the Snap Ring of the Aerial Safety Rope. The distance of the Snap Ring from the wide edge of one team's side of the Battlefield is approximately 14 m. When a Drone is at the farthest position, the Pilot should not fly it forward any further.

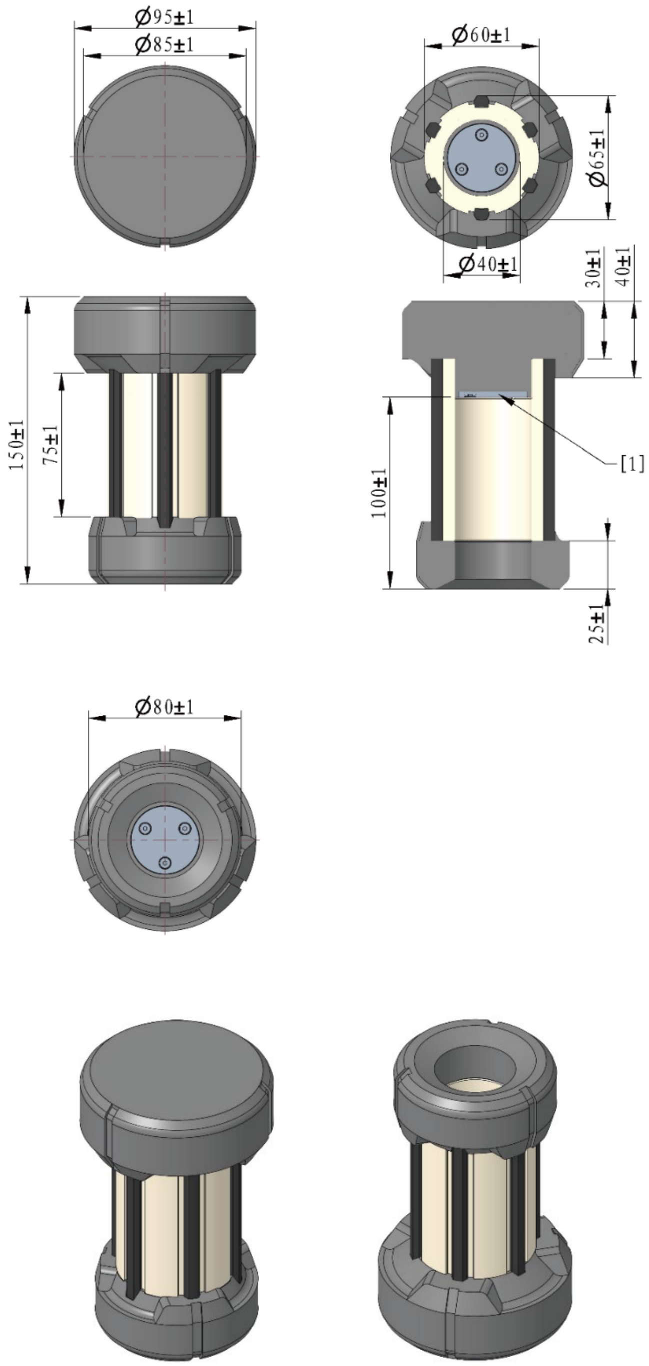
- Drones must be connected to a Drone Withdrawal System with an Aerial Safety Rope during competition to ensure the safety of Drones. A Drone Withdrawal System consists of a wire rope, a withdrawal device, and a retractable tether box. The withdrawal device and the retractable tether box are suspended on the wire rope. A Snap Ring is set on the wire rope about 14 m away from the edge of a team's side of the field. The retractable tether box cannot pass the Snap Ring. The retractable tether box weighs approximately 800 g and contains a 2.4 m soft tether with a constant elastic force. The static elastic force of the tether is less than 5 N. The end of the tether is connected to a Drone with a hook.
- When the Drone is flying normally, the withdrawal device that hangs on the wire rope directly above the Pilot Operation Room does not move with the Drone. At the same time, only the retractable tether box moves with the Drone. When the Drone crashes or experiences other situations where the Referee determines that it is possible and necessary to recover the Drone, the withdrawal device will be activated and move to the position of the retractable tether box to recover the Drone.
- The Retractable Tether Box has no rotary interface. If the Drone performs too many rotations in a single direction, the Retractable Tether Box may jam.

4.6 Miscellaneous

4.6.1 Mobile Component

An Energy Unit is a Mobile Component, which can be grabbed and carried by robots. An Energy Unit is a hollow cylindrical component weighing approximately 400 g ± 50 g, with asymmetrical ends.

As the competition progresses, the surface of Mobile Components may suffer mild damage or gather dust. The Participating Teams must adapt to such conditions, but Mobile Components that are severely stained or damaged will not be used in the official competition.



[1] Iron plate

Figure 4-38 Energy Unit

4.6.2 Projectile

Robots deal damage by firing Projectiles to attack Armor Modules. The parameters and scenarios of use for Projectiles in the competition are as follows:

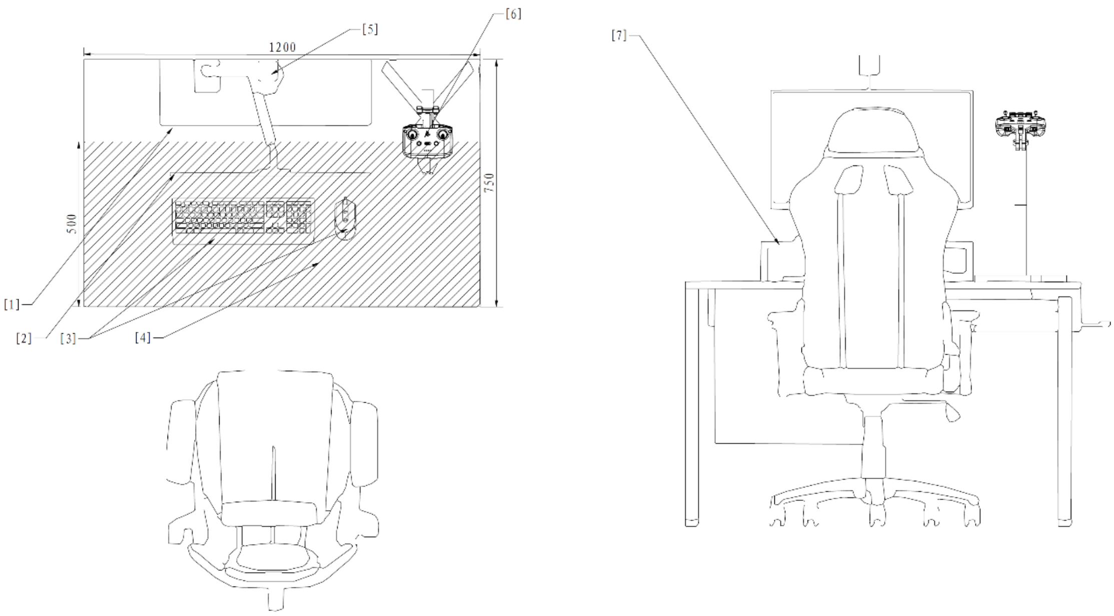
Table4-1 Projectile Parameters and Scenarios of Use

Parameters	42mm Luminous Projectile	17mm Fluorescent Projectile
Appearance	Similar to a golf ball	Sphere
Color	Semilucent	Yellow-green
Size	42.5mm ± 0.5mm	16.8mm ± 0.2mm
Weight	44.5g ± 0.5g	3.2g ± 0.1g
Shore Hardness	90A ± 5A	90A ± 5A
Material	Plastic (TPE)	Plastic (TPU)
Scenarios of Use	RMUC 2026 throughout the entire process	RMUC 2026 throughout the entire process
Rated Quantity	100 Projectiles per round	Limited, but no fewer than 4,000 Projectiles per round

4.6.3 Operation Room

The Operation Room is located near the perimeter of the Battlefield, and serves as the area for the Operators during the competition. The Operation Room consists of the Main Operation Room and the Pilot Operation Room. Each Operator's desk in the Main Operation Room measures 120 cm in length and 75 cm in width, and is equipped with the corresponding number of computers. Each computer comes with a matching monitor, mouse, keyboard, Video Transmitter Module (Receiver) and its mount, three USB Type-A Receptacle ports, one RS232 Plug port, one 1GbE (RJ45) Ethernet port, a wired headset, and other Official Equipment. The Pilot Operation Room is equipped only with goggles to protect against laser exposure from the opponent's Radar.

The Main Operation Room desktop layout is shown below. The Operation Room desktop features a monitor, broadcast camera, Video Transmitter Module (Receiver) and its stand, keyboard, mouse, and equipment area. The monitor, broadcast camera, and interface are fixed. The stand for the Video Transmitter Module (Receiver) is mounted on the desk, and the Operator can operate the Video Transmitter Module (Receiver) within the stand's movable range. The keyboard and mouse can be moved freely, and the Operator can place the Custom Controller and Custom Client in the designated area. The layout is shown below.



[1]	Interface	[2]	Monitor	[3]	Keyboard and mouse	[4]	Equipment area
[5]	Broadcast camera	[6]	Video Transmitter Module (Receiver) and its stand	[7]	Shelf		

Figure 4-39 Layout of the Main Operation Room

If a participating team removes the Video Transmitter Module (Receiver), resulting in video transmission interruption or any other effects, they must bear full responsibility.

5 Competition Mechanism

5.1 HP Loss and Penalty for Exceeding Limit

5.1.1 Attack Damage or Collision

Ground Robots, Base, and Outpost may lose HP when their Armor Modules are attacked or collide with other objects. The mechanism is as follows:

The Dart Detection Module detects attacks from Darts and 42mm Projectiles through the Armor Modules and the phototubes.

The Armor Module can detect Projectile attacks. The minimum detection interval for 17mm Projectiles is 50ms. The minimum detection interval for 42mm Projectiles is 200ms.

The Projectile needs to come into contact with the attack surface of the Armor Module at a certain speed in order to be successfully detected. The speed ranges in which Armor Modules can effectively detect different types of Projectiles are shown in the table below.

Table 5-1 Speed Ranges of Different Types of Projectiles for Effective Detection by Armor Modules

Armor Module	17 mm Projectile	42 mm Projectile
● Large Armor Module and Small Armor Module ● Base Armor Module ● Outpost Armor Module	Higher than 12 m/s	Higher than 10 m/s
● Power Rune Armor Module	Higher than 12 m/s	-

- In an actual match, the normal speed of the Projectile that touches the Armor Module attack surface is different from its initial launching speed due to the Projectile's speed decay and its incident angle not being normal to the Armor Module attack surface. Damage detection is based on the normal component of the Projectile's speed upon contact with the Armor Module attack surface.
- The Participating Teams are not allowed to hit a Power Rune with 42mm Projectiles or Darts.
- If a Hero has not fired a 42mm Projectile for four consecutive seconds, the opponent's robots, Outpost, and Base will become immune to 42mm Projectile damage until the Hero fires a 42mm Projectile again.
- To ensure detection accuracy and reduce false positives, the Referee System needs to wait for the launch information of a 42mm Projectile to match in some cases. Consequently,

after an Armor Module detects a strike from a 42mm Projectile, damage calculation may be slightly delayed.

When the Armor Module of the Ground Robots, the Base, or an Outpost has a collision, there is a chance that the collision is identified as Projectile damage. However, causing damage through collisions (including ramming with robots or throwing objects) is not allowed.

Without buffs, the original damage is shown in the table below.

Table 5-2 HP Loss Mechanism for Attack Damage or Collision

Damage Type / Target	42 mm Projectile	17 mm Projectile	Dart	Collision
Robot Armor Module	200	20	-	2
Base Armor Module	200	● Upper front Armor Module: 5 ● Remaining five Armor Modules: 20	-	-
Base Dart Detection Module	-	-	● Fixed target: 200 ● Random fixed target: 300 ● Random moving target: 625 ● Terminal moving target: 1,000	-
Outpost Middle Armor Module	200	20	-	-
Outpost Dart Detection Module	-	-	750	-

5.1.2 Exceeding Initial Launching Speed Limit

The Launching Mechanisms of Hero, Infantry, Drone, and Sentry will be locked for a certain duration if the robots exceed the initial launching speed limit. The duration does not accumulate.

Locks and unlocks due to exceeding the initial launching speed limit are independent from locks and unlocks caused by other reasons. Assume the initial launching speed limit is V0 (m/s), and the actual speed detected by the Referee System is V1 (m/s). The specific lock durations are shown in the table below.

Table5-3 Penalty Mechanism for Exceeding Initial Launching Speed Limit

17 mm Projectile	42 mm Projectile (not in Deployment Mode)	42 mm Projectile (in Deployment Mode)	Lock Duration
$0 < V1 - V0 < 5$	$V0 < V1 \leq 1.1 \times V0$	$V0 < V1 \leq 18$	15s
$5 \leq V1 - V0 < 10$	$1.1 \times V0 < V1 \leq 1.2 \times V0$	-	20s
$10 \leq V1 - V0$	$1.2 \times V0 < V1$	$18 < V1$	The remainder of the current round

5.1.3 Exceeding Barrel Heat Limit and Cooling

The Launching Mechanisms of Hero, Infantry, Drone, and Sentry will be locked when the robots exceed the Barrel Heat Limits. Locks and unlocks due to exceeding the Barrel Heat Limits are independent from locks and unlocks caused by other reasons. The mechanism is as follows:

Set Q_0 as the Barrel Heat Limit and Q_1 as the current Barrel Heat. For each 17 mm Projectile detected by the Referee System, the current Barrel Heat Q_1 is increased by 10 (regardless of the 17 mm Projectile's initial launching speed). For each 42 mm Projectile detected, the current Barrel Heat Q_1 is increased by 100 (regardless of the 42 mm Projectile's initial launching speed). The Barrel Heat cools at a frequency of 10 Hz. The cooling value per detection cycle is equal to the cooling rate/10.

When $Q_2 > Q_1 > Q_0$, video transmission of the robot Operator's FPV operating interface will be disconnected, and the Launching Mechanism of the robot will be locked. These restrictions will remain in place until $Q_1 = 0$.

When $Q_1 > Q_2$, the Launching Mechanism of the robot will be locked for the remainder of the current round.

For the 17 mm Launching Mechanism, $Q_2 = Q_0 + 100$.

For the 42 mm Launching Mechanism, $Q_2 = Q_0 + 200$.

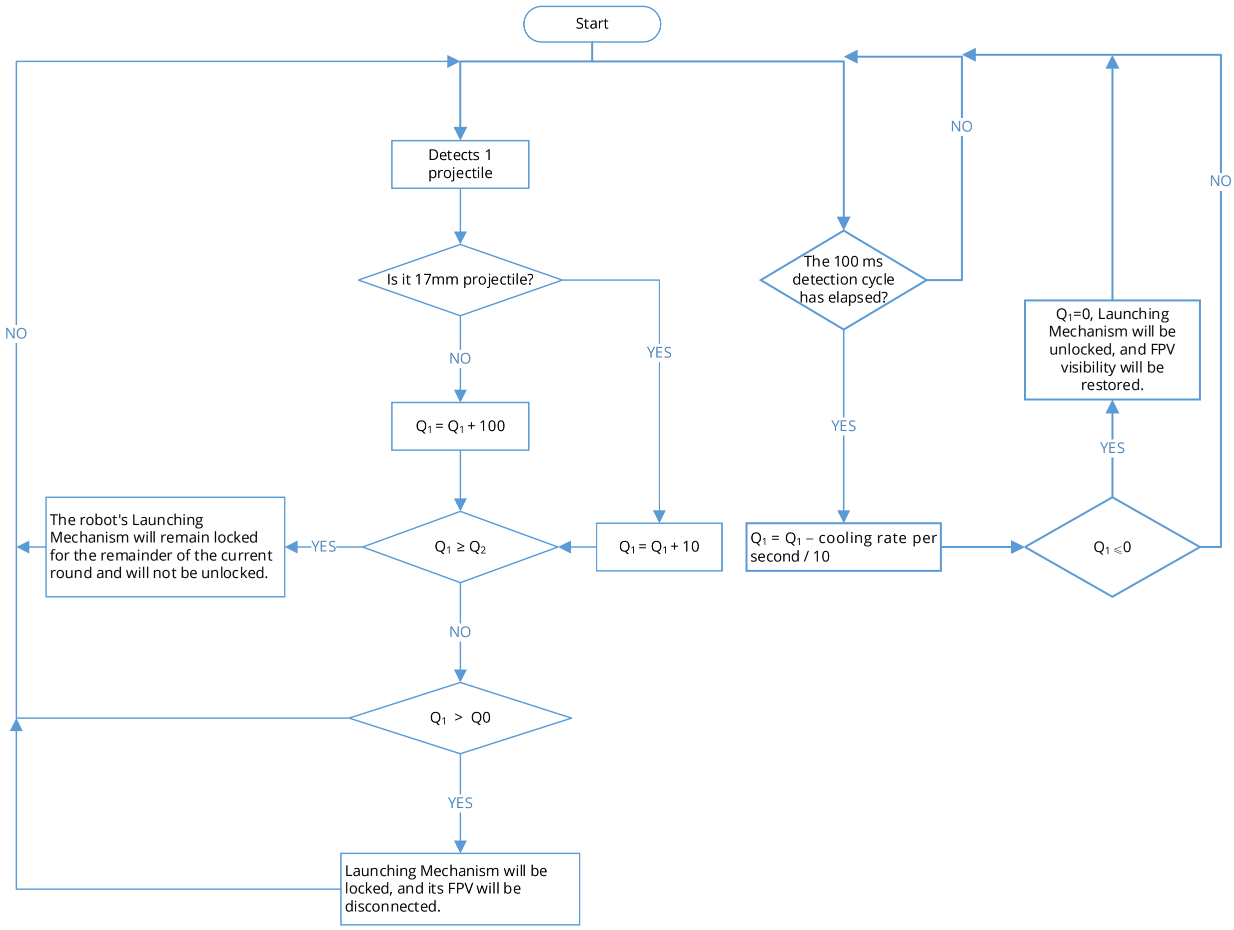


Figure 5-1 Exceeding Barrel Heat Limit and Cooling Logic

When video transmission is disconnected due to overheating, the Mini-map and custom UI remain available.

5.1.4 Exceeding Chassis Power Consumption Limit

The chassis power of robots will be continuously monitored by the Referee System, and the robot chassis needs to run within the Chassis Power Limit. When a Ground Robot exceeds the Chassis Power Consumption Limit, Buffer Energy will be deducted from that robot. Upon the exhaustion of Buffer Energy, if the chassis power is still above the limit, the chassis will be powered off for five seconds. The Referee System calculates chassis power consumption at a frequency of 10 Hz.

Let Z represent Buffer Energy, Q represent the Buffer Energy Limit, P_r represent instantaneous Chassis output power, and P_l represent the power limit. Q = 60 J

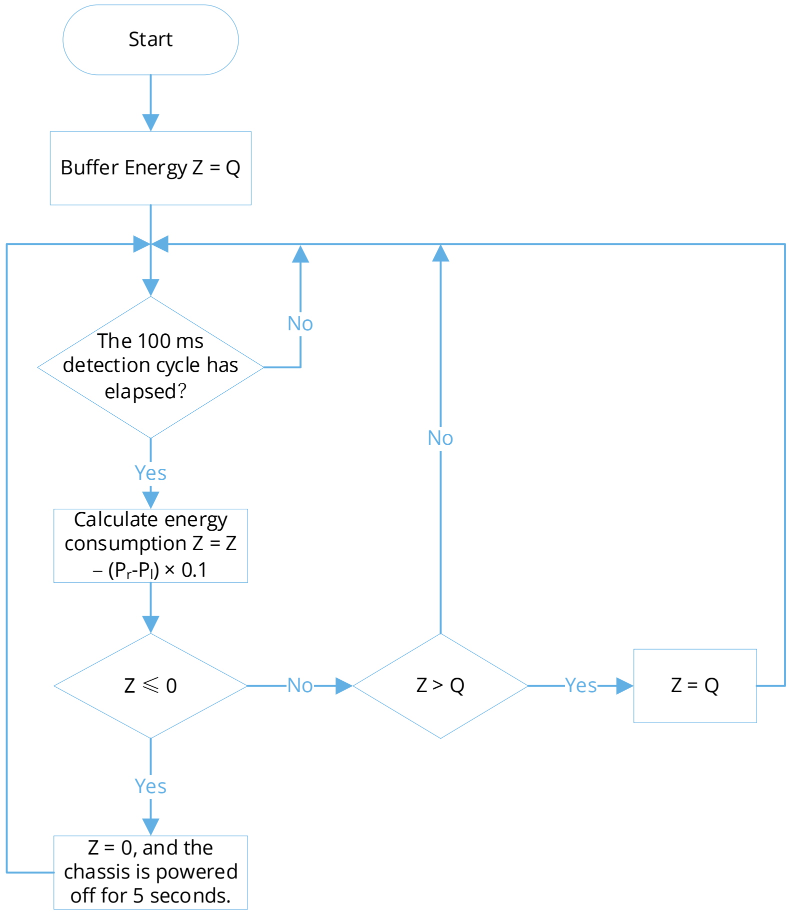


Figure 5-2 Robot Chassis Power Detection and Penalty for Exceeding Limit

5.1.5 Referee System On-board Module Disconnection

The Participating Teams must mount the corresponding Referee System Module on their robots in accordance with the requirements of the RoboMaster 2026 University Series Robot Building Specifications Manual, and ensure stable connection between each module of the Referee System and the server throughout the competition. The Referee System server detects the connectivity of each module at a frequency of 2 Hz.

5.1.5.1 Irregular Disconnection

If a robot's Main Control Module gets disconnected from the server during a competition, it can reconnect to the competition. Experience Points and the Level will continue to be calculated during the disconnection period.

Table5-4 Penalty Mechanism for Robots in Irregular Disconnection State

Robot Type	**Consequences of Abnormal Disconnection**
Ground Robot	● When the Launching Mechanism (if any), Gimbal, and Chassis are powered off, 5% of the Maximum HP is dealt for each second elapsed until it drops to 0.

Robot Type	Consequences of Abnormal Disconnection
	● The RFID Interaction Module is disabled. ● The robot no longer detects any damage caused by collision and Projectile strikes and any HP Loss. ● The Respawn Timer stops. ● Inter-robot Communication is disconnected.
Drone	● The Launching Mechanism is powered off and Air Support cannot be requested. ● The system does not detect illumination from the laser launching device. ● Video transmission is disconnected. ● Inter-robot Communication is disconnected.
Radar	● Inter-robot Communication is disconnected. ● The laser launching device and its actuator are powered off.
Dart	● If it is disconnected when the Dart Launching Station gate closes, the Dart Launching Station gate cannot be opened. ● If it is disconnected when the Dart Launching Station gate is open, the Dart Launching Station gate will immediately close and cannot be opened.

5.1.5.2 Armor Module or Supercapacitor Management Module Disconnection

If a Ground Robot's design or structural issues cause the disconnection of its Armor Module or Supercapacitor Management Module, the robot will lose HP.

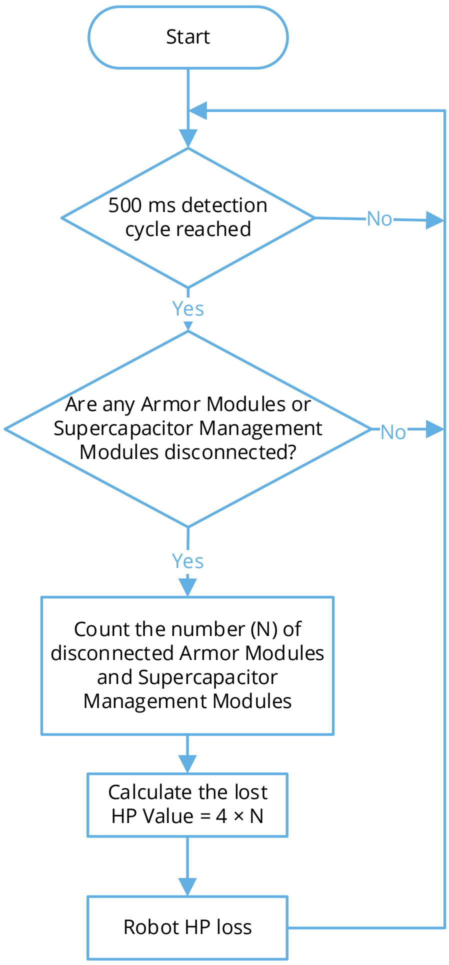


Figure 5-3 HP Loss Mechanism for Armor Module and Supercapacitor Management Module Disconnection

5.1.5.3 Speed Monitor Module Disconnection

If a Speed Monitor Module (17 mm or 42 mm Projectile) mounted on a Hero, Infantry, Sentry, or Drone goes disconnected, the robot's Launching Mechanism (17 mm or 42 mm Projectile) will be immediately locked. Locks and unlocks caused by the Speed Monitor Module going disconnected are independent from locks and unlocks caused by other reasons.

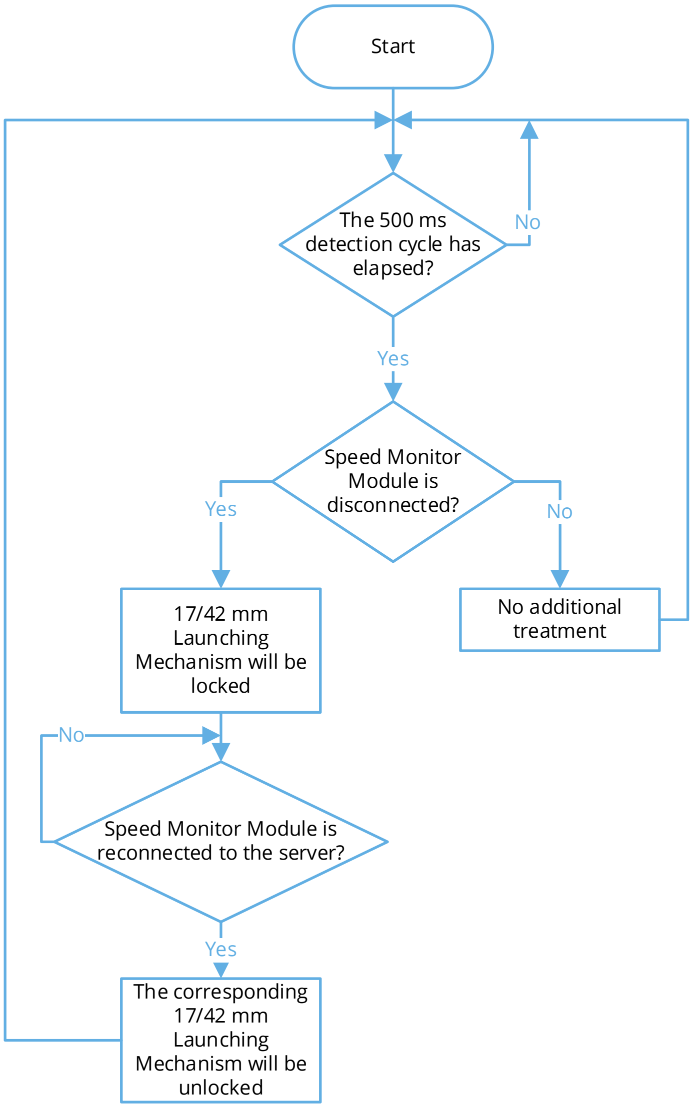


Figure 5-4 Power-off Mechanism for Speed Monitor Module Disconnection

5.1.5.4 UWB Positioning Module Disconnection

If a UWB Positioning Module mounted on a Hero, Engineer, Infantry, Drone, or Sentry goes disconnected, the opponent Radar will always consider its identifying mark as “accurate”, while the Radar on the robot's own side will always consider its identifying mark as “inaccurate”.

5.1.5.5 Laser Detection Module Disconnection

If a Laser Detection Module mounted on a Drone is disconnected, the Drone will be considered to be continuously illuminated by the opponent's laser launching device throughout the Air Support period.

5.1.6 Penalty Damage

When a Ground Robot receives a Yellow Card, Red Card, or other Penalty, it may lose HP. For details, see “7.1 Penalty System“.

5.2 HP Recovery and Respawn Mechanism

Except when in the Irregular Disconnection state or after being ejected, Ground Robots can recover their HP and respawn.

5.2.1 HP Recovery Mechanism

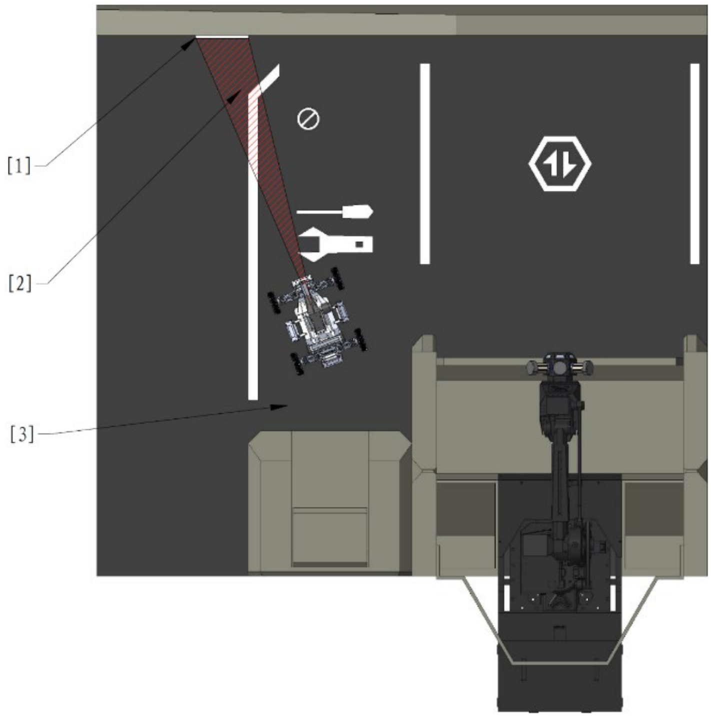
HP Recovery in the Resupply Zone

A Ground Robot that occupies the Resupply Zone Buff Point of its team will receive an HP Recovery Buff, which is 10% of its Maximum HP per second. After four minutes into the match, when robots are out of combat and occupy their own Resupply Zone Buff Point, they will receive an HP Recovery Buff equal to 25% of their maximum HP per second; if the robots are not out of combat, the 25% per second HP Recovery Buff will immediately expire.

Remote Exchange for HP

When a Hero, Infantry, or Sentry is out of combat, they have the option to remotely exchange Gold Coins for HP. Six seconds after a remote HP exchange is confirmed, the robot will receive an additional 60% of its current HP, without exceeding the Maximum HP.

- If the robot is defeated within six seconds after a remote HP exchange is confirmed, the exchange will lapse and the Gold Coins will not be returned.
- When a robot's RFID Interaction Module detects the RFID Interaction Module Card of the Resupply Zone, or its Video Transmitter Module recognizes a Visual Marker of the Resupply Zone when the robot is located in a Resupply Zone defined by the rule, the robot will be deemed to be in the Resupply Zone. An alive robot in its team's Resupply Zone is deemed to have occupied the Resupply Zone Buff Point.
- The Video Transmitter Module is only able to recognize a Visual Marker when the Visual Marker is in the image center of the Video Transmitter Module and the Video Transmitter Module has no obvious roll angle.



[1] Visual Marker [2] Video Transmission Range of Robots [3] Recognizable Zone

Figure 5-5 Relative Location of the Robot's Video Transmitter Module and Visual Marker

5.2.2 Respawn Mechanism

When a robot is defeated, its Barrel Heat is reset to 0, and its Buffer Energy is reset to the Buffer Energy Limit. A defeated robot can respawn by completing the Respawn Timer or exchanging Gold Coins for Instant Respawn before the Respawn Timer is completed.

- When seeking to complete the Respawn Timer, the length of the Respawn Timer is related to the remaining time of the competition when the robot is defeated and the robot's cumulative number of Instant Respawn exchanges. Its formula is as follows:

$$\text{Respawn Timer}=10+\frac{420\text{-Remaining Time}}{10}+20\times\text{Cumulative Respawn Times}$$

The remaining competition time is measured in seconds. The number is rounded off to the nearest integer.

Example: If a robot is defeated with 120 seconds remaining in the match, and it has already used Instant Respawn twice before, the length of its Respawn Timer for this time will be as follows:
$10+\frac{420-120}{10}+20\times2=80$·

The Respawn Timer of a robot starts immediately after it is defeated. From the moment the robot is defeated, the Respawn Timer increases by 1 per second; if the robot is in the Resupply Zone Buff Point or the Base HP of its side is below 2,000, the Respawn Timer increases by 4 per second.

When seeking to complete the Respawn Timer:

- The respawned robot retains its Level and Experience Points from before its defeat and remains invincible for 30 seconds.
- Its HP is restored to 10% of its Maximum HP.
- The robot enters the "Weakened" state.
  - The Launching Mechanism is locked.
  - The robot cannot occupy any Buff Points or rebuild Outposts.

Thereafter, when its RFID Interaction Module detects the RFID Interaction Module Card of an occupiable Outpost, Base Buff Point, or Resupply Zone Buff Point, its "Weakened" state will be lifted. If the Invincible state gained from the current respawn has already lasted for more than 10 seconds, the Invincible state ends immediately. If the Invincible state gained from the current respawn has not yet reached 10 seconds, it ends when the 10-second duration has elapsed.

- When exchanging Gold Coins for Instant Respawn:
  - The respawned robot retains its Level and Experience Points from before its defeat and remains invincible for three seconds.
  - Its HP is restored to 100% of the Maximum HP.
  - Its Launching Mechanism is immediately unlocked.
  - Its Chassis Power Limit is doubled (up to 200 W), for 4 seconds.

5.3 Economy System

5.3.1 Overview of the Economy System

During the Three-Minute Setup Period, Ground Robots can pre-load 42 mm and 17 mm Projectiles while Drone can pre-load 17 mm Projectiles only.

At the start of a round, each team receives a certain quantity of initial Gold Coins. As the competition progresses, both sides passively gain Gold Coins, as detailed in the table below. During the match, additional Gold Coins can also be obtained by installing energy units. Gold Coins can be exchanged for Air Support, 17 mm and 42 mm Projectile Allowance, remote HP Recovery, and Instant Respawn.

Table5-5 Regular Gold Coin Gains

Game Countdown	Red Team	Blue Team
06:59	400 (initial)	400 (initial)
05:59	50	50
04:59	50	50

Game Countdown	Red Team	Blue Team
03:59	50	50
02:59	50	50
01:59	50	50
00:59	150	150

The ratings obtained by a participating team for its “Project Documents” and “Technical Proposal” during the Final Robot Assessment will impact the team's initial Gold Coin quantity for each round during the Regional Competition. The correspondence between the impact and the ratings is as follows:

Table5-6 Impact of Project Document Ratings

Project Document Rating	Degree of Impact
S	+50
A	+25
B	0
C	–25
D	–50

Table5-7 Impact of Technical Proposal Ratings

Technical Proposal Rating	Degree of Impact
S	+150
A	+75
B	0
C	–25
D	–50

Table5-8 Exchange Rules

Exchange Item	Exchange Ratio	Exchange Limit
17 mm Projectile Allowance	● Remote Exchange: 150 Gold Coins / 100 rounds ● Non-Remote Exchange: 10 Gold Coins / 10 rounds	1,000 rounds per team

Exchange Item	Exchange Ratio	Exchange Limit
42 mm Projectile Allowance	● Remote Exchange: 150 Gold Coins / 10 rounds ● Non-Remote Exchange: 10 Gold Coins / 1 round	100 rounds per team
Air Support Time	1 Gold Coin / 1 second	No restrictions
HP (Remote Exchange)	$50+\text{ROUNDUP}(\frac{420\text{-Remaining Time}}{60}\times 20)$ Gold Coin per use	Unlimited
Instant Respawn	$\text{ROUNDUP}\left(\frac{420\text{-Remaining Time}}{60}\right)\times 80+\text{Robot Level}\times 20$ Gold Coin / 1 unit	Unlimited

- Time is measured in seconds.
- ROUNDUP means to round up to the nearest integer.
- The exchange limit of the 17 mm Projectile Allowance refers to the maximum Projectile Allowance obtained through the exchange of Gold Coins.

5.3.2 Projectile Allowance Mechanism

For every round of Projectiles fired by a robot, the Projectile Allowance corresponding to the type of Projectiles fired is reduced by one round.

In the following special circumstances, the Armor Modules of the opponent's robots, Outpost, and Base will become immune to 42 mm Projectile damage until the Hero is alive and has a Projectile Allowance greater than 0:

- Three seconds after the Hero becomes defeated or abnormally disconnected.
- When the Hero launches a 42 mm Projectile despite having a Projectile Allowance of 0 (it launches more Projectiles than its allowance).
- After the Hero launches the third 42 mm Projectile after it is defeated.

Each robot's Initial Projectile Allowance and its mechanism are shown in the table below.

Table5-9 Projectile Allowance for Robots

Robot	Initial Projectile Allowance	Projectile Allowance Mechanism
Hero	0	Projectile Allowance can be obtained via exchange in the Resupply Zone, Base Buff Point, and Outpost Buff Point, or via remote exchange.
Infantry	0	

Robot	Initial Projectile Allowance	Projectile Allowance Mechanism
Sentry	300	Projectile Allowance can be obtained via exchange in the Resupply Zone, Base Buff Point, and Outpost Buff Point, via remote exchange, or from the Resupply Zone.
Drone	750	Extra Projectile Allowance is not allowed.

Resupply Zone, Base Buff Point, or Outpost Buff Point Exchange

When a Hero, Infantry, or Sentry occupies a Resupply Zone, Base Buff Point, or Outpost Buff Point, the Operator can exchange Projectile Allowance via the Referee System Player's Client. A Sentry can also independently exchange Projectile Allowance via the Referee System Serial Port. The specific Projectile types and minimum quantities that can be exchanged for are shown in the table below.

Table5-10 Minimum Exchange Units of Projectile Allowance for Different Types of Projectiles in the Resupply Zone, Base Buff Point, or Outpost Buff Point

17 mm Projectile Allowance	42 mm Projectile Allowance
10 rounds	1 round

Remote Exchange

If a robot is out of combat, the Operator can remotely exchange for Projectile Allowance through the Referee System Player's Client. A Sentry can remotely exchange for Projectile Allowance through the Referee System Serial Port on its own. The specific Projectile types and minimum quantities that can be exchanged for are shown in the table below.

Table5-11 Minimum Exchange Units of Projectile Allowance for Different Types of Projectiles during Remote Exchange

17 mm Projectile Allowance	42 mm Projectile Allowance
100 rounds	10 rounds

When a remote exchange is completed, the Projectile Allowance will be effective after six seconds.

Obtained from the Resupply Zone

After the competition starts, every minute, a Sentry can obtain 100 rounds of Projectile Allowance by occupying the Resupply Zone Buff Point on its own side. The allowance not obtained through this method can be accumulated.

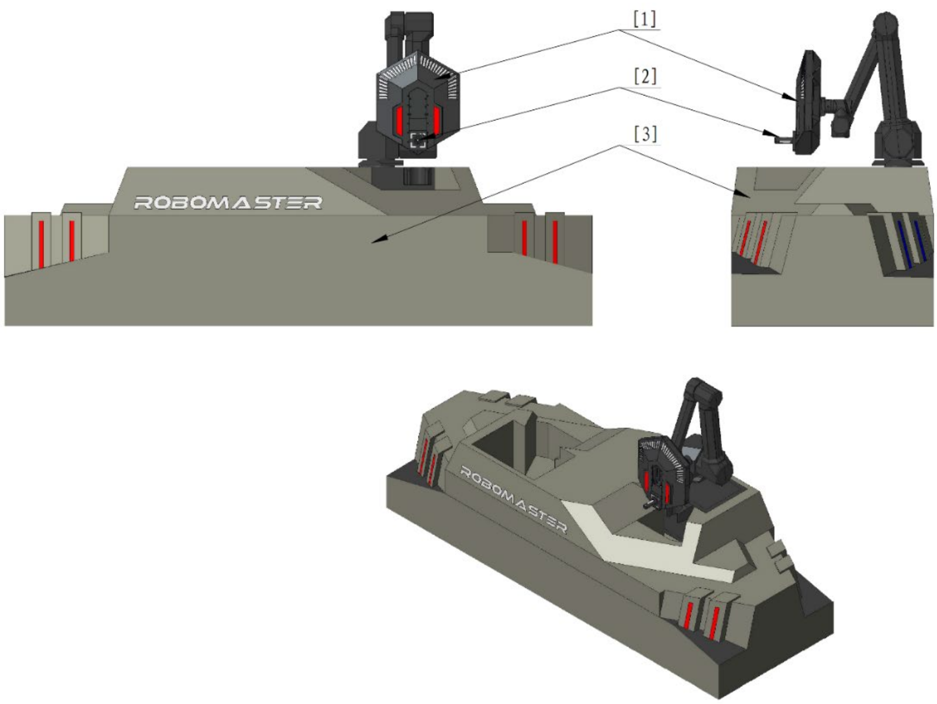
Example: If a Sentry has never occupied the Resupply Zone Buff Point on its own side before and occupies it with 30 seconds remaining in the competition, the Sentry's Projectile Allowance increases by 600.

5.3.3 Tech Core – Energy Unit Mechanism

During the competition, the Engineer installs its carried Energy Units into the Tech Core in exchange for Gold Coins and team buffs.

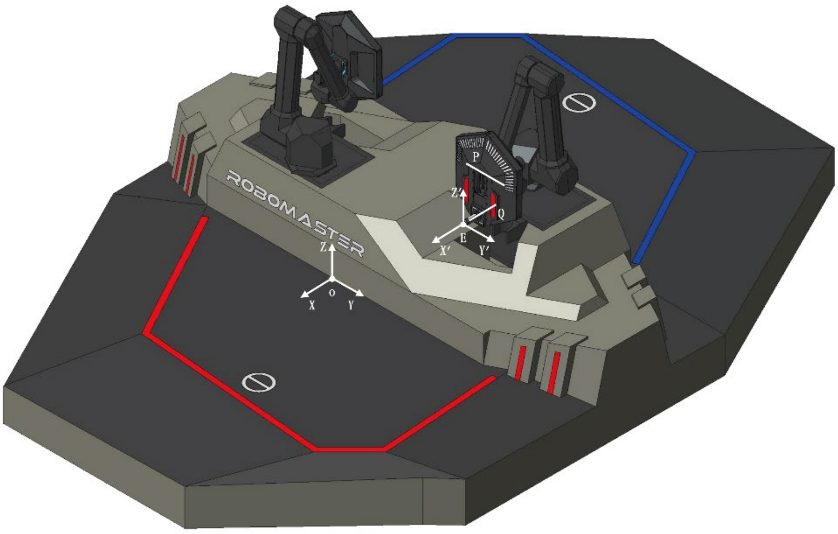
For each Tech Core, a right-handed Cartesian coordinate system named Assembly Coordinate System OXYZ is established using the midpoint of the intersection line between the front of the Base and the Battlefield surface as the origin O. The negative X-axis is defined by the direction of the Base front normal pointing toward the Tech Core, and the positive Z-axis points vertically upward. A right-handed Cartesian coordinate system EX'Y'Z' is established using the geometric center of the front surface of the Energy Unit assembly post as the origin E. The X', Y', and Z' axes align parallel to and share the same positive directions as the X, Y, and Z axes. The unit normal vector of the front surface of the Energy Unit assembly post is $\vec{e}$. A spherical coordinate system (r, θ, φ) is established based on the coordinate system of the Tech Core. Here, θ is the angle between the projection of the unit normal vector $\vec{e}$ on the X'Y' plane and the positive X'-axis, where θ ∈ [–90, 90]; φ is the angle between $\vec{e}$ and the positive Z'-axis, where φ ∈ [0, 90]. Additionally, α is defined as the rotation angle of the Tech Core around the direction of $\vec{e}$. A counterclockwise rotation is considered positive for α, where α ∈ [–45, 45].

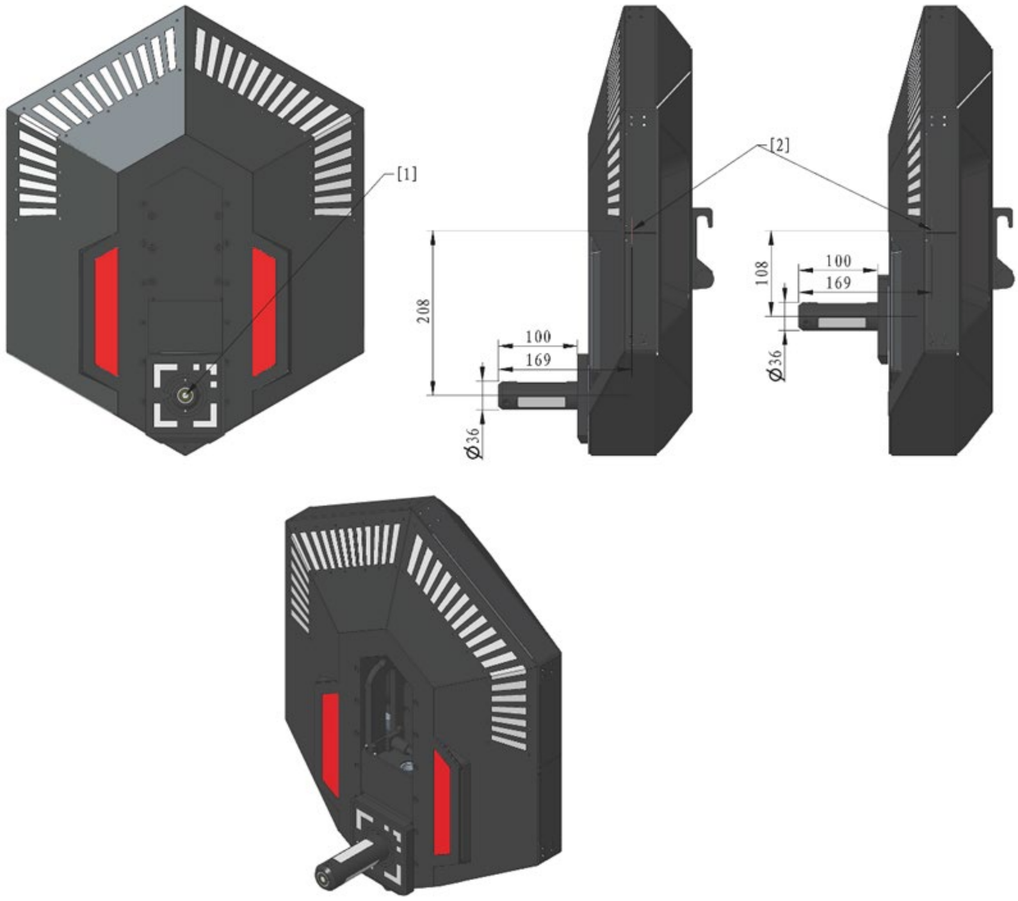
The pose of the Tech Core is defined as the combination of the coordinates (x, y, z) of the center point E on the front surface of the Energy Unit assembly post in the OXYZ coordinate system, the orientation coordinates (θ, φ) of the unit normal vector $\vec{e}$ in the spherical coordinate system of the Tech Core, and the rotation angle α of the Tech Core around the direction of $\vec{e}$, forming (x, y, z, θ, φ, α).

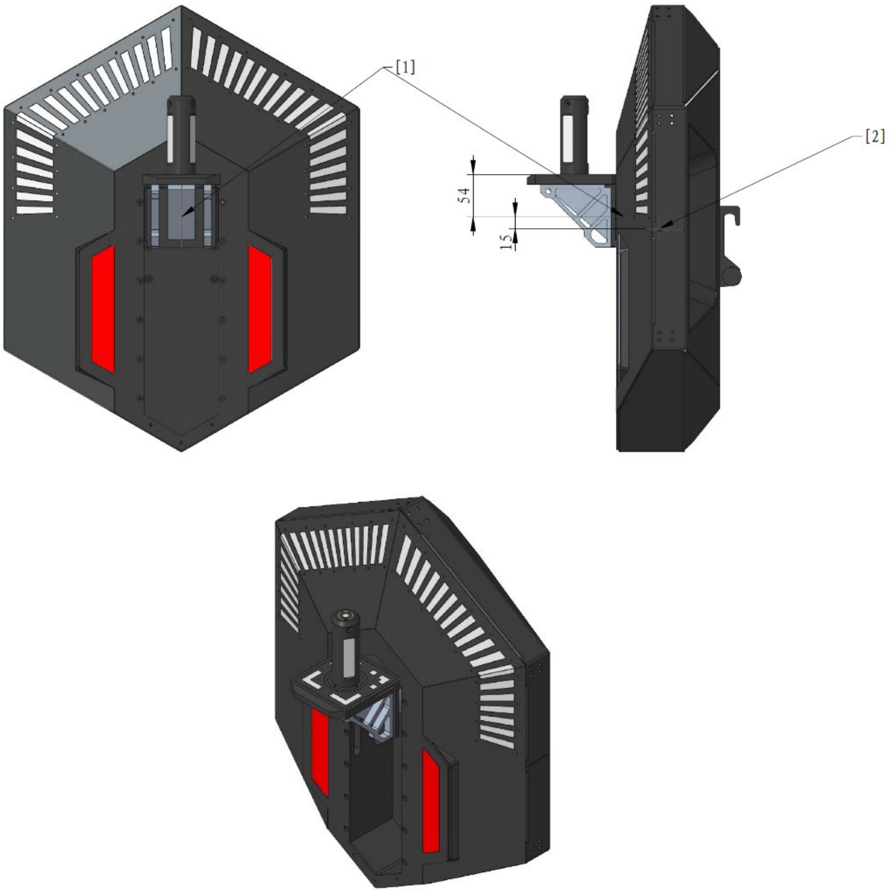


[1] Tech Core [2] Assembly Post [3] Foundation

Figure 5-6 Tech Core


P
Q
Z
X
Y
O
E
X'
Y'
Z'
ROBOMASTER


[1]
[2]
208
100
169
Ø36
108
100
169
Ø36



[1] Q-axis [2] P-axis

Figure 5-7 Tech Core Coordinates

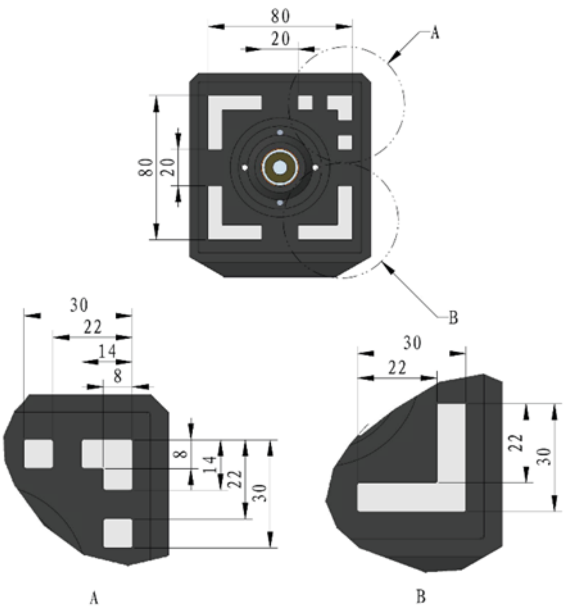


Figure 5-8 Tech Core MARK Light Size Graph

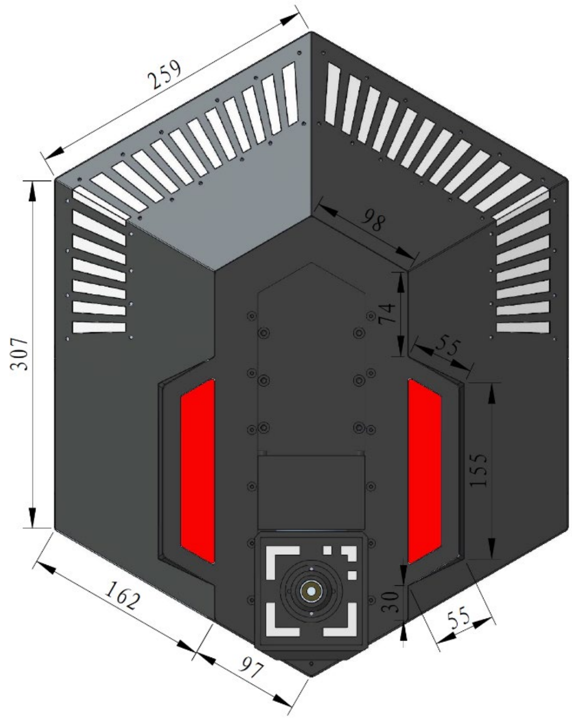


Figure 5-9 Tech Core size Graph

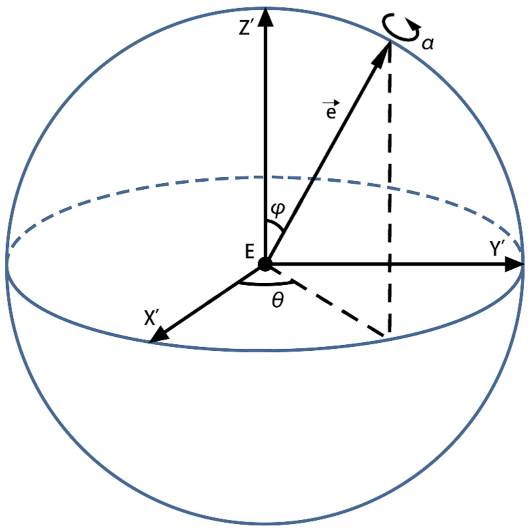


Figure 5-10 Spherical Coordinate System

The initial pose of the Tech Core is: x = –200, y = 200, z = 800, θ = 0, φ = 90, α = 0 (length unit: mm; angle unit: °).

The pose range of the Tech Core must meet the following conditions:

- No part of the Tech Core can extend beyond the plane of the front face of the foundation.
- The pose of the Tech Core is related to the difficulty level. Their relationship is shown below.

Difficulty	x	y	z	θ	φ	α	Panning Requirements	Spinning Requirements
Level 1	[–100, 0]	[100, 300]	[500, 700]	0	90	0	No panning requirements	No spinning requirements
Level 2	[–100, 0]	[100, 300]	[500, 700]	[–90, 90]	[0, 90]	[–45, 45]	100 mm movement after insertion	
Level 3	[–100, 0]	[100, 300]	[500, 700]	[–90, 90]	[0, 90]	[–45, 45]	100 mm movement after insertion	After the translation is complete, use the Q-axis as the axis of rotation, as detailed in “Figure 5-7 Tech Core Coordinates“, and rotate it to the designated angle within a rotation range of at most 90° clockwise or counter-clockwise around the Q-axis, with 5° precision.
Level 4	[–100, 0]	[100, 300]	[500, 700]	[–90, 90]	[0, 90]	[–45, 45]	100 mm movement after insertion	

In each round, the red and blue teams' Tech Cores have the same pose changes at the same difficulty level. The pose value range and panning and spinning requirements are the same for Difficulty Level 3 and Difficulty Level 4. However, at Difficulty Level 4, a team's Tech Core will be temporarily moved to the other team's location. At this point, the team's Tech Core coordinates x'y'z' relate to the other team's Tech Core coordinates xyz as follows: $x' = x$, $y - y' \leq 400$, $z' = z$, with random rotation requirements. The assembling side must complete the assembly for both teams' Tech Cores. After selecting Difficulty Level 4, the Engineer must complete assembly within 45 seconds after the Tech Core is in place. A failed assembly at Difficulty Level 4 will prevent that side from attempting Difficulty Level 4 again within the next 90 seconds.

Priority for Difficulty Level 4 Assembly

- If neither side is assembling, and the other side selects Difficulty Level 4, the Tech Core will immediately move to the designated location.
- If one side is attempting assembly at a difficulty level other than Level 4, and the other side selects

Level 4, the side currently assembling will have a buffer period of 15 seconds to tap OK or withdraw from the assembly. Afterwards, the Tech Core will immediately move to the designated location. In this case, if the side attempting Difficulty Level 4 fails to assemble, they will not be able to attempt Difficulty Level 4 again in this round, and the Gold Coin earnings every 10 seconds will be reduced by 25.

If the Engineer does not leave the Tech Core during the buffer period, it may cause mechanical interference or damage.

- If one side is attempting assembly at Difficulty Level 4, the other side cannot select Difficulty Level 4.

Engineers can select the difficulty level independently, but must complete the assembly for the previous difficulty level before moving on to the next one. The available difficulty levels are subject to certain conditions, as shown in the table below.

Match Time	**Available Difficulty Levels**
At the start of the competition	Both sides can only select Difficulty Level 1.
One minute after the competition begins	Both sides can select Difficulty Level 1 or 2.
Two minutes after the competition begins	Both sides can select Difficulty Level 1, 2, or 3.
Three minutes after the competition begins	Both sides can select Difficulty Level 1, 2, 3, or 4.

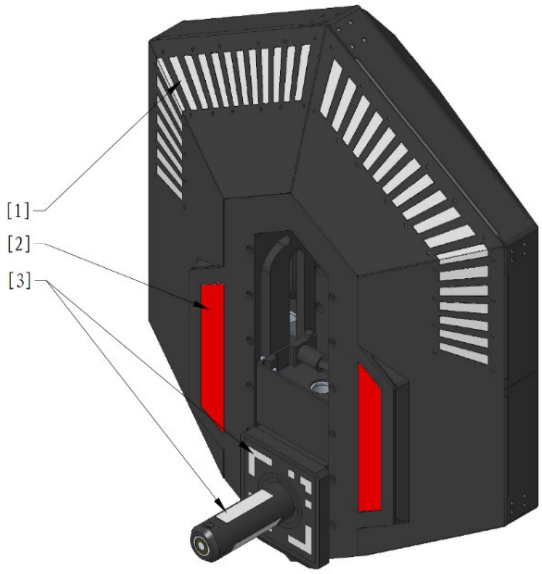
After a successful assembly, teams will receive different rewards depending on whether it is their first time completing the assembly. Details are shown in the table below.

Level Completed	**First-time Completion Reward**	**Non-first Completion Reward**
Level 1	Afterwards, the side gains 50 Gold Coins every 10 seconds.	Afterwards, the side gains additional 5 Gold Coins every 10 seconds.
Level 2	• Unlock Robots Level Cap: Before the assembly task at this difficulty level is completed, the level cap for robots on the side is Level 5. Upon completion, the level cap increases to Level 7. • Afterwards, the side gains additional 25 Gold Coins every 10 seconds.	Afterwards, the side gains additional 10 Gold Coins every 10 seconds.

Level Completed	First-time Completion Reward	Non-first Completion Reward
Level 3	● Ground Robots, Outpost, and Base on the side gain a 25% Defense Buff, lasting until the End of Round. The buff remains even after the robot is defeated. ● The level cap for robots on the side increases to Level 10. ● Afterwards, the side gains additional 25 Gold Coins every 10 seconds.	Afterwards, the side gains additional 15 Gold Coins every 10 seconds.
Level 4	● Ground Robots, Outpost, and Base on the side gain a 50% Defense Buff, lasting until the End of Round. The buff remains even after the robot is defeated. ● Base HP increases by 2,000, and any excess HP will be converted into an equivalent amount of Virtual Shield. When the Base takes Damage afterwards, the Virtual Shield will be depleted first, followed by the Base HP. ● Afterwards, the side gains additional 50 Gold Coins every 10 seconds.	Can only be completed once.

Once robots reach their level cap, they will no longer gain experience.

Example: If a team successfully completes five assembly tasks in one round, at Difficulty Levels 1, -2, 3, 2, and 1 respectively, they will receive a total of 115 (50 + 25 + 25 + 10 + 5) Gold Coins every 10 seconds.



[1] First group of lights [2] Second group of lights [3] Third group of lights

Figure 5-11 Tech Core Light Effects

Table5-12 Summary of Tech Core Light Effects

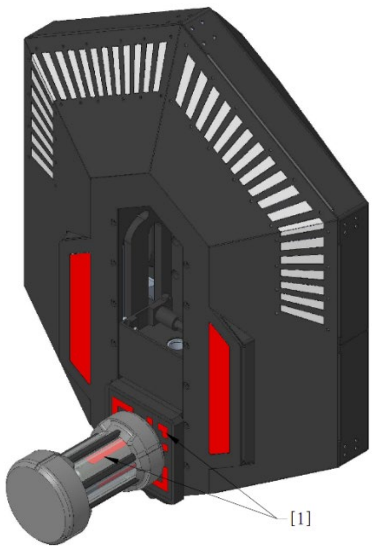
Status	Light Effect
The competition is in progress and has not yet entered the assembly phase.	● The first group of lights are all off. ● The second and third groups of lights are solid on, in the color corresponding to the team (red or blue).
The competition is in progress, the assembly difficulty level has been selected, and the Robotic Arm has not yet moved into position.	● The first group of lights present a flowing light effect. ● The second and third groups of lights are solid on, in the color corresponding to the team (red or blue).
The competition is in progress, the assembly difficulty level has been selected, and the Robotic Arm has moved into position.	● The first group of lights flash white at a frequency of 1 Hz. ● The second and third groups of lights flash at a frequency of 1 Hz, in the color corresponding to the team (red or blue).

Status	Light Effect
Step 2 is completed.	● The first group of lights flash white at a frequency of 1 Hz. ● The second group of lights flash at a frequency of 1 Hz, in the color corresponding to the team (red or blue). ● The third group of lights flash at a frequency of 3 Hz, in the color corresponding to the team (red or blue).
Step 3 is completed.	● The first group of lights flash white at a frequency of 1 Hz. ● The second and third groups of lights flash at a frequency of 3 Hz, in the color corresponding to the team (red or blue).
Step 4 is completed.	● The first group of lights flash white at a frequency of 1 Hz. ● The second and third groups of lights are solid on, in the color corresponding to the team (red or blue).
Step 5 is in progress.	● The light effect for the first group of lights is as follows: ➢ The designated angular position illuminates in the color corresponding to the team (red or blue) without flashing. ➢ Another angular position matching the Energy Unit's current rotation angle illuminates in white without flashing. ● The second and third groups of lights are solid on, in the color corresponding to the team (red or blue).

Status	Light Effect
Step 5 is completed.	● In the first group of lights, the designated angular position illuminates in green without flashing. ● The second and third groups of lights are solid on, in the color corresponding to the team (red or blue).
Assembly is confirmed, and the Energy Unit is being reclaimed.	● The first group of lights flash white at a frequency of 3 Hz. ● The second and third groups of lights flash at a frequency of 3 Hz, in the color corresponding to the team (red or blue).

To assemble an Energy Unit, the following steps must be completed:

1. After the Engineer occupies the Assembly Zone Buff Point, the Operator selects the assembly difficulty level on the Referee System Player's Client.
2. Once the Tech Core moves to the designated pose (ready for assembly), the corresponding light effect will appear. Then, the Engineer aligns the bottom opening of the Energy Unit with the Energy Unit assembly post, and inserts it by 100 mm along the negative X-axis. Upon reaching the bottom, the corresponding light effect will appear, and step 2 is completed, as shown in the figure below.

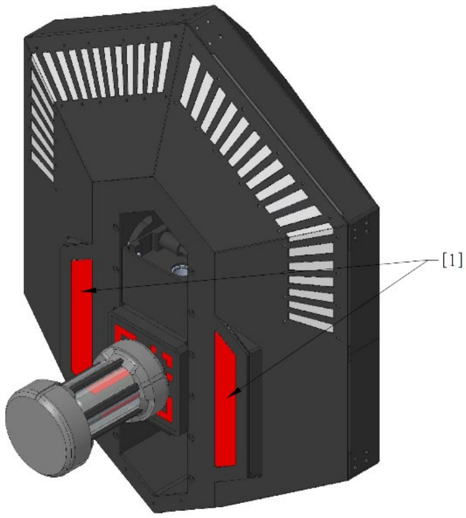


[1] Flash at 3 Hz

Figure 5-12 Step 2

- The Energy Unit is actually opaque; the Light Effect shown here is for illustration only.
- In this step, aside from the force required to overcome the gravity of the energy unit, the moving resistance is approximately 10N.

3. The Engineer moves the Energy Unit, which has been secured on the Energy Unit assembly post, by 100 mm along the positive Z-axis. Upon reaching 100 mm, the corresponding light effect will appear, and step 3 is completed, as shown in the figure below.



[1] Flash at 3 Hz

Figure 5-13 Step 3

4. The Engineer rotates the Energy Unit upward by 90° with the P-axis as the axis of rotation. Upon reaching 90°, the corresponding light effect will appear, and step 4 is completed, as shown in the figure below.

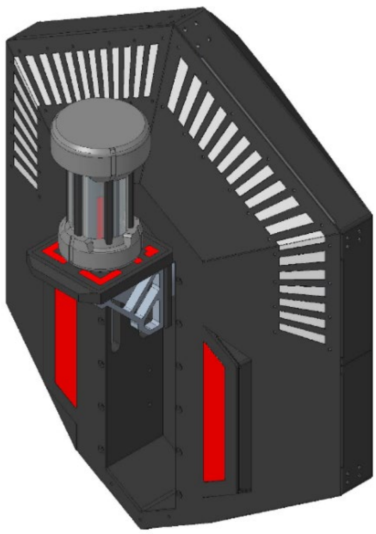


Figure 5-14 Step 4

5. The Engineer rotates the Energy Unit around the Q-axis to the designated angle until the corresponding light effect appears(it will appear after the unit stays in the correct Location for up to 0.2 s). At this point, step 5 is completed, as shown in the figure below.

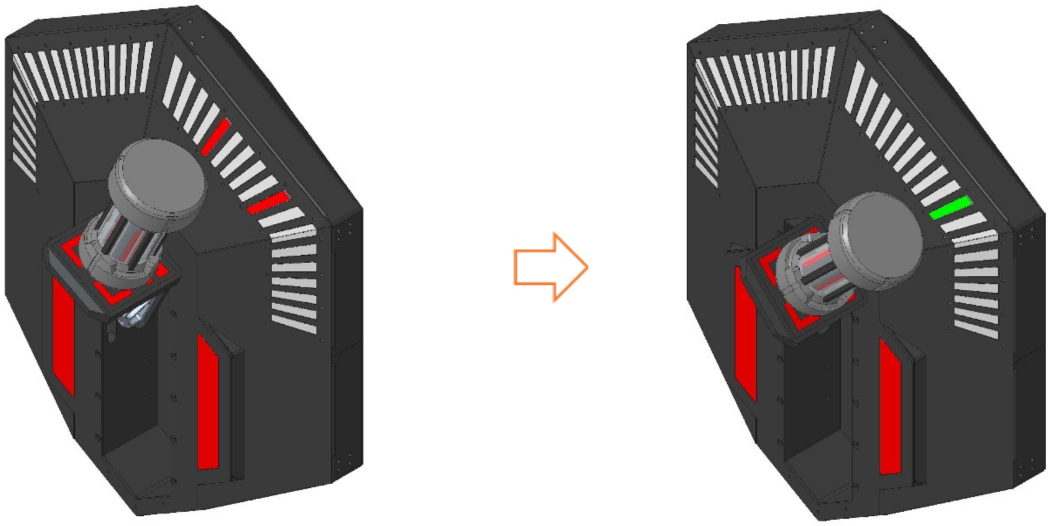


Figure 5-15 Step 5

- If after the operator presses the "OK to Assemble" button, the Tech Core does not detect that the energy unit has been correctly assembled as required, it will not accept any "Confirm The Assembly " command inputs for the following 1 s.
- For safety reasons, during this step, if the position is maintained correctly for more than 2 seconds, the rotation mechanism will lock. At this point, the Engineer Mechanism can detach from the Energy Unit, while the Energy Unit will remain in place.
- In this step, aside from the force required to overcome the gravity of the energy unit, the rotational resistance torque is approximately 0.3~0.6 N·m.

6. The Operator presses the corresponding button to confirm the assembly. If the Tech Core detects that the Energy Unit has been fully inserted and assembled correctly according to the panning and spinning requirements, the Energy Unit will be reclaimed and the corresponding rewards will be granted to that side.

- For Difficulty Level 1, completing steps 1, 2, and 6 is considered a successful assembly.
- For Difficulty Level 2, completing steps 1, 2, 3, and 6 is considered a successful assembly.
- For Difficulty Level 3, completing steps 1, 2, 3, 4, 5, and 6 is considered a successful assembly.
- For Difficulty Level 4, in addition to completing steps 1, 2, 3, 4, 5, and 6, the Engineer repeat each assembly step (except steps 1 and 4) at another tech core within five seconds after completing that

step. Then, the Operator presses the corresponding button to confirm the assembly. If the Tech Core detects that both Energy Units have been fully inserted and assembled correctly according to the panning and spinning requirements, and the time requirement is met, the Energy Units will be reclaimed and the corresponding rewards will be granted to that side; otherwise, the assembly will be deemed failed.

In addition, the Operator should pay attention to the following during the assembly process:

- After the Operator selects the assembly difficulty level, if an Energy Unit is present inside the Tech Core, the assembly difficulty level cannot be changed. In addition, before the Energy Unit is successfully assembled, the pose of the Tech Core will remain unchanged for the same assembly difficulty level.
- After the Operator confirms the assembly, they must not perform any further operation on or interfere with the Energy Unit; otherwise, the assembly may fail.
- During the assembly process, the Engineer can remove the Energy Unit. This process requires overcoming gravity and applying an additional force of approximately 10 N outward along the rotational symmetry axis of the Energy Unit. When the Tech Core's sensor detects that the Energy Unit has been removed, the assembly will be immediately deemed failed.
- During the assembly process, if the Engineer fails to detect the Interaction Module Card in the Assembly Zone for 15 consecutive seconds, the current assembly will be deemed failed and automatically terminated. Afterwards, the Tech Core will immediately reclaim the Energy Unit.
- The Tech Core does not support obstacle detection or active obstacle avoidance during operation. If the movement path is blocked, the system will trigger an overload alert and briefly hover, and then attempt to resume movement. If the Engineer's mechanisms suffer damage as they come into contact or collide with the Tech Core, any resulting losses will be the responsibility of the participating team.
- If the Engineer is defeated after grabbing the Energy Unit, the current assembly will be immediately deemed failed. After the Operator manually cancels the assembly, the Tech Core will reclaim the Energy Unit.
- If one side's Engineer is defeated during the assembly and the other side selects Difficulty Level 4 at that moment, the defeated robot will be temporarily activated, and the Operator can control the defeated robot to move away from the Tech Core.
- When Engineer perform assembly, they must operate the Energy Unit rather than the movable structures of the Tech Core and other body structures. If field props are damaged due to direct operation of the Tech Core, the referee will impose corresponding penalties based on the severity of the impact.

- Facing the Tech Core during assembly, the energy unit can rotate 90° around the Q-axis, either Clockwise or Counterclockwise.

5.4 Experience and Performance Systems

5.4.1 Experience System

At the start of the competition, Hero, Infantry, and Drone all begin at Level 1, and can level up by gaining Experience Points. The Experience System does not apply to other robots.

During the competition, a robot can gain Experience Points in various ways, as shown in the table below.

Behavior Type	Experience Points Gained
Launching Projectiles	● Infantry and Drone: 1 Experience Point for each Projectile launched. ● Hero: 10 Experience Points for each Projectile launched.
Dealing Attack Damage	● Dealing Attack Damage to robots: The attacking side gains four Experience Points for every one point of damage. ● Dealing Attack Damage to an Outpost Armor Module: The attacking side gains two Experience Points for every one point of damage. ● Dealing Attack Damage to a Base Armor Module: The attacking side gains one Experience Point for every two points of damage. For an odd number of points of damage, it is rounded up. When a team's robot, Base, or Outpost receives Attack Damage, but the Referee System cannot detect the specific source of the damage, the source of the damage cannot gain Experience Points, or the source of the damage is a penalized robot or a robot of the same team: if the damage is from Projectiles, the Experience Points will be equally divided among all of the other team's alive robots that can deal Projectile damage of that type and gain experience; if the damage is not from Projectiles, the Experience Points will be equally divided among all Hero, Infantry, and Drone robots of the other team that are alive at the moment. The average is rounded up and must be accurate to one decimal place.

Behavior Type	Experience Points Gained
	Example: An Infantry of the Blue Team received 120 points of damage from a 17 mm Projectile, but the Referee System is unable to identify the source of the damage. At this time, the Red Team has two alive Infantries, one Drone providing Air Support, and one alive Hero. The Infantries and the Drone each receive $\frac{4\times120}{3}$=160 Experience Points, while the Hero does not receive any Experience Points.
Defeating robots	In the following calculations, both Engineer and Sentry are considered to be at Level 1. ● If a robot is destroyed and the destroyer is eligible to gain experience, the number of Experience Points gained is as follows: ➢ When the Level of the destroyed robot is higher than or equal to that of the destroyer, the number of Experience Points is calculated as follows: Number of Experience Points gained by the destroyer = 50 × Level of the destroyed robot × (1 + 0.2 × Difference between the Level of the destroyed robot and that of the destroyer) ➢ When the Level of the destroyed robot is lower than that of the destroyer, the Level difference is considered to be 0. The number of Experience Points is calculated as follows: Number of Experience Points gained by the destroyer = 50 × Level of the destroyed robot ● If the Referee System does not identify a destroyer or the destroyer is ineligible to gain experience: ➢ If the damage is from Projectiles, the Level of the destroyer is considered to be the Level corresponding to the average Experience Points of the other team's robots that are alive and can deal the Projectile damage of that type, and the Experience Points will be equally divided among these robots. ➢ If the damage is not from Projectiles, the Experience Points will be equally divided among all Hero, Infantry, and Drone robots of the other team that are alive at the moment. The average is rounded up and must be accurate to one decimal place.

Behavior Type	Experience Points Gained
	If a robot is defeated or goes disconnected irregularly, or the Referee System is unable to detect a destroyer for reasons other than suffering a hit on its Armor Module, it will be deemed that no destroyer has been found. **Example 1: When a Level 2 Infantry destroys a Level 6 Infantry of the opponent, the number of Experience Points gained by the destroyer is as follows: 50 × 6 × (1 + (6 – 2) × 0.2) = 540.** **Example 2: One Level 9 Infantry from the Red Team is defeated, and no destroyer is detected. At this point, the Blue Team has two Infantries with total Experience Points of 1,250 and 4,000, and one alive Drone with total Experience Points of 3,000. The average number of Experience Points is 2,750, corresponding to robots at Level 6. Therefore, the number of Experience Points gained by each of the Blue Team's two Infantries and one Drone is as follows:** $\frac{50\times9\times(1+(9-6)\times0.2)}{3}$ **=240**
Activating the Small Power Rune	For details, see “5.5.2 Power Rune Mechanism“.
Activating the Large Power Rune	For details, see “5.5.2 Power Rune Mechanism“.
Dealing damage in Deployment Mode	A robot receives 100 Experience Points for each instance of damage it deals in Deployment Mode. For details, see “5.6.1 Special Mechanism of Hero“.
Gaining Terrain Crossing Buff	Robots receive 300 Experience Points the first time they acquire each of the Terrain Crossing Buff (Road), Terrain Crossing Buff (Elevated Ground), Terrain Crossing Buff (Launch Ramp), and Terrain Crossing Buff (Tunnel).
Hitting the Outpost/Base using a Dart	For details, see “5.5.1 Base and Outpost Mechanisms“.

Table5-13 Levels and Experience Points of Hero, Infantry, and Drone

Level	Total Experience Points Required to Level Up
1	0
2	550

Level	Total Experience Points Required to Level Up
3	1,100
4	1,650
5	2,200
6	2,750
7	3,300
8	3,850
9	4,400
10	5,000

5.4.2 Performance System

After the Three-Minute Setup Period begins, the Hero Operator can select the robot type, while the Infantry Operator can select the Chassis and Launching Mechanism types for their robots. For Drones, the Launching Mechanism type cannot be selected. The Performance System does not apply to other robots.

Once the Seven-Minute Round has begun, the selected Chassis and Launching Mechanism types cannot be changed during the entire round. If the robot, Chassis, or Launching Mechanism type is not selected, after the start of the Seven-Minute Round, the robot type will be automatically set to “Long-Range Attack-Focused”, the Chassis type will be automatically set to “HP-Focused”, and the Launching Mechanism type will be automatically set to “Cooling-Focused”. After a robot levels up, any increase in Maximum HP is immediately added to the robot's current HP.

Table5-14 Hero Robot Specifications

Launching Mechanism Type	Level	Maximum HP	Chassis Power Limit (W)	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
Melee-Focused	1	200	70	140	12
	2	225	75	150	14
	3	250	80	160	16
	4	275	85	170	18

Launching Mechanism Type	Level	Maximum HP	Chassis Power Limit (W)	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
	5	300	90	180	20
	6	325	95	190	22
	7	350	100	200	24
	8	375	105	210	26
	9	400	110	220	28
	10	450	120	240	30
Long-Range attack-Focused	1	150	50	100	20
	2	165	55	102	23
	3	180	60	104	26
	4	195	65	106	29
	5	210	70	108	32
	6	225	75	110	35
	7	240	80	115	38
	8	255	85	120	41
	9	270	90	125	44
	10	300	100	130	50

Table5-15 Infantry Chassis Specifications

Chassis Type	Level	Maximum HP	Chassis Power Limit (W)
Power-Focused	1	150	60
	2	175	65
	3	200	70
	4	225	75

Chassis Type	Level	Maximum HP	Chassis Power Limit (W)
	5	250	80
	6	275	85
	7	300	90
	8	325	95
	9	350	100
	10	400	100
HP-Focused	1	200	45
	2	225	50
	3	250	55
	4	275	60
	5	300	65
	6	325	70
	7	350	75
	8	375	80
	9	400	90
	10	400	100

Table5-16 Infantry 17 mm Launching Mechanism Specifications

Launching Mechanism Type	Level	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
Burst-Focused	1	170	5
	2	180	7
	3	190	9
	4	200	11
	5	210	12

Launching Mechanism Type	Level	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
	6	220	13
	7	230	14
	8	240	16
	9	250	18
	10	260	20
Cooling-Focused	1	40	12
	2	48	14
	3	56	16
	4	64	18
	5	72	20
	6	80	22
	7	88	24
	8	96	26
	9	114	28
	10	120	30

Table5-17 Drone 17 mm Launching Mechanism Specifications

Level	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
1	100	20
2	110	30
3	120	40
4	130	50
5	140	60

Level	Barrel Heat Limit	Barrel Heat Cooling Rate (per second)
6	150	70
7	160	80
8	170	90
9	180	100
10	200	120

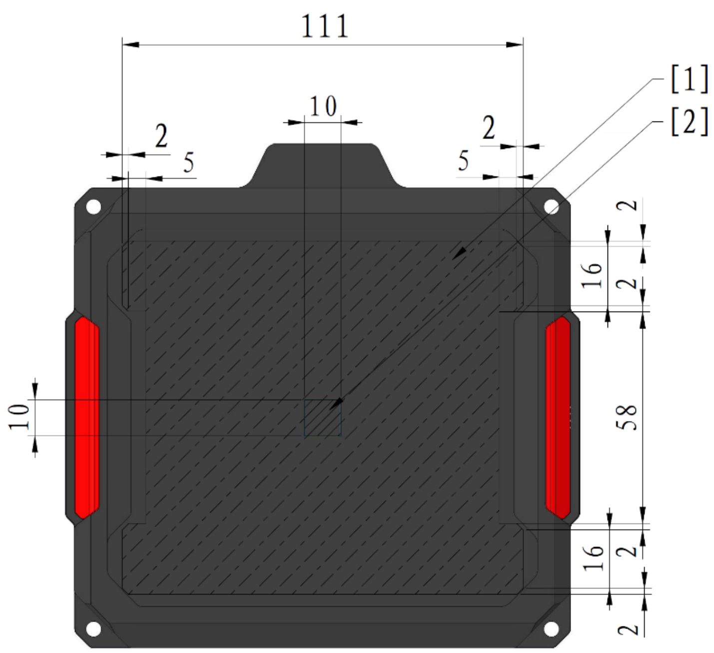
5.5 Battlefield-related Mechanism

5.5.1 Base and Outpost Mechanisms

Base HP is 5,000 and starts with a 150 HP Virtual Shield. The Outpost HP is 1,500.

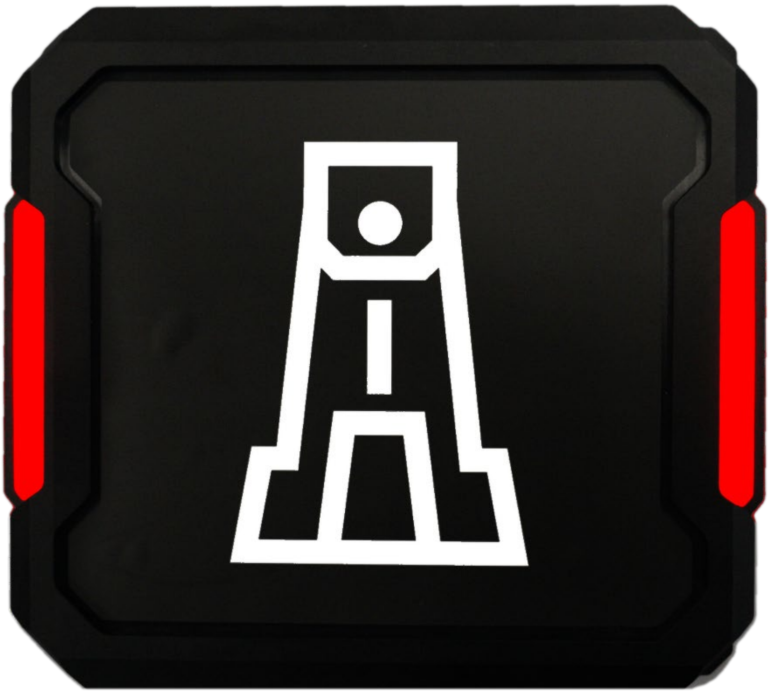
The deduction of Virtual Shield is not considered as a loss of Base HP, but it is counted towards the total Damage dealt by the Opponent.

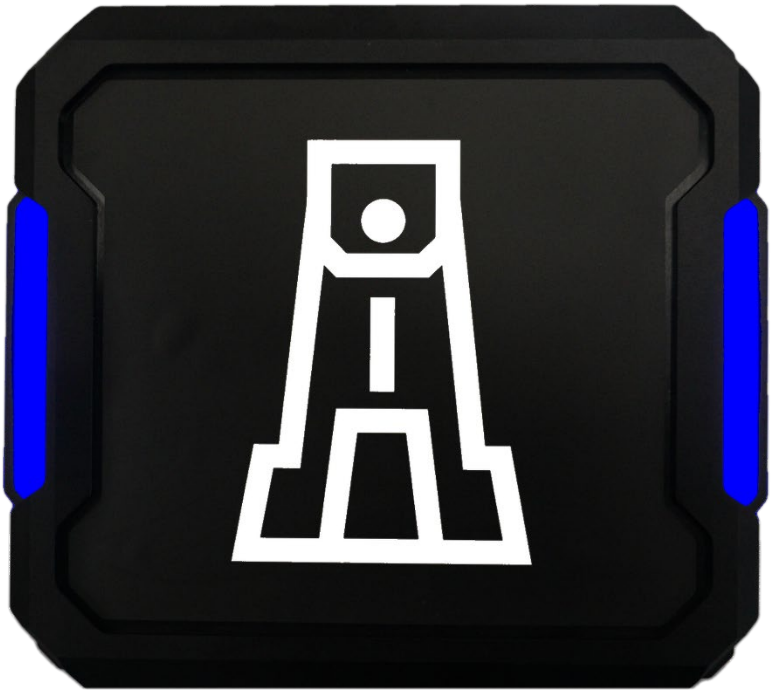
The center of the Armor Module of the Base and Outpost (excluding the Dart Detection Module) features a 10 mm × 10 mm rectangular area, as shown in the figure below. If a Projectile hits this area, the attack receives a 150% buff.



[1] Detailed dimensions of effective detection area [2] Critical hit area dimensions

Figure 5-16 Rectangular Area





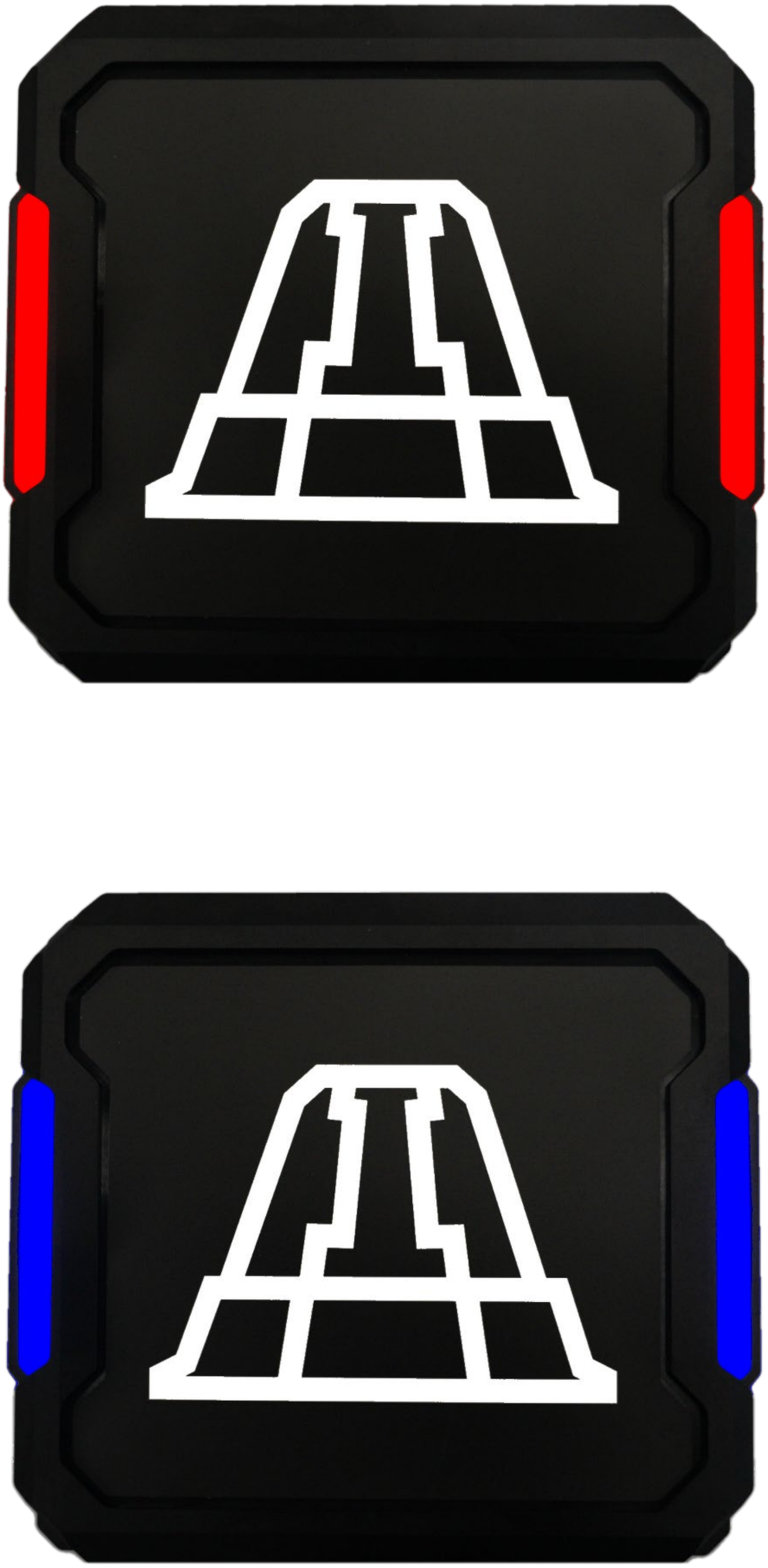


Figure 5-17 Base, Outpost Armor Modules

When a team's Outpost is alive, the Base is invincible. When its Base HP declines to 2,000 or below, the

Base Protective Armor will be enabled.

After a team's Outpost is destroyed, the other team can occupy the Buff Point associated with the destroyed Outpost. For details, see “5.5.3.6 Outpost Buff Point Mechanism“.

Each time a team loses 1,000 Base HP, they gain one opportunity to rebuild the Outpost. The opportunities are accumulative. Hero, Infantry, and Sentry can rebuild a destroyed Outpost by continuously scanning their Outpost Buff Point RFID Interaction Module Card for 10 seconds. Engineer can do so by scanning for 5 seconds. Occupation durations for different robots are calculated independently. This process restores the Outpost to the alive state with 750 HP. The Outpost cannot be rebuilt after 5 minutes into the match.

The middle armor of the Outpost can rotate. For its initial position, see “Figure 4-33 Outpost“. After the match begins, the middle armor starts rotating, reaching a speed of 0.8π rad/s within 5 seconds, then maintains a constant speed in a random direction. In each round, the Outposts of both the red and blue teams will rotate in the same, fixed direction.

A team's Outpost armor will stop rotating if any of the following conditions is met:

- The team's Outpost is destroyed for the first time.
- The other team's Base Protective Armor is expanded.
- The competition has been in progress for three minutes.

The Outpost armor will not rotate again in the current round after it has stopped rotating.

When the Outpost is alive, the process for the middle armor to stop rotating is as follows: The middle armor decelerates immediately and returns to its initial position within 10 seconds.

- When the Outpost middle armors of both teams stop rotating simultaneously, the time when they return to their initial positions may differ slightly.
- When an Outpost is alive, the Dart Guiding Light on the Outpost is solid on, and the Dart Guiding Light on the Base is off. When the Outpost is destroyed, the Dart Guiding Light on the Outpost is off, and the Dart Guiding Light on the Base is on.

5.5.2 Power Rune Mechanism

Robots can activate Power Runes by launching Projectiles. Both teams can strike the Power Rune at the same time. Once the Red Team's Power Rune is activated, it provides buff exclusively to the Red Team; the same applies to the Blue Team.

The Power Rune continuously rotates during the match, but its rotation characteristics change depending on its status:

The Power Runes of both teams rotate on the same axis, i.e., the Red Team's Power Rune rotates in the clockwise direction while the Blue Team's Power Rune rotates in the counterclockwise direction (as per the rotation direction when facing the respective team's Power Rune). Before the start of a round, the Power Runes rotate in a random direction. During the round, the Power Rune will maintain the same direction throughout.

- The rotating speed of a Small Power Rune is set at 1/3 π rad/s.
- The rotating speed of a Large Power Rune that is not activating is set at 1/3 π rad/s.
- The rotating speed of the Large Power Rune in the activating state varies periodically according to a trigonometric function. The Target Function for Speed is $spd=a*\sin(\omega*t)+b$, where spd is measured in rad/s, t is measured in s, the value range for a is 0.780 to 1.045, the value range for ω is 1.884 to 2.000, and b always satisfies b = 2.090 − a. Each time when the Large Power Rune becomes available, all parameters will be reset, where t shall be 0, and a and ω will be any value within their respective ranges. The actual rotating speed of the Power Rune has a time deviation within 500 ms from the Target Function for Speed.

After a team activates its Power Rune, all alive robots of the team will receive a certain buff while the Power Rune is activated. After the buff effect of the Power Rune ends, the Power Rune will be unavailable. Both teams can have Power Rune buffs at the same time.

According to the time, the Power Rune is divided into two phases: Small Power Rune and Large Power Rune. From the start of the match until the first three minutes, the Small Power Rune stage applies; after three minutes from the start of the match, the Large Power Rune stage applies.

- **Small Power Rune:** The status of each team's Small Power Rune is independent from each other. At the start of the match and at 1 minute and 30 seconds, each team receives one opportunity to activate their Small Power Rune, and these opportunities can be accumulated. The Infantry Operator can trigger an inactive Small Power Rune to the Activating state through the Referee System Player's Client, or the Sentry can trigger it via the Referee System Serial Port. If the Small Power Rune remains unactivated 20 seconds after entering the Activating state, it will revert to the inactive state. After a team's robot activates its Small Power Rune, all robots, Outpost, and Base of the team will receive a 25% Defense Buff that lasts 45 seconds. During the buff period of the Small Power Rune, all Hero, Infantry, and Drone robots experience a 100% boost in earned experience. The whole team can gain a maximum of 1,200 extra Experience Points in one buff period of the Small Power Rune.
- **Large Power Rune:** The status of each team's Large Power Rune is independent from each other. At 3 minutes, 4 minutes and 15 seconds, and 5 minutes and 30 seconds after the match begins, each team will receive one opportunity to activate the Large Power Rune, and these opportunities can be accumulated. The Infantry Operator can trigger an inactive Large Power Rune to the Activating state through the Referee System Player's Client, or the Sentry can trigger it via the

Referee System Serial Port. If the Large Power Rune remains unactivated 20 seconds after entering the Activating state, it will revert to the inactive state. Each Armor Module of the Large Power Rune is divided into rings 1–10. After the Large Power Rune is activated by one team, the system will grant varying durations and effects of attack, Defense Buffs, and heat cooling buffs to all robots of that team, based on the average ring struck and the number of activated Light Arms. The team's Outpost and Base will also receive corresponding Defense Buffs. For details, see "Table5-18 Average Ring and Corresponding Buffs". Meanwhile, when the Large Power Rune is activated, 750 Experience Points will be equally assigned to the team's Hero, Infantry, and Drone robots that are alive.

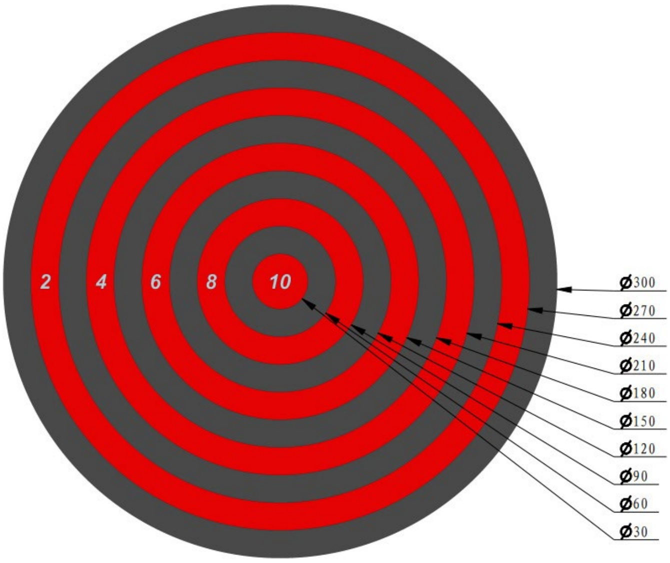


Figure 5-18 Power Rune Strike Area

The Power Rune Armor Module detects impact positions at an accuracy of 1 mm in the radial direction.

Table5-18 Average Ring and Corresponding Buffs

Average Ring Range	Attack Buff	Defense Buff	Heat Cooling Buff
[1, 3]	150%	25%	None
(3, 7]	150%	25%	2 times
(7, 8]	200%	25%	2 times
(8, 9]	200%	25%	3 times
(9, 10]	300%	50%	5 times

Table5-19 Number of Activated Light Arms and Corresponding Buff Duration

Number of Activated Light Arms	Buff Duration (seconds)
5	30
6	35
7	40
8	45
9	50
10	60

5.5.2.1 Power Rune Status

The Power Rune has four possible states: Inactive, Activating, Activated, and Activation Failed.

1. Inactive

The Power Rune enters the Inactive state, as shown below.

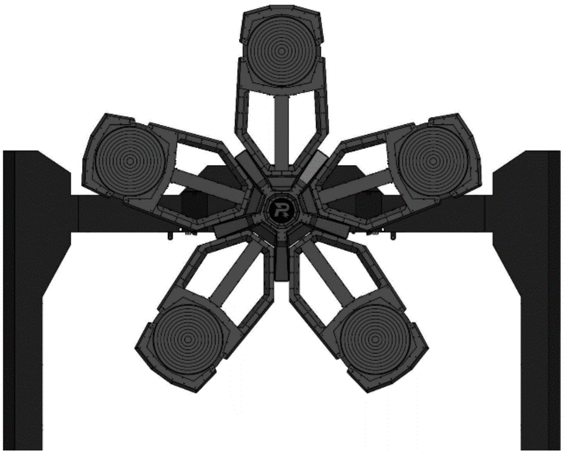


Figure 5-19 Power Rune in the Inactive State

2. Activating

Small Power Rune

When the Small Power Rune enters the Activating state, the Power Rune will randomly illuminate one of the five Armor Modules. The illuminated Armor Module displays a special light effect, and the corresponding Light Arm's central shaft shows a Flowing Arrow Light effect. At this time, if a Projectile hits any illuminated Armor Module within 2.5 seconds, the corresponding Light Arm will be fully lit with the corresponding Light Effect. Meanwhile, the Power Rune will randomly light up another Armor Module from the remaining four, and the process repeats.

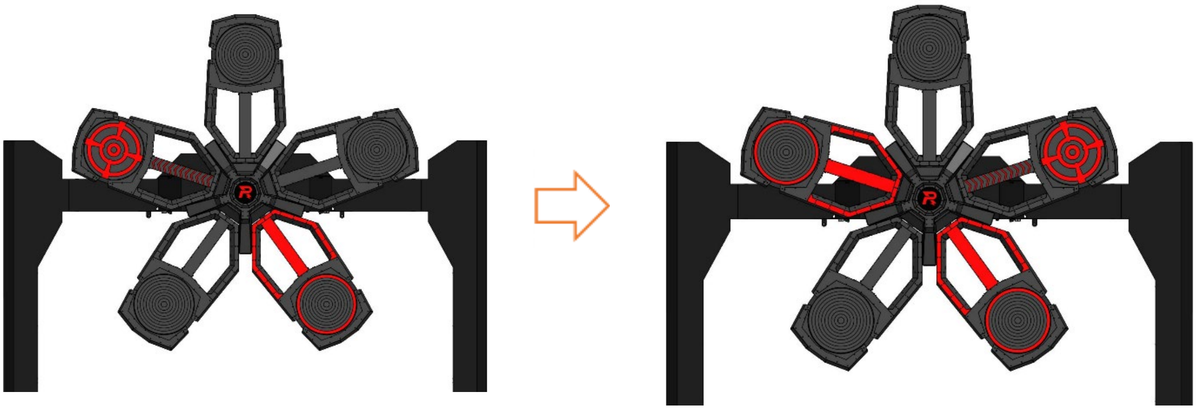


Figure 5-20 Small Power Rune in the Activating State

Large Power Rune

When the Large Power Rune enters the Activating state, the Power Rune will randomly illuminate two of the five Armor Modules. Each illuminated Armor Module displays a special light effect, and the corresponding Light Arm's central shaft shows a Flowing Arrow Light effect. At this time, if a Projectile hits any illuminated Armor Module within 2.5 seconds, the corresponding Light Arm will change its Light Effect. Within the following 1 second, robots can hit another illuminated Armor Module. Regardless of success, the Power Rune will then randomly light up any two of the five Armor Modules again, and the process repeats. The light effect of the Large Power Rune's middle Light Arm indicates the activation progress. For example, if the first group has been activated and the second group is activating, approximately 1/5 of the middle Light Arm will be lit. After the second group is successfully activated (hitting the first illuminated Armor Module in the second group counts as successfully activating the second group, regardless of whether the second Armor Module in that group is hit; the light effect changes when the third group begins activating), approximately 2/5 of the middle Light Arm will be lit.

The light effects are shown in the figure below.

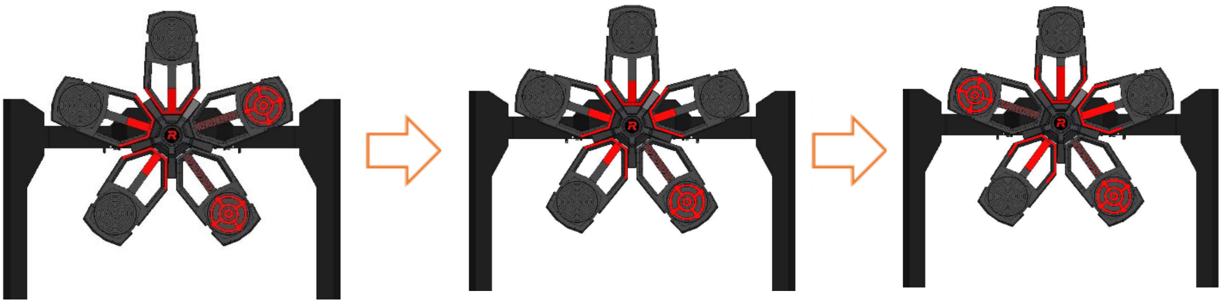


Figure 5-21 Large Power Rune in the Activating State

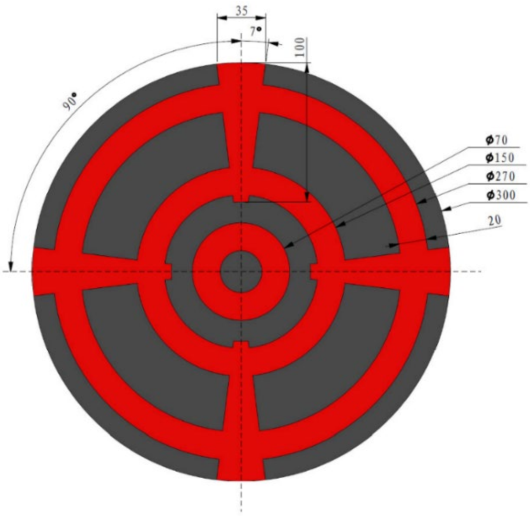


Figure 5-22 Power Rune in the Available State

3. Activated

If all five Light Arms are activated, the Power Rune is activated, as shown below.

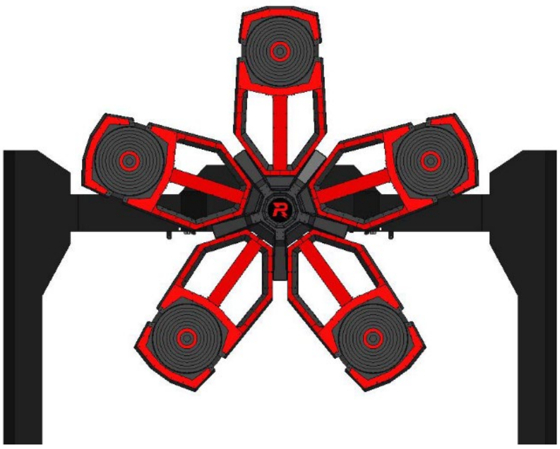


Figure 5-23 Power Rune in the Activated State

4. Activation Failed

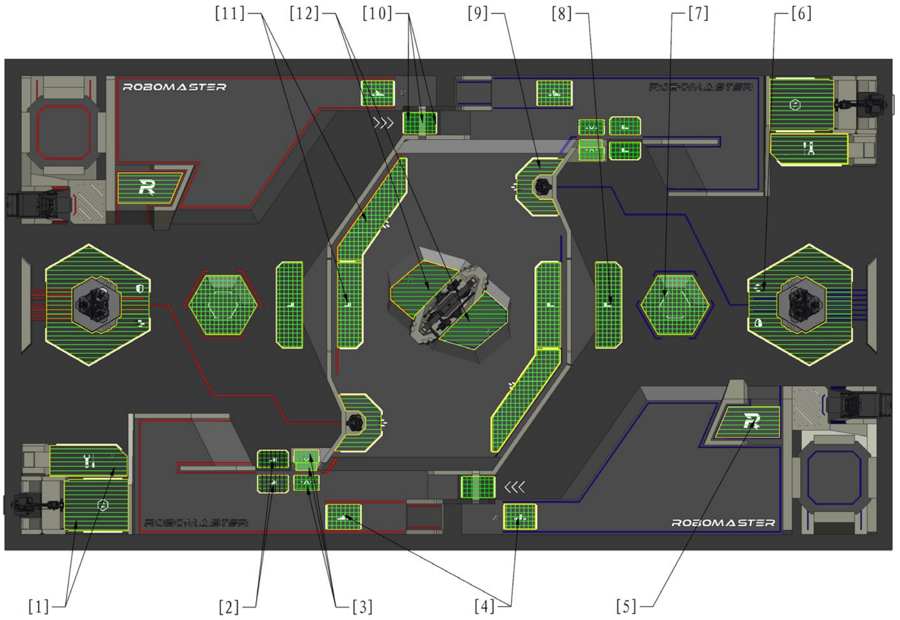
If any of the following conditions occurs during activation, the activation will fail and the Power Rune will be reset to the Activating state:

- Failure to hit an illuminated Armor Module within 2.5 seconds
- Hit an Armor Module that is not randomly illuminated

5.5.3 Battlefield Buff Mechanism

5.5.3.1 Battlefield Buff Points

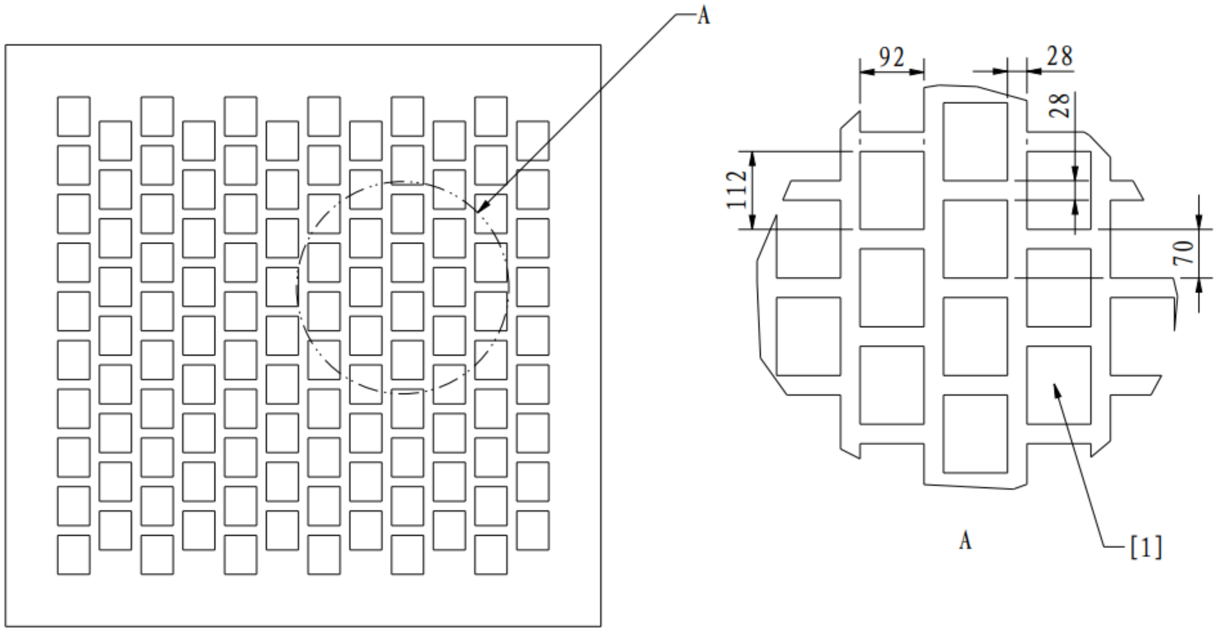
The Battlefield Buff Points are shown in the figure below. All Buff Points in the Battlefield are laid with RFID Interaction Module Cards, and there may be Deadbands for the RFID Interaction Module Cards. The Participating Teams should make adjustments and adapt accordingly. Detection by the RFID Interaction Module may experience a delay, and the expiration of robots' Occupy status is also subject to a delay of two seconds. If the robot that occupies the Buff Point is defeated, the buff gained will expire.



[1]	Resupply Zone Buff Point	[2]	Terrain Crossing Buff Point (Road)	[3]	Terrain Crossing Buff Point (Tunnel)	[4]	Terrain Crossing Buff Point (Launch Ramp)
[5]	Trapezoid-Shaped Elevated Ground Buff Point	[6]	Base Buff Point	[7]	Fortress Buff Point	[8]	Terrain Crossing Buff Points (Elevated Ground)
[9]	Outpost Buff Point	[10]	Terrain Crossing Buff Point (Tunnel)	[11]	Central Elevated Ground Buff Point and Terrain Crossing Buff Point (Elevated Ground)	[12]	Assembly Zone Buff Point

Figure 5-24 Distribution of Battlefield Buff Points

In actual competitions, the coverage area of the Terrain Crossing Buff Point (Tunnel) RFID Interaction Module will be slightly larger than the illustrated range, but due to terrain constraints, it will still be significantly smaller than the coverage of other Buff Zones. It is recommended that robots maintain a speed below 0.5 m/s when passing through to ensure stable detection.



[1] Locations where RFID Interaction Module Cards are lodged

Figure 5-25 RFID Interaction Module Card Layout

Table5-20 Buff Types

Type	**Notes**
Attack Buff	Increases damage caused by Projectile attacks.
Defense Buff	Changes the damage suffered from a Projectile attack or collision. Negative Defense Buff is also known as “Vulnerability”. Defense Buff is not applicable to HP loss caused by violation penalties, Referee System on-board module disconnection, and Dart influence.
Barrel Heat Cooling Buff	Increases the Barrel Heat Cooling Rate.
Buffer Energy Buff	The robot receives extra Buffer Energy for Chassis Power.

Type	Notes
HP Recovery Buff	The robot recovers a certain amount of HP every second until it reaches the maximum HP.

Attack Buff changes the Projectile damage dealt by robots to: Original Damage × Attack Buff.

Defense Buff changes the Attack Damage received by robots, the Base, and the Outpost to: Original Damage × (1 – Defense Buff).

For more information on original damage, see "Table 5-2 HP Loss Mechanism for Attack Damage or Collision".

Example 1: The Red Team's robot has a 200% Attack Buff and the Blue Team's robot has a 25% Defense Buff, and the progress marked by the opponent Radar exceeds 100. If the Red Team's robot launches a round of 42 mm Projectiles and hits the Blue Team's robot, the damage suffered by the latter should be: 200 × 200% × (1 – 25% + 15%) = 360.

Example 2: The Red Team's robot has a 200% Attack Buff and the progress of the Blue Team's Hero marked by the opponent Radar exceeds 100. If the Red Team's robot launches a round of 42 mm Projectiles and hits the Blue Team's Hero, the damage suffered by the latter should be: 200 × 200% × (1 + 15%) = 460.

When a robot receives more than one buff of the same type, the maximum buff effect should be applied, including Attack Buff, Defense Buff, Vulnerability, HP Recovery, and Barrel Heat Cooling Buff.

Example 1: The Red Team's Infantry obtains a 25% Defense Buff by activating the Large Power Rune. If the robot then occupies the Base Buff Point on its own side, the Defense Buff rises to 50%. However, if a subsequent 15% Vulnerability is applied, the net Defense Buff is 35%.

Example 2: The Blue Team's Infantry obtains a 25% Defense Buff by activating the Large Power Rune. If the robot then occupies the opponent's Fortress Buff Point, it will have a 75% Vulnerability. If it subsequently receives another 15% Vulnerability, it still retains a 75% Vulnerability.

If the final damage or HP recovery after buff calculation is a decimal number, it will be rounded off.

5.5.3.2 Base Buff Points Mechanism

A team's Base Buff Point can be occupied by the team's Ground Robots. Multiple robots from one team can occupy the Base Buff Point simultaneously.

During the Seven-Minute Round, robots that occupy the Base Buff Point on their own side will receive a 50% Defense Buff, exchange for Projectile Allowance, and remove the "Weakened" state. For more information on the "Weakened" state, see "5.2.2 Respawn Mechanism".

5.5.3.3 Central Elevated Ground Buff Point Mechanism

Hero, Infantry, and Sentry can all occupy the Central Elevated Ground Buff Point. Unconnected Central Elevated Ground Buff Points are occupied separately, and the laying range of their RFID Interaction Module Cards is consistent with that in the higher areas of two Terrain Crossing Buff Points (Elevated Ground). Multiple robots from one team can occupy the Central Elevated Ground Buff Point simultaneously. If a robot of one side occupies a Central Elevated Ground Buff Point, no robots of the other side can occupy it at the same time.

Robots that occupy the Central Elevated Ground Buff Point receive a 25% Defense Buff.

5.5.3.4 Trapezoid-Shaped Elevated Ground Buff Point Mechanism

All Ground Robots can occupy the Trapezoid-Shaped Elevated Ground Buff Point on their own side. Multiple robots from one team can occupy the Trapezoid-Shaped Elevated Ground Buff Point simultaneously.

If a Trapezoid-Shaped Elevated Ground Buff Point is occupied by a robot from the same team, the robot occupying the Trapezoid-Shaped Elevated Ground Buff Point will receive a 50% Defense Buff.

5.5.3.5 Terrain Crossing Buff Point Mechanism

There are two Terrain Crossing Buff Points (Elevated Ground) on each side of the Central Elevated Ground. Each team's Road Zone contains two Terrain Crossing Buff Points (Road), two Terrain Crossing Buff Points (Launch Ramp), and six Terrain Crossing Buff Points (Tunnel).

All Ground Robots can occupy the Terrain Crossing Buff Points. Multiple robots from one team and robots from both teams, can occupy a Terrain Crossing Buff Point simultaneously.

A robot must detect the RFID Interaction Module Card at every Terrain Crossing Buff Point of the same type on one side within X seconds, and must not detect any other RFID Interaction Module Card along the way. For Terrain Crossing Buff Points (Road/Elevated Ground), the robot must detect the lower Buff Point first and then the higher one to trigger the Terrain Crossing Buff. For Terrain Crossing Buff Points (Tunnel), it must detect the Buff Points at one end, the middle, and the other end in sequence to trigger the buff. The X values are as follows: 3 seconds for Terrain Crossing Buff Points (Road) and (Tunnel), 5 seconds for Terrain Crossing Buff Points (Elevated Ground), and 10 seconds for Terrain Crossing Buff Points (Launch Ramp). The specific buffs are as follows:

Terrain Crossing Buff (Launch Ramp)

- 25% Defense Buff for 30 seconds

Terrain Crossing Buff (Elevated Ground)

- 25% Defense Buff for 30 seconds

Terrain Crossing Buff (Road)

- 25% Defense Buff for 5 seconds
- A robot that receives a Terrain Crossing Buff (Road) cannot receive the same buff again within 15 seconds.

Terrain Crossing Buff (Tunnel)

- 50% Defense Buff for 10 seconds
- Double Barrel Heat Cooling Buff for 120 seconds

When Robots already have a terrain crossing buff (Launch Ramp, high ground, or Road) and gain the same type of buff again, they will receive a 50% Defense Buff. At the same time, the remaining duration of this Defense Buff will be updated to the longer of the remaining time of the current buff and the initial duration of the newly gained buff.

Example:

If a robot first gain a terrain traversal buff (Launch Ramp) with an initial duration of 30 seconds, and at 10 seconds (with 20 seconds remaining) gain another terrain traversal buff (Road) with an initial duration of 5 seconds, the remaining duration will be the larger value, which is 20 seconds.

If a robot first gain a terrain traversal Buff (Launch Ramp) and, at 29 seconds (with 1 second remaining), gain another terrain traversal Buff (Road, with an initial duration of 5 seconds), the remaining duration is updated to 5 seconds.

After a robot is defeated, all previously acquired Terrain Crossing Buffs will be removed.

5.5.3.6 Outpost Buff Point Mechanism

When a team's Outpost is alive, the team's Hero, Engineer, Infantry, and Sentry robots can all occupy its Outpost Buff Point. Within the first five minutes of the match, if a team's Outpost is alive and the opponent's Outpost is destroyed, the team's Hero, Engineer, Infantry, and Sentry can occupy the opponent's Outpost Buff Point. Multiple robots from one team can occupy the Outpost Buff Point simultaneously.

Robots that occupy the Outpost Buff Point can exchange for Projectile Allowance (see “5.3.2 Projectile Allowance Mechanism“), gain a 25% Defense Buff, and remove the “Weakened” state along with its corresponding Invincible state. For more information on the “Weakened” state, see “5.2.2 Respawn Mechanism”.

5.5.3.7 Assembly Zone Buff Point Mechanism

Only Engineers of a team can occupy the team's Assembly Zone Buff Point.

An Engineer that occupies the Assembly Zone Buff Point will become invincible for up to 180 seconds in total. The period is not reset if it leaves the Buff Point and then occupies it again.

5.5.3.8 Resupply Zone Buff Point Mechanism

All Ground Robots can occupy their own side's Resupply Zone Buff Point. Multiple robots from one team can occupy the Resupply Zone Buff Point simultaneously.

A robot that occupies its team's Resupply Zone Buff Point can speed up the respawn timer or receive an HP Recovery Buff. For detailed implementation methods and figures, see “5.2 HP Recovery and Respawn Mechanism“.

After occupying the Resupply Zone Buff Point, Engineer will become invincible.

5.5.3.9 Fortress Buff Point Mechanism

Only Infantry or Sentry from a team can occupy the team's Fortress Buff Point. A team's Fortress Buff Point can be occupied by robots from both teams, but can only be occupied by one robot of the team at any given time. A team's Fortress Buff Point can be occupied by multiple opponent robots at the same time.

A team's Fortress Buff Point remains unavailable until its Outpost is destroyed. A team's Fortress Buff Point becomes available when its Outpost is destroyed.

Occupation by Own-Side Robots

Define Δ as the difference between the maximum HP and current HP of the own-side Base.

Robots on their own side that occupy the Fortress Buff Point receive a 50% Defense Buff, as well as a Barrel Heat Cooling Buff equal to w, where w is related to Δ and follows the formula: $w = \frac{\Delta}{40}$ (rounded down). The upper limit for w is 75. The Barrel Heat Cooling Buff from the Fortress Buff Point is applied at the highest rate among all available Barrel Heat Cooling Buffs.

Example:

A robot with a Barrel Heat Cooling Rate of 40/second has received a 75 Barrel Heat Cooling Buff from the Fortress Buff Point and a 5-time Barrel Heat Cooling Buff at the same time. Since 40 × 5 > 40 + 75, the actual Barrel Heat Cooling Rate of the robot is 200/second.

After a robot occupies the Fortress, the Fortress Buff Point will provide the robot with a “reserved Projectile Allowance,” which is independent from the robot's own Projectile Allowance, with an upper limit of N. When a robot occupying the Fortress Buff Point launches Projectiles, it will prioritize consuming the “reserved Projectile Allowance”. Once the reserved Projectile Allowance is depleted, the robot's own Projectile Allowance will be used.

$$N=100+2\times\frac{\Delta}{15}\text{(rounded down), with N up to 500.}$$

Occupation by Opponent Robots

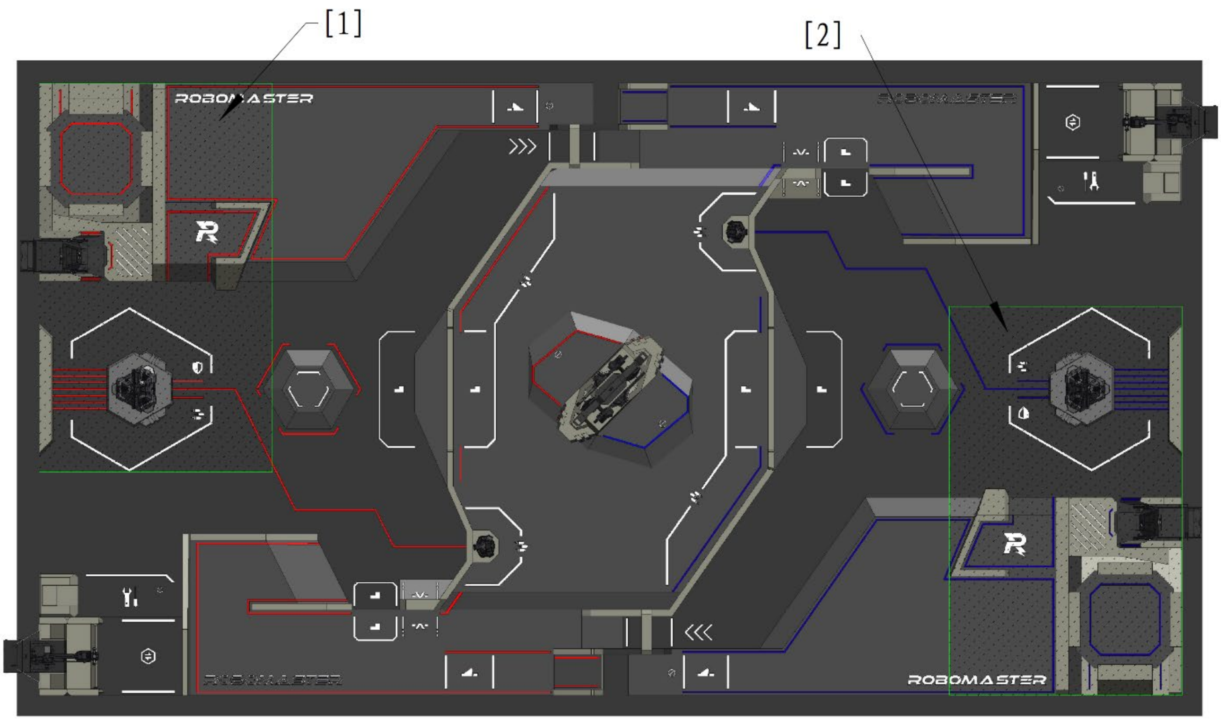
After the match has been underway for three minutes and the opponent's Outpost has been destroyed, Infantry and Sentry can occupy the opponent's Fortress Buff Point.

Robots occupying the opponent's Fortress Buff Point become 100% more susceptible to damage. When a single robot accumulates 20 seconds of uninterrupted occupation, the opponent's Base Protective Armor will be expanded. The timer for occupying the opponent's Fortress Buff Point can be retained for 3 seconds after the robot is defeated or the Occupy status expires, and the timer will pause during this retention period. After the opponent's Base Protective Armor is expanded, the robots occupying the opponent's Fortress Buff Point will no longer become more vulnerable.

5.6 Special Mechanism of Robots

5.6.1 Special Mechanism of Hero

When a Hero selects the “Long-Range Attack-Focused” type and is located within the deployable area of its team (see “Figure 5-26 Own-Side Deployable Area“), it can choose to enter Deployment Mode. The Hero enters Deployment Mode two seconds after confirmation. In this mode, the Hero's chassis powers off, video transmission is lost, a 25% Defense Buff is gained, and the maximum initial launching speed increases to 16.5m/s. When it launches 42 mm Projectiles to attack the Base, a 150% Attack Buff is granted.



[1] Red Team's Deployable Area [2] Blue Team's Deployable Area

Figure 5-26 Own-Side Deployable Area

- Criteria for determining whether the Hero is within the deployable area of its team:

  ① The Hero's RFID Interaction Module does not detect an RFID Interaction Module Card of the opponent team's deployable area.

  ② The Hero's RFID Interaction Module detects an RFID Interaction Module Card in the deployable area of its team, or the coordinates from the UWB Positioning Module are within the deployable area of its team.

The Hero is considered to be within its team's deployable area when conditions ① and ② are all met. When either condition ① or ② is not met, the Hero will automatically exit Deployment Mode.

- For information on how a Hero enters and exits Deployment Mode, see the Referee System Player's Client Interface Instructions.
- When a Hero is in Deployment Mode, video feeds of both the Official Client and Custom Client will be cut off, but custom data streams of the Custom Client will not be affected.
- After video transmission is lost due to Hero's Deployment Mode, the Mini-map and custom UI remain available.

5.6.2 Special Mechanism of Engineer

During the first three minutes of the match, as long as the Engineer remains alive, it has a 50% Defense Buff.

5.6.3 Special Mechanisms of Drone

Air Support

In each round of the competition, the maximum Projectile Allowance for a Drone is 750 rounds of Projectiles. During the Seven-Minute Round, Drones cannot obtain Projectiles from any source.

At the start of the competition, Drone has 30 seconds of Air Support time, with an additional 20 seconds added every minute thereafter.

The Gimbal Operator can call for or pause Air Support via the Referee System Player's Client. During Air Support, Drones will receive the FPV of the Battlefield and can launch Projectiles while not in contact with the Landing Pad. During Air Support, the robot's Launching Mechanism is unlocked; otherwise, it is locked.

If the Gimbal Operator does not pause Air Support after the Air Support time runs out, each additional second of Air Support will consume one Gold Coin. For details, see “Table5-8 Exchange Rules“.

Radar Countering Drone

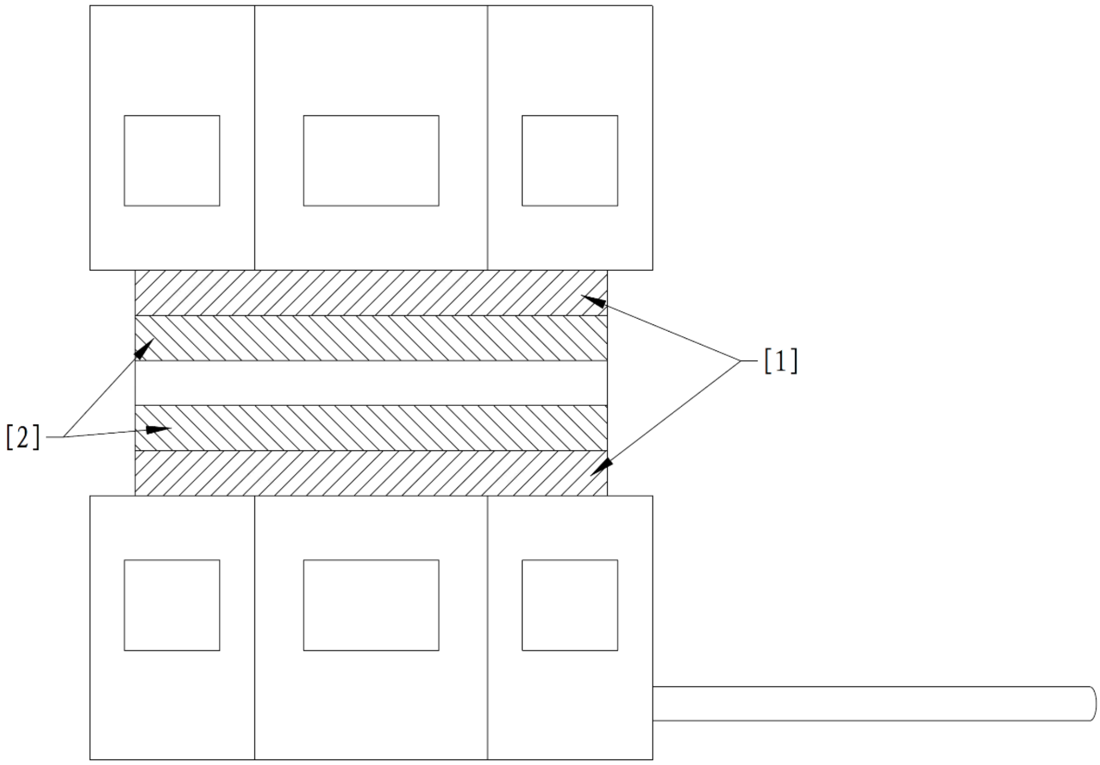
A laser emitter may be installed on the Radar System to aim at and illuminate the Laser Detection Module on the Drone. The laser emitter and its actuator on the Radar System shall be powered on only when the opponent Drone initiates Aerial Support. The Drone possesses a "Targeting Progress" (P), ranging from 0 (initial and minimum value) to 100 (maximum value). When P reaches P_0, the Drone's launching mechanism becomes locked, and P immediately resets to zero. This locked state persists for 45 seconds before being lifted. The Dron's launching mechanism can be locked a maximum of 3 times per match via this mechanism.

Each time this Mechanism is used to lock the Opponent Drone's Launching Mechanism, the Sighting difficulty of the laser detection module will increase, as follows:

Before the first successful lock: the laser detection module remains unchanged, and P_0 is 50 for this round.

Before the second successful lock: the laser detection module's detectable area is reduced vertically to 3/5 of its original size, and P_0 is 100 for this round.

Before the second successful lock: the laser detection module's detectable area is reduced vertically to 3/5 of its original size, and P_0 is 100 for this round.



[1] First reduction of the area [2] Second reduction area

Figure 5-27 Graph of the reduced area of the laser detection module

The calculation logic for “Targeting Progress” is as follows:

Parameter Definitions:

- t: Duration of current continuous laser illumination.
- Δt: Determination interval, fixed at 0.1 seconds.
- n: Consecutive determination count (n=0 at the start of illumination; n increments by 1 for each triggered determination).

Determination Rules:

When the Drone’s Laser Detection Module is not illuminated by the laser, Progress P decays at a constant rate of 0.5/s, but will not drop below 0. Meanwhile, t and n reset to zero immediately.

When the Laser Detection Module is illuminated by the laser, P stops decaying. Every time t accumulates 0.1 seconds, an increase in progress is triggered:

- The 1st 0.1 seconds: P = P + 1
- The 2nd 0.1 seconds: P = P + 2
- The n-th 0.1 seconds: P = P + n

If a single continuous illumination lasts less than 0.1 seconds before being interrupted, t and n reset to zero immediately.

5.6.4 Special Mechanism of Sentry

After the Three-Minute Setup Period begins and before the 15-Second Referee System Initialization Period, the Gimbal Operator can select either the fully automatic control or semi-automatic control mode for the corresponding Sentry. If the control mode is not selected within the specified time, it will default to fully automatic control at the start of the Seven-Minute Round. The Sentry can exchange for Projectile Allowance, remotely exchange for Projectiles and HP, confirm respawn, exchange for Instant Respawn, and switch modes, by communicating directly with the Referee System server.

In addition, the Gimbal Operator can use relevant commands from the Referee System Player's Client to intervene in the actions of the Sentry. The specific commands include:

- Forward the Mini-map marker
- Purchase Projectile Allowance in the Resupply Zone, Base Buff Point, or Outpost Buff Point.
- Confirm respawn
- Perform remote exchange for Projectiles
- Perform remote exchange for HP
- Perform remote exchange for Instant Respawn
- Switch modes

For semi-automatic Sentry, the above operations do not require any Gold Coin. For fully automatic Sentry, the Gimbal Operator must spend 50 Gold Coins for each operation performed.

The Sentry features a “pose” system that enhances certain abilities while reducing others. Sentry starts each round in Mobility mode by default. Mode switching has a cooldown of five seconds. If a Sentry remains in the same mode for more than three minutes during a round, the effect of that mode will decrease, as detailed in the table below.

Mode	Default Effect	Decreased Effect
Offense	Gain triple Heat Cooling Buff, have Chassis Power halved, and become 25% more vulnerable.	Gain double Heat Cooling Buff, have Chassis Power halved, and become 25% more vulnerable.
Defense	Gain a 50% Defense Buff, have Chassis Power halved, and have the heat cooling rate reduced to 1/3 of the original.	Gain a 25% Defense Buff, have Chassis Power halved, and have the heat cooling rate reduced to 1/3 of the original.
Mobility	Have the Chassis Power Limit increased to 1.5 times the original, become 25% more vulnerable, and have the heat cooling rate reduced to 1/3 of the original.	Have the Chassis Power Limit increased to 1.2 times the original, become 25% more vulnerable, and have the heat cooling rate reduced to 1/3 of the original.

- All of the calculations above are rounded to the nearest integer.
- Changes to heat cooling and Chassis Power Limit are applied multiplicatively based on the Sentry's values after calculations of Heat Cooling Buff and other Chassis Power.

Example: In its default Defense mode, a fully automatic Sentry's basic heat cooling rate is 30 per second. It then gains a triple Barrel Heat Cooling Buff by activating a Large Power Rune, and also gains an additional Barrel Heat Cooling Buff of 75 per second by occupying the Fortress Buff Point. First, take the larger one of the two values (i.e., +75 per second). At this point, the fully automatic Sentry's heat cooling rate is 30 + 75 = 105. Then, apply the cooling rate decay factor (1/3) in Defense mode, resulting in the fully automatic Sentry's actual heat cooling rate: 105 × 1/3 = 35.

5.6.5 Special Mechanism of Dart System

In each round, the Dart System can load up to four Darts, the gate of a Dart Launching Station can be opened twice, once after 30 seconds of the competition have elapsed, and the other after 4 minutes. The Gimbal Operator can choose when to open the gate, and any unused opportunities are cumulative.

Opening and Closing of the Dart Launching Station Gate

During the match, the Gimbal Operator will use the keyboard and mouse on the Referee System Player's Client to open and close the gate of the Dart Launching Station. The Gimbal Operator is not allowed to launch Darts when the gate is opening or closing. The Referee System Player's Client will display the status of the gate.

It takes around seven seconds for the gate to open completely.

When the gate of the Dart Launching Station is fully open, the Gimbal Operator can then launch Darts by controlling the Dart System within a 30-second period. Meanwhile, the Dart Detection Module on the Outpost or Base of the other team will update the Detection Window, which lasts for 40 seconds. The launched Dart needs to hit the Dart Detection Module within the Detection Window, or the attack will be invalid.

When the gate closes for the first time, the Dart Launching Station will enter a 15-second cooling period. The gate can only open for the second time after the end of the cooling period. During the cooling period, the Dart Launching Station's gate cannot open.

Selecting a Dart Target

After an Outpost is destroyed, the Gimbal Operator can choose the Dart's target as a “Fixed Target”, “Random Fixed Target”, “Random Moving Target”, or “Terminal Moving Target” via the Referee System Player's Client before the Dart Launching Station's gate opens.

- If “Fixed Target” is selected, the Dart Detection Module will remain in its initial location.
- If “Random Fixed Target” is selected, the Base Dart Detection Module will shift to another random location before the gate fully opens, and move back to the initial location when the Dart Detection Window ends. In the Detection Window, if a Dart hits the target, the Dart Detection Module will shift to a new location.
- If “Random Moving Target” is selected, the Base Dart Detection Module will begin moving as the gate starts to open. After a launched dart is detected, the module shifts to a new location distinct from where it was at the time of the launch within 600 ms and then remains stationary at the new location. In the Detection Window, after a Dart hits the target or 10 seconds after the first dart launch is detected, the Dart Detection Module resumes moving until the next launch is detected.
- If “Terminal Moving Target” is selected, when the gate of the Dart Launching Station opens, the Base Dart Detection Module will begin to move. After 1.2 seconds from detecting the launch of any Dart, it will start to move to a new location different from where it was at the moment the Dart was launched, completing the move and coming to a stop within 600 ms. In the Detection Window, after a Dart hits the target or 10 seconds after the first dart launch is detected, the Dart Detection

Module resumes moving until the next launch is detected.

The Dart Detection Module and Dart Guiding Light maintain their relative locations while moving, i.e., moving within the same plane of their original locations. Their movement extends parallel to the field's width, ranging from –280 mm to 280 mm from their starting point.

Gains for Successful Dart Hits

The following are the additional gains for a successful Dart hit, apart from the Damage dealt to the target itself:

- When a Dart hits the other team's Outpost, the Fixed Target or the Random Fixed Target of the other team's Base for the first, second, third, and fourth time, the operating interface of all the team's Operators, the location of robots on the Mini-map, and the custom UI will be blacked out for 10 seconds (first time), 5 seconds (second time), 3 seconds (third time), and 2 seconds (fourth time), respectively. When a Dart hits the other team's Base, the alive Hero, Infantry, and Drone of the attacking side share 200 (Fixed Target) or 600 (Random Fixed Target) Experience Points.
- When a Dart hits a Random Fixed Target at the Opponent's Base, any Terrain Crossing or Power Rune Buff Effect currently held by the other team will temporarily expire. Timing continues as normal, but no Buff Effect is granted during this period. The expiration duration matches the blackout duration mentioned above.
- When a Dart hits a Random Moving Target or Terminal Moving Target at the other team's Base, the operating interface of all the team's Operators, the location of robots on the Mini-map, and the custom UI will be blacked out for 10 seconds. The alive Hero, Infantry, and Drone of the attacking side will share 2,500 Experience Points, and any Terrain Crossing or Power Rune Buff Effect currently held by the other team will expire.
- If the Dart Detection Module of the Base is hit after “Random Moving Target” or “Terminal Moving Target” is selected, all alive Ground Robots of the other team will immediately lose 10% (for Random Moving Target) or 25% (for Terminal Moving Target) of their maximum HP respectively. This HP loss is counted in the total damage for the other team, but this HP loss itself and any resulting defeated robots will not generate any experience. If the target is hit continuously, the duration when the operating interface is blacked out will increase accordingly.
- If four Darts all hit the other team's Base when “Fixed Target” or “Random Fixed Target” is selected, or if any of the Darts hits the other team's Base when “Random Moving Target” or “Terminal Moving Target” is selected, the Base Protective Armor of the attacked Base will be expanded immediately.
- When the Dart Guiding Light on the Base or Outpost is illuminated, its Buff Points will expire temporarily for 30 seconds if the Base or Outpost is hit by a Dart; if it is hit successively, the

expiration period will be refreshed. After each Dart hit, the Detection Window for the corresponding target will close for 2 seconds.

- During periods when the Operator's interface, the location of robots on the Mini-map, or the custom UI are blacked out, the Operator can choose to control the robots in semi-automatic mode. Specifically, the Referee System Player's Client will display a large map interface similar to what the Gimbal Operator sees. The Operator can send commands to control the robots by tapping the large map interface, with at least 3 seconds required between consecutive commands.
- The Custom Client can still normally receive video transmission and other data.

5.6.6 Special Mechanism of Radar

The functions of the Radar include radar detection, radar countermeasures, and Radar analysis of information waves, as detailed below:

Radar Detection

A team's Radar can detect the locations of both teams' Ground Robots and Drones, and send the coordinates of the robots to the Referee System server. Each Ground Robot and Drone have a “Tracking Progress” value P, ranging from 0 (initial and minimum value) to 150 (maximum value). The Referee System updates P in real time based on the recognition accuracy of the Radar. Consistently accurate tracking will cause P to rise more quickly, while consecutive tracking errors or data interruptions will cause P to decay more rapidly. When P reaches a certain threshold, the corresponding target will receive a specific effect. In addition, by continuously tracking and keeping the opponent in a vulnerable state, the Radar can also unlock a “Double Vulnerability” effect.

If the wireless environment of the competition site is poor, or a large part of the UWB Positioning Module is blocked, the coordinates detected by a robot's UWB Positioning Module may drift intermittently or continuously.

Parameter settings:

- d: recognition error, i.e., the straight-line distance between the coordinates sent by the Radar and the robot's actual planar coordinates, divided into three levels:
  - Accurate: d < 0.8 m
  - Semi-accurate: 0.8 m ≤ d < 1.6 m
  - Inaccurate: d ≥ 1.6 m
- x: current tracking impact value (i.e., change to the progress), with an initial value of 0.
- Determining frequency: decided by the Radar's transmission frequency.

Logic for calculating the progress P:

Whenever the Referee System receives Radar data or encounters a data interruption, it updates x first, then updates P.

Table5-21 Logic for Calculating the “Tracking Progress” P

Tracking Status	Logic for Updating x	Logic for Updating P
Accurate	If previous tracking is accurate or semi-accurate, x = x + 1.0; otherwise, x = 1.0.	P = P + x If the calculated value of P is less than 0, set P to 0; if the calculated value of P is greater than 150, set P to 150.
Semi-accurate	If previous tracking is accurate or semi-accurate, x = x + 0.5; otherwise, x = 0.5.	
Inaccurate	If previous tracking is inaccurate or interrupted, x = x − 0.8; otherwise, x = −0.8.	
Data interruption	If the Referee System does not receive any data for a cumulative period of 0.5 seconds, it is considered to trigger an “Inaccurate” judgment.	

Example: If a team's Radar sends the coordinates of the other team's Engineer at a frequency of 5 Hz and an accuracy of 100% continuously to the Referee System server, the Tracking Progress of the Engineer within 0 to 1 second at an interval of 0.2 seconds will be 1, 3, 6, 10, and 15 respectively. At the 3rd second, the robot's Tracking Progress will reach 120. Thereafter, if the Radar sends the robot's coordinates at a frequency of 5 Hz and an accuracy of 0% to the Referee System server, the Tracking Progress of the robot within 3 to 4 seconds at an interval of 0.2 seconds will be 119.2, 117.6, 115.2, 112, and 108 respectively. Thereafter, if the Radar does not send the Engineer's coordinates for 1 seconds, the Tracking Progress of the robot within 4 to 5 seconds will be 103.2 and 97.6 respectively (determined as two additional and consecutive errors based on the previous error). Thereafter, if the Radar sends the correct coordinates of the Engineer twice consecutively, the robot's Tracking Progress will be 98.6 and 100.6 respectively.

Thresholds of P and corresponding effects:

Table5-22 Thresholds of P and Corresponding Effects on the Target

Target Type	Value Range for P	Mini-map Display Effect	Buff (Ground Robots only)
Opponent robot	[0, 100)	Location corresponding to the robot's coordinates sent by own-side Radar	No additional effect

Target Type	Value Range for P	Mini-map Display Effect	Buff (Ground Robots only)
	[100, 120)	● Actual location detected by the robot' UWB Positioning Module ● Special marker	15% more vulnerable
	[120, 150]	● Actual location detected by the robot' UWB Positioning Module ● Special marker	20% more vulnerable
Own-side robot	[0, 50)	Location corresponding to the robot's coordinates sent by own-side Radar	No additional effect
	[50, 150]	● Actual location detected by the robot' UWB Positioning Module ● Special marker	No additional effect

Double Vulnerability Mechanism:When the Radar accumulates a vulnerability effect on the opponent robot for 1 minute (when multiple robots are affected, the duration of the buff applied to these robots will be calculated separately), the Radar will receive a chance to trigger "double vulnerability". The Radar can send a command to the Referee System, consume the chance, and double the vulnerability of all vulnerable robots, which will last 30 seconds. This "double vulnerability" effect can be triggered twice at most for each round.

Radar Countermeasures

For details, see "5.6.3 Special Mechanism of Drone".

Radar Analysis of Information Waves

During the match, an official signal emitter will be placed on the Radar Base of both teams. This emitter will transmit electromagnetic waves containing information waves carrying the opponent team's data, as well as interference waves in adjacent frequency bands. The Participating Teams are permitted to design and install antennas and demodulation circuits on their Radar to decode and identify these information and interference waves, thereby gaining information and strategic advantages for their team.

The modulation method of the electromagnetic wave is 2-GFSK, which stands for binary Gaussian Frequency Shift Keying; the FSK mapping relationship is: 0 → -1, 1 → +1; the relevant parameters for each level of electromagnetic wave are shown in the table below:

Table5-23 Frequency, Bandwidth, and Strength of Information Waves

Wave Source	Center Frequency (MHz)	Bandwidth (MHz)	Power Strength (dBm)	Gaussian filter parameters
Red Team broadcast source	433.2	0.54	-60	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:1.5756 rad/sample
Red Team Level 1 interference source	432.2	0.94	-10	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:2.8323 rad/sample
Red Team Level 2 interference source	432.5	0.86	-10	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:2.5809 rad/sample
Red Team Level 3 interference source	432.8	0.25	-10	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:0.6646 rad/sample
Blue Team broadcast source	433.92	0.54	-60	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:1.5756 rad/sample
Blue Team Level 1 interference source	434.92	0.94	-10	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:2.8323 rad/sample
Blue Team Level 2 interference source	434.62	0.86	-10	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:2.5809 rad/sample

Wave Source	Center Frequency (MHz)	Bandwidth (MHz)	Power Strength (dBm)	Gaussian filter parameters
Blue Team Level 3 interference source	434.32	0.25	-10	SPS:52,BT:0.35,Sample Rate:1M, Sensitivity:0.6646 rad/sample

Parameter Relationships:

- SPS (Samples Per Symbol) = Sample Rate / Symbol Rate
- Sample Rate = 1M (Sample Rate refers to the number of samples per second)
- Symbol Rate = Baud Rate = = Bit Rate (1 bit per symbol in 2FSK)
- BT (Gaussian filter bandwidth time product) = 0.35
- Sensitivity (frequency modulation sensitivity) = 2 * pi * Δf / Sample Rate (unit: rad/sample)
- Frequency offset Δf = signal_bandwidth / 2 - Sample Rate / SPS
- Bandwidth B = signal_bandwidth (PlutoSDR RF filter bandwidth)

Detailed spectrum distribution, air interface frame structure, time domain characteristics, and other information can be referred to in Appendix 1: Radar Information Wave Characteristics.

The information wave will include the following details about the opponent team:

- Each robot's location
- Each robot's HP
- Remaining Projectile allowance for each robot
- Remaining economy of the team
- Current buffs held by each robot
- Total economy of the team
- Buff Points status

For the specific structure of the communication protocol, please refer to the RoboMaster 2026 University Series Communication Protocol.

The interference wave does not contain specific information; however, it includes a 6-character key (alphanumeric combination) customized by the opponent. The initial key at the start of the match is a random value.

The Radar can interpret the key and send it to the RoboMaster Champion server. If the interpretation is successful, the subsequent opponent interference wave difficulty increases by one level. When the difficulty level of the opponent interference wave difficulty is higher than that of the team, it will not cause vulnerability or accumulate “double vulnerability” progress even if the opponent Radar accurately tracks the team's robots.

5.6.7 Energy Mechanism

In each round, every Hero, Infantry, or Sentry has an initial chassis energy of 20,000 and a chassis energy limit of 40,000. After the chassis energy is depleted to 0, the robot will enter the power-saving mode. Robots in power-saving mode will have their basic Chassis Power Limit reduced to 35 W. When the chassis energy is greater than or equal to 25,000, the basic Chassis Power Limit for robots will increase to 125% of the basic value corresponding to the relevant Performance System and Level, but it will not exceed 200 W.

Chassis Energy Consumption

Each time the Power Management Module Chassis Port outputs 1 J of energy, the chassis energy decreases by 1. Chassis energy will not become negative.

Replenishing Chassis Energy

When occupying the Resupply Zone Buff Point, robots can recharge chassis energy wirelessly.

When the Supercapacitor Management Module Port (input) is transferring energy to the supercapacitor, the chassis energy increases by 8 for every 1 J difference between the energy output from the Supercapacitor Management Module Port (input) and the Power Management Module Chassis Port.

5.7 Custom Client

A Custom Client is a multi-purpose device developed by the Participating Teams to receive competition-related information and robot video feeds sent by the Referee System's RoboMaster Champion server, and to transmit competition-specific interaction commands to the RoboMaster Champion server.

A typical Custom Client may consist of a display device, a computing platform (including a memory, controller, and processor), and a set of input devices. It can also be integrated into a single unit, such as a handheld tablet device with a network port.

Example: A team developed a Custom Client capable of receiving robots' video transmission and displaying it on a self-made monitor, while simultaneously overlaying a custom UI shown via an unofficial data link. The robots can be controlled through a Custom Controller or a homemade joystick. As a result, the team can complete the competition without having to use the official Referee System Player's Client.

The Custom Client can connect to the RoboMaster Champion server via the RJ45 interface configured in the Operation Room to send and receive relevant data, and can also communicate with the Custom Controller via wired connection.

A team's Custom Client can receive competition information accessible to the team from the Referee System Player's Client, such as both teams' HP, the team's economy, remaining match time, and announcements of key match events.

The Custom Client can send competition commands supported by the Player's Client to the Referee System, such as selecting a Performance System before the match, exchanging Projectile Allowance during the match, and initiating Air Support or other active commands.

5.8 Match Format and Winning Criteria

The official competition of RMUC 2026 consists of two stages: Group Stage and Knockout Stage. The Group Stage follows the BO2 or BO3 match format, while for Knockout Stage it is BO3 or BO5.

The winner of a round is determined item by item in the following numbered order of priority, until a final result is reached:

1. When the round has ended and if the Base of either team is destroyed, the team with the higher Base Remaining HP will be the winner.
2. When the round has ended and if the Base Remaining HP of both teams is the same and neither Outpost has been destroyed, the team whose Outpost has higher Remaining HP will be the winner.
3. When the round has ended and if the Base Remaining HP of both teams is the same, but only one team's Outpost has been destroyed, the team whose Outpost remains intact will be the winner.
4. When the round has ended and if the Base Remaining HP of both teams is the same and the Outposts have the same Remaining HP or have all been destroyed, the team that has inflicted the higher total Attack Damage will be the winner.
5. When the round has ended and if the Base Remaining HP of both teams is the same, the Outposts have the same Remaining HP or have all been destroyed, and the total Attack Damage inflicted by each team is the same, the team with the higher total Remaining HP will be the winner.
6. If neither team fulfills these criteria, the round is considered a draw. In BO3 and BO5 rounds, any draw will result in an immediate tie-breaker round, until a winner emerges.

6 Competition Process

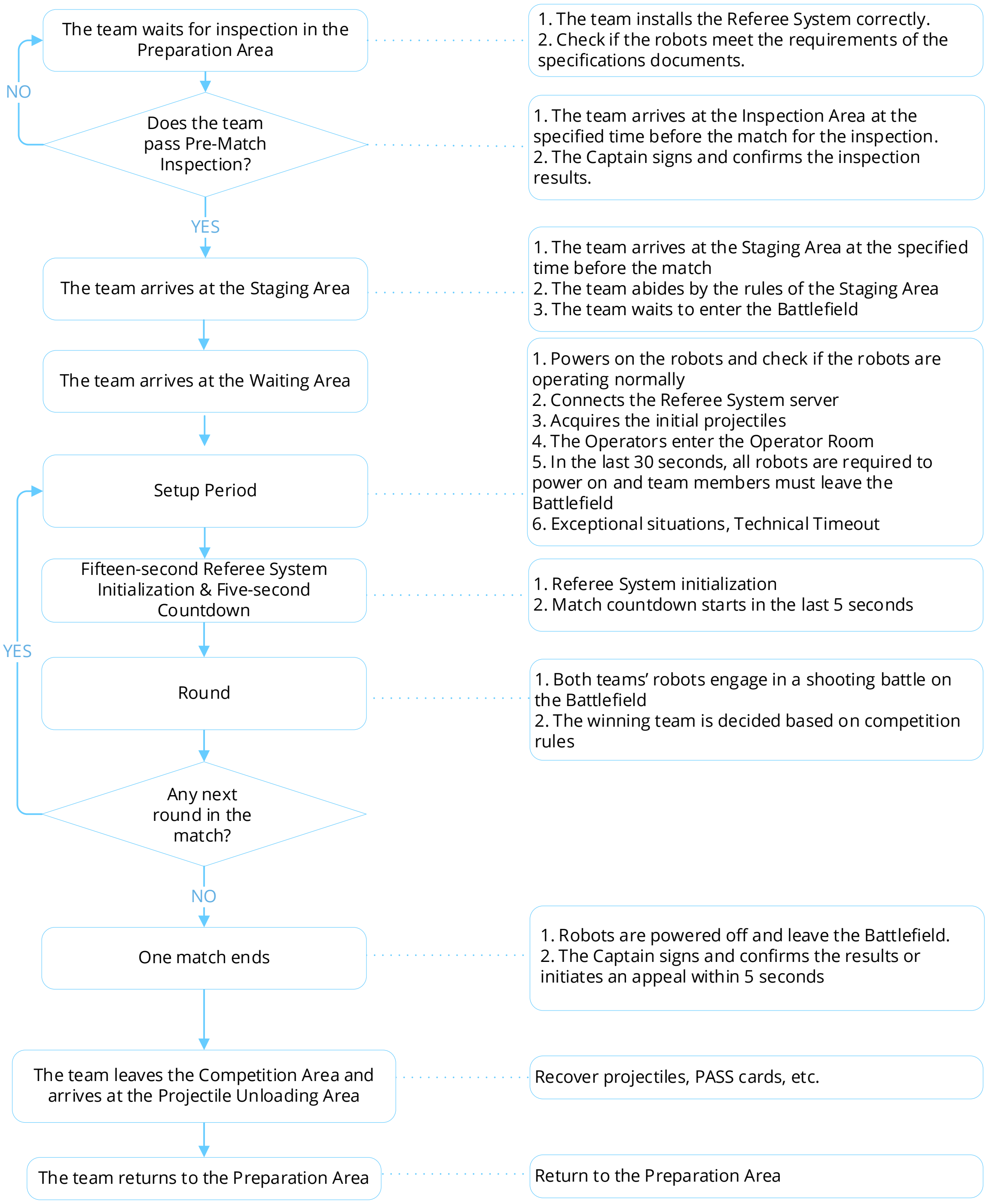


Figure 6-1 Process of a Single Match

6.1 Pre-Match Inspection

- The inspection results of the Mock Inspection and Practice Match are for reference only and are not taken into account for the inspection in the actual competition.

- The inspection results during the competition are only applicable to the current match.
- Passing of inspection only means that the robot meets the specifications at the time of inspection. The Participating Teams are required to ensure that their robots fulfill the requirements of the Building Specifications Manual at all times. During inspection, teams must proactively demonstrate all robot functions to inspection personnel.

To ensure that robots built by the Participating Teams meet the requirements outlined in the RoboMaster 2026 University Series Robot Building Specifications Manual, the Participating Teams are advised to enter the Inspection Area for Pre-Match Inspection 60 minutes before each match begins. If a participating team does not begin Pre-Match Inspection 55 minutes prior to the start of the match, their Backup Robot will lose eligibility to enter the field. If Pre-Match Inspection has not started 50 minutes before the beginning of the match, one Ground Robot will be disqualified from entering the field. Any robots that have not arrived at the Inspection Area 45 minutes before the start of the match will not be permitted to enter the Inspection Area. The inspection process is as follows:

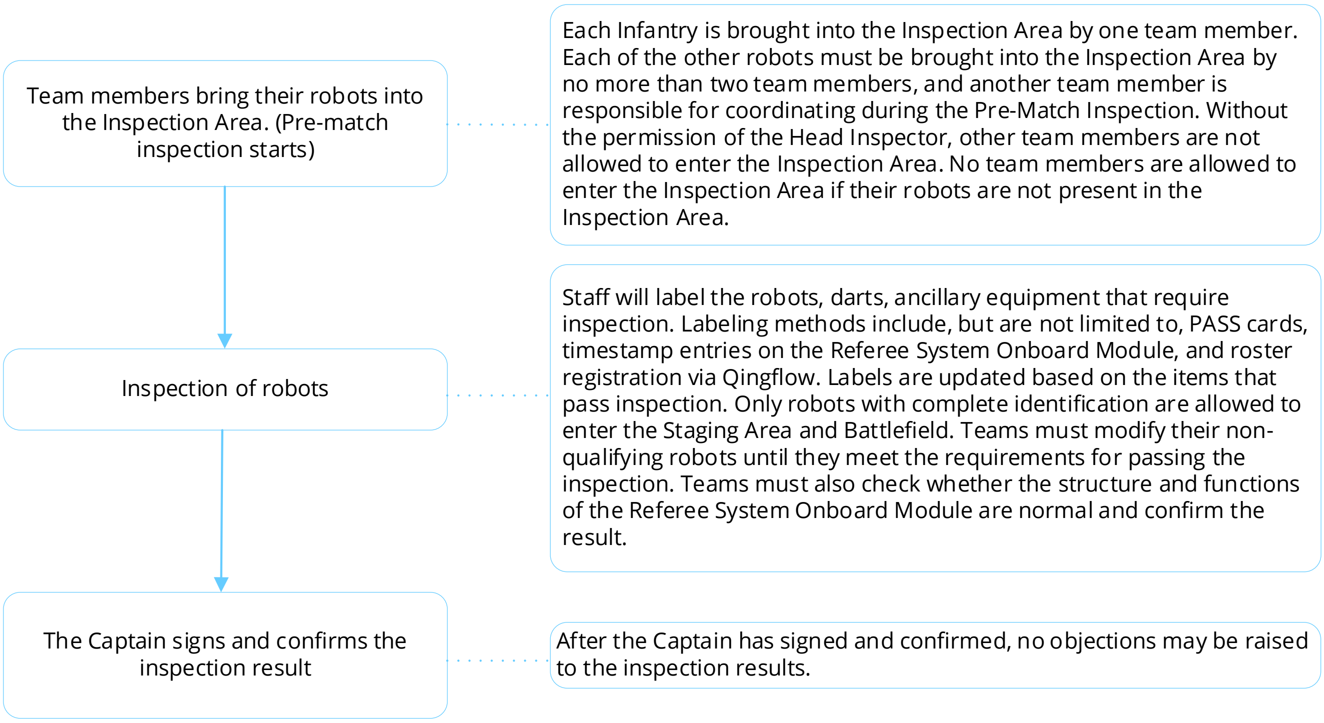


Figure 6-2 Pre-Match Inspection Process

The rules regarding Backup Robots are as follows:

- During the Mock Inspection, each team can register up to 10 robots. Throughout subsequent matches in the same competition stage, both the robots and Backup Robots of the Participating Teams must be listed on the registered roster, and no substitutions are allowed within the same competition stage. Unique identifiers, such as tamper-evident labels or laser engraving (to be determined onsite), are used to mark the primary structures of registered robots. During competition inspections, these identifiers will be verified. If a marked robot requires maintenance during the competition, the unique identifier will be affected, and any changes must be reported through the designated channel.

After the Wild Card competition ends, the list of robots can be re-registered, and the specific schedule will be subject to the latest notice from the RMOC.

- Each participating team can have a maximum of 2 Backup Robots, 10 Custom Controllers, 10 Custom Clients, and 2 wireless charging devices (Transmitter) for each match. Regardless of whether the Backup Robots is the Dart System, a maximum of 4 backup Darts are allowed in each BO2 and BO3 match, while a maximum of 8 backup Darts are allowed in each BO5 match.
- Team members are required to declare the types of Backup Robots they are carrying during the Pre-Match Inspection. Backup Hero and Engineer must have an Armor Sticker affixed in the Inspection Area. If the Backup Robot needs to enter the field as an Infantry or a Sentry, the Pit Crew must promptly obtain the appropriate Armor Sticker from the Referee. Armor stickers must be affixed in accordance with the requirements stated in the RoboMaster 2026 University Series Robot Building Specifications Manual.
- Each participating team can borrow the Referee Systems for no more than two Backup Robots.

6.2 Staging

The Participating Teams must arrive at the Staging Area 10 minutes before each match. The staff at the Staging Area will verify the Pass Cards of participating robots and details of Pit Crew.

The number of Pit Crew members allowed for each team is limited as follows:

- 1 Supervisor
- 2 Team Coaches
- 20 Regular Members/Reserve Members, with no more than 2 Reserve Members.

The number of Pit Crew members will increase in the 2026 season, so there will no longer be additional entry activities for veteran members and publicists.

One Pit Crew Member should wear the designated Captain label and undertake the Captain's role. If any participating team needs to repair their robots after entering the Staging Area, they must obtain the permission of the Referee. A robot can leave the Staging Area for repair only after the staff at the Staging Area have removed the Pass Card on the robot. When repair is finished, the robot needs to be brought back to the Inspection Area for another Pre-Match Inspection before re-entering the Staging Area. If the team is unable to arrive at the Staging Area in time as a result of this delay, the robot will not be able to enter the field, and the team will bear the consequences.

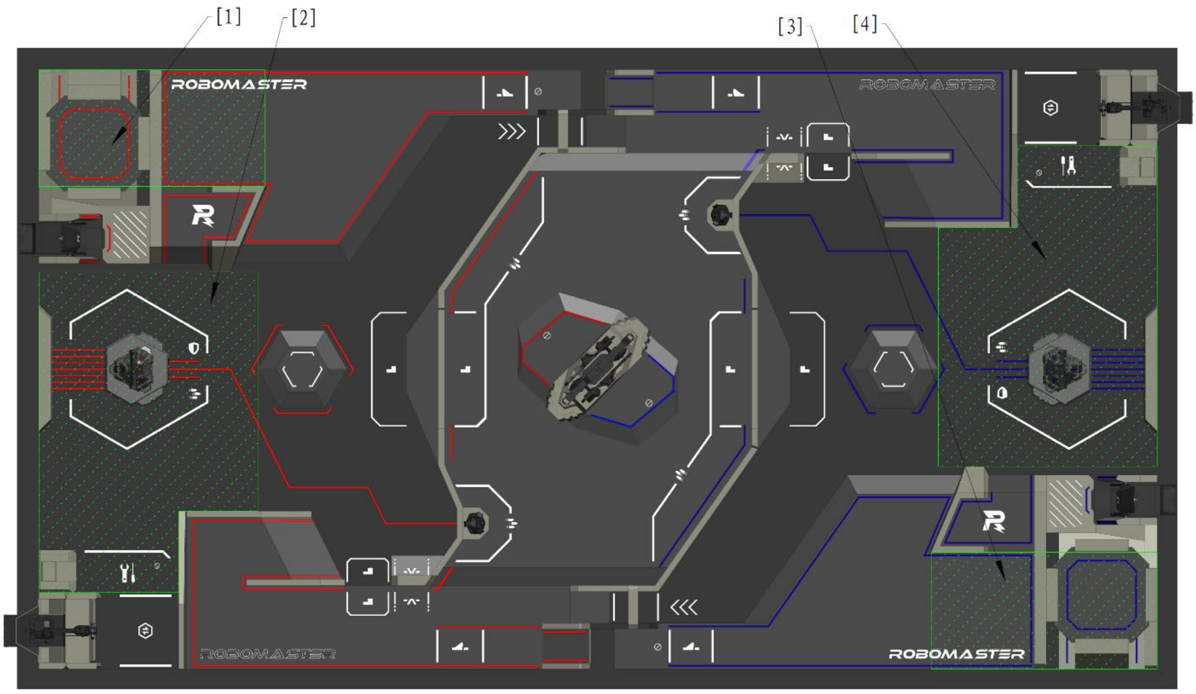
Designated Captain label: Any Pit Crew member wearing the designated Captain label will assume the Captain's role during the match. The Captain's duties include managing the team's competition process, confirming results, and raising appeals.

After leaving the Staging Area, the Participating Teams will enter the Waiting Area to place their robots. With the permission of the Referee, the Participating Teams will wait at the entrance of the Battlefield with their robots for further instructions. The Referee will follow the competition process and open the door and lead the team members into the Competition Area. The countdown for the Three-Minute Setup Period begins once the door is open.

6.3 Three-Minute Setup Period

- After the end of the second and fourth rounds of a BO5 match, both teams have 10 minutes to commission their robots. When 10 minutes run out, the Three-Minute Setup Period of the next round begins.
- When the Participating Teams arrive at the Competition Area, they can collect all the required 42 mm Luminous Projectiles from the designated location. For information on the Projectile quantity limit, see “4.6.2 Projectile ”. For example, under the BO3 Match Format, a participating team can claim up to 300 42 mm Luminous Projectiles at once.
- After a match has ended and the Radar has been removed from the Radar Foundation, the Participating Teams can place the Radar on the Radar Foundation in preparation for the next match.

During the Three-Minute Setup Period, the Pit Crew will place robots in their respective Initial Zones, check whether the Referee System is operating normally, pre-load their robots with Projectiles, load Darts into the Dart System, mount the Radar on the Radar Foundation, and connect the Aerial Safety Rope to the Drone. The Pit Crew can repair robots or replace equivalent parts, provided the requirements of the specifications documents are met. The Pit Crew are required to commission their robots near their team's Initial Zone. Only one Pit Crew member from each team is allowed to leave the zone to commission their robots, and they must not cross the Central Elevated Ground. The Initial Zones are shown in the figure below.



[1] Red Team Landing Pad and its Initial Zone [2] Red Team Base and its Initial Zone

[3] Blue Team Landing Pad and its Initial Zone [4] Blue Team Base and its Initial Zone

Figure 6-3 Commissioning Area during the Setup Period

- Equivalent Part: Standard modules or components having the same material, form, and functions, for example, motors of the same model and self-built Friction Wheel modules. The main structure supporting the robot's body, such as a robot's chassis or the body of a Drone, cannot be replaced as equivalent parts.
- If the equivalent part includes the Referee System Module, the participating team must ensure that the Referee System Onboard Module works properly, including version number, sensor calibration, and Mounting Specifications. When the Referee System Onboard Module in the equivalent part experiences an abnormality, all consequences arising therefrom will be borne by the participating team and it will not trigger an Official Technical Timeout.

When there are 1 minute and 30 seconds remaining in the Three-Minute Setup Period, it is recommended that the Operator enter the Operation Room to complete the commissioning of the keyboard and mouse (both of which can be brought in). Any USB keyboard or mouse connected to the computer in the Operation Room must be produced by a certified manufacturer and comply with the USB HID protocol specifications. Check and confirm that robots are operating properly and that official equipment is functioning normally. If any official equipment in the Operation Room cannot operate normally, the Pit Crew must raise the issue before the final 15 seconds of the Three-Minute Setup

Period. Otherwise, the Referee will not announce technical timeout. During the Three-Minute Setup Period, only one person, in addition to the Operators and Gimbal Operators for the fielded robots, is allowed to enter the Operation Room to make tactical deployments. Personnel other than the Operators and Gimbal Operators for the fielded robots must leave the Operation Room before the Three-Minute Setup Period ends.

When there are 30 seconds remaining in the Three-Minute Setup Period, all robots on the Battlefield must be powered on and personnel must leave the Battlefield in an orderly manner; any robots not powered on when 15 seconds remain must be removed from the Battlefield. After the Three-Minute Setup Period ends, the Pit Crew must place the Sentry and Radar Remote Controllers in the designated area at the Battlefield entrance or the Operation Room entrance.

6.3.1 Official Technical Timeout

During the Three-Minute Setup Period, if the Referee System, official equipment, or another module related to the Referee System experiences any fault, or a robot needs to be inspected urgently (see below for details), the Head Referee may announce an Official Technical Timeout and pause the setup countdown. The starting time of the Timeout is decided by the Head Referee based on the situation.

During an Official Technical Timeout, Pit Crew members cannot fix the malfunctions of the Referee System or other official equipment unless the fix is required by the Referee. If a replacement is needed for a malfunctioning Referee System, replace the Referee System Onboard Module within the time limit specified in the RoboMaster 2026 University Series Robot Building Specifications Manual. When the relevant malfunctions of the Referee System or official equipment have been eliminated and the Head Referee has resumed the countdown, Pit Crew members are required to follow the set procedures for the Three-Minute Setup Period and leave the Battlefield within the specified time.

Table 6-1 Malfunctions

Regulation	Description
1	A fault occurs with the official equipment in the Operation Room, and any key competition component in the Battlefield experiences structural damage or functional irregularity.
2	During the Three-Minute Setup Period of the first round, a Referee System Onboard Module experiences faults, such as damage to the Armor Module, Speed Monitor Module disconnection, etc.
3	During the Three-Minute Setup Period, the main controller of the Referee System is unable to connect to the server or a robot cannot transmit images to the Operation Room.

Regulation	Description
4	Other situations where the Head Referee deems it necessary to call an Official Technical Timeout.

As it is impossible to determine whether the malfunction stems from the Referee System Module itself, flaws in the robot's electrical or structural design, or damage caused during previous matches, an Official Technical Timeout will not be called for regular battle damage, except for the situations mentioned in Regulation 3 above. The Referee will provide backup Referee System modules. The Participating Teams can request a Team Technical Timeout to repair their robots.

If the Referee determines that the malfunction referred to in Regulations 2 and 3 above is caused by the team, the Referee will explain the situation and end the Official Technical Timeout.

6.3.2 Team Technical Timeout

If the mechanical structure of a robot, a software system, the keyboard or mouse in the Operation Room, or other equipment experiences any fault, the Pit Crew can make a request to the Referee in the Battlefield or Operation Room for “Team Technical Timeout” only before the 15-second countdown in the Three-minute Setup Period, where they should also provide the reasons for the request. Team Technical Timeout once requested and conveyed to the Head Referee, this Timeout cannot be canceled or revised.

After the Team Technical Timeout is confirmed by the Head Referee, the Referee will notify both teams at the same time regardless of which team initiated the timeout. During the Team Technical Timeout, both teams can continue to commission and repair their robots as usual.

The Head Referee can end the Technical Timeout once they determine that the teams are ready. Even if the participating team does not enter the Battlefield or ends the Technical Timeout early, the opportunity consumed is still the opportunity corresponding to the time declared by the participating team when applying for the timeout.

To ensure that subsequent matches begin on time, only one Team Technical Timeout is allowed in each Three-Minute Setup Period on a first-come-first-served basis. The Technical Timeout usage is recorded in the Match Results Confirmation Form.

The Team Technical Timeout arrangements for different phases in a competition stage are as follows:

Table 6-2 Team Technical Timeout Arrangements

Competition Stage	Phase	Arrangement
Regional Competition	Group Stage	2 Technical Timeouts for 2 minutes each.

Competition Stage	Phase	Arrangement
Regional Competition	Knockout Stage	1 Technical Timeout for 3 minutes. Technical Timeout opportunities not used in the Group Stage can be carried over to the Knockout Stage.

6.4 15-Second Referee System Initialization Period

After the Three-Minute Setup Period ends, the competition enters a 15-Second Referee System Initialization Period. During the Initialization Period, the Referee System server automatically detects the connection status of the Referee System Player's Client, the status of Referee System Modules, and the status of Battlefield Components, and reset the HP of all robots to ensure their HP is full when the match officially begins.

6.5 Five-Second Countdown

After the 15-Second Referee System Initialization Period, the match enters a Five-Second Countdown. At this time, the Referee System Player's Client will not respond to control commands from robots (including Custom Controllers and Custom Clients). Once the countdown ends, the Referee System Player's Client will resume responding to control commands from robots, and the competition starts.

6.6 Seven-Minute Round

During the Seven-Minute Round, robots from both teams will engage in a shooting battle on the Battlefield.

6.7 End of Round

When the time of one round of competition runs out or one team triggers the winning criteria in advance, this round of competition ends. For winning criteria, see "5.8 Match Format and Winning Criteria". The match is over when a winner has emerged or all rounds have ended.

6.8 Result Confirmation

During a match, the Referee will record on the Match Results Confirmation Form the penalties issued for each round, the key competition data at the end of the match, the winning teams, the use of Technical Timeout opportunities by the teams, and other relevant details.

Within five minutes after the end of a match, the Captains of both teams must sign and confirm the match results. If a team Captain does not sign and confirm the results within five minutes or has not requested an appeal, it is deemed that the team agrees with the match results.

6.9 Exit the Field

After a match is over, members from both teams must power off all their robots, remove them from the Battlefield, and proceed to the Projectile Unloading Area to unload their Projectiles. At the Projectile Unloading Area, the Participating Teams must follow the instructions from the staff, return all related materials, empty the Projectiles in their robots, and return all Projectiles used in the competition.

7 Violations and Penalties

In order to ensure the fairness of the competition and maintain competition discipline, the the Participating Teams, participants, and participant robots must strictly follow the competition rules. If there is a violation, the Referee will issue a corresponding penalty. Some violation penalties issued before the official start of the competition will be enforced after the official start of the competition. Serious violations and all appeals in the competition will be publicized.

Penalty for violation stated in this section will be determined by the Head Referee according to the actual situation. If an event that affects competition fairness happens during the competition but it does not involve penalty rules or serious violations, the Head Referee will make a decision based on the actual situation.

If a team's actions have directly caused the other team to commit a violation, the other team is not penalized but must cease its violating behavior immediately.

7.1 Penalty System

7.1.1 Forms of Penalties

During the competition, the Referee System or Referees will issue penalties against participants and robots that violate competition rules. The forms of penalties are as follows:

Table 7-1 Forms of Penalties

Form of Penalty	Description
Automatic penalties by the Referee System	All HP losses in “5.1 HP Loss and Penalty for Exceeding Limit“, except those caused by attacks or collision and the penalty damage, are automatic penalties by the Referee System.
Manual penalties via the Referee System	Penalties issued by the Referee using the server against robots for violation of rules.
Manual penalties by Referees	Penalties cannot be issued via the Referee System, for example, issuing a verbal warning or disqualifying a team.

7.1.2 Types of Penalties

The types of manual penalties are shown in the table below.

Table 7-2 Types of Penalties

<table>
<tr><th>Type of Penalty</th><th>Description</th></tr>
<tr><td>Verbal Warning</td><td>Verbal Alert</td></tr>
<tr><td>Yellow Card</td><td>

One team receives a Yellow Card:

If the offending robot is a Sentry, its chassis is powered off for 2 seconds while the operating interface of other alive robots is blacked out for 2 seconds.
If the offending robot is not a Sentry, the operating interface for the offending robot is blacked out for 5 seconds and those for other robots are blacked out for 2 seconds.
The Referee System will automatically deduct the offending robot's HP by 15% of its current Maximum HP, while the remaining alive robots will have their HP deducted by 5% of their current Maximum HP. If the robot receives a Yellow Card again within 30 seconds after it receives a Yellow Card, the percentage of their current Maximum HP lost will be twice that of the previous loss for that robot, and 5% for the other alive robots of the team.
In each round, an Engineer that has received two Yellow Cards will also receive a Red Card. For all other robots, a Red Card will be issued when a total of three Yellow Cards have been received.

Both teams receive a Yellow Card:
The operating interface of all Operators is blacked out for 2 seconds and the HP of all robots loses by 5% of their Maximum HP, without taking into account the cumulative number of Yellow Cards received.

<hr>

If a robot receives successive Yellow Cards, the time that the operating interface is blacked out will add up accordingly. The 30-second period will be reset.
If a robot's remaining HP is less than or equal to that needs to be deducted from penalty, this robot's HP reduces to 1.
The duration of chassis power off, blockage of the operating

</td></tr>
</table>

Type of Penalty	Description
	interface, and other such situations caused by violation penalties is independent from the Competition Mechanism and will not be combined. ● When a Yellow Card causes the interface to be blacked out, video transmission, the Mini-map, and the custom UI are all unavailable.
Red Card (Ejection)	● Ejection of robots: ➢ If a robot is ejected before entering the 15-Second Referee System Initialization Period, the offending robot will not be allowed to enter the field and must be removed from the Battlefield, nor can it be substituted by other robots in any round of the current match. ➢ If a robot is ejected during the 15-Second Referee System Initialization Period, the Red Card should be issued after the competition starts. ➢ If a Drone is ejected during the competition, its Launching Mechanism will be powered off, the image transmission data link will be disconnected, the Pilot cannot start the Drone's propellers, and the Gimbal Operator cannot call for Air Support; if the Drone is flying, the Operator must immediately land it onto the designated location. ➢ If the Dart System is ejected during the competition, the Dart launching button will be hidden from view, and the gate of the Dart Launching Station can no longer be opened; if the gate is already open, it will close immediately. ➢ If a Radar is ejected during a match, its laser launching device and actuator will be powered off, and Radar’s Inter-robot Communication will be disconnected (it can only receive one way and can no longer send).. ➢ If a robot other than a Drone, Dart, and Radar is ejected during the competition, the robot's HP will become 0 and the transmitted images will become monochrome. ● Ejected ancillary equipment is not allowed to enter the field. ● Pit Crew members ejected by a Referee must leave the Competition Area immediately and cannot be substituted by other Pit Crew members for all

Type of Penalty	Description
	rounds in the current match. If an Operator is ejected, all robots controlled by them must also be ejected for the current round and will not be allowed to enter the field nor can they be substituted by other robots for all rounds in the current match. If a Gimbal Operator, Drone, and Dart System are all ejected, they will no longer be allowed to compete, nor can they be substituted by other robots in all rounds of the current match.
Forfeiture	● If a Forfeiture is issued for a round (hereinafter referred to as "Round Forfeiture"), the following rules will apply: ➢ If a Forfeiture is issued before the Seven-Minute Round, including the Three-Minute Setup Period and 15-Second Referee System Initialization Period, the HP of the offending team's Base will be 0, and the HP of the team's other robots will remain full. The HP of the other team's Base and robots will be full. The Round time is considered as 1 second, and the total damage of both teams is considered as 0. ➢ If a Forfeiture is issued during the Seven-Minute Round, the round will end immediately. The HP of the offending team's Base will be 0, and the team's other robots will maintain their HP level at the end of the round. The HP of the other team's Base and robots will remain at the level when the round ends. ➢ If a Forfeiture is issued after the Seven-Minute Round, the HP of the offending team's Base will become 0, and the team's other robots will maintain their HP level at the end of the round. The HP of the other team's Base and robots will remain at the level when the round ends. ● If a Forfeiture is issued in a match (hereinafter referred to as "Match Forfeiture"), it applies to all rounds in the match, and the HP for each round should be calculated according to the above descriptions.
Exclusion from Awards	● A participant is excluded from awards. ● A participating team is excluded from awards.
Disqualification from the Competition	● A participant is disqualified from the competition and excluded from awards. ● A participating team is disqualified from the competition and excluded from awards, but its results so far in this season will be maintained as a basis for other teams' advancement.

7.2 Penalty Rules

This section outlines the Penalty Rules for violations. The rules numbered R# clearly specify the regulations that the Participating Teams, participants, and participating robots must follow.

7.2.1 Personnel

7.2.1.1 General Specifications

R1 The Participating Teams must comply with the requirements outlined in the RoboMaster 2026 University Championship Participant Manual.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R2 Participants and their actions must not interfere with the normal operation of the official equipment, competition processes, or the regular work of the RMOC staff.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R3 The Participating Teams are not allowed to set up wireless devices or use walkie-talkies in any competition-related areas, including but not limited to the Preparation Area, Inspection Area, Staging Area, and Competition Area.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R4 The Participating Teams must not damage any official equipment, including but not limited to equipment in the Competition Area, Staging Area, Preparation Area, and Inspection Area.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition and the offender must compensate as per the price.

R5 Apart from Pit Crew who have entered the Staging Area and Competition Area due to match-related reasons, no participants are allowed to enter either area without special reasons.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R6 Any Pit Crew member who has entered the Staging Area and Competition Area for competition may not leave either area or be substituted by another Pit Crew member without the permission of the Referee.

Penalty: Offenders are not allowed to enter the Staging Area and Competition Area. The highest penalty that can be imposed on the offender is disqualification from the competition.

R7 Except for the Projectiles preloaded in the Inspection Area, the Participating Teams are not allowed to bring any Projectiles intended for use in the competition into the Staging Area or Competition Area.

Penalty: Projectiles will be confiscated. The highest penalty that can be imposed on the offender is disqualification from the competition.

R8 After a match is over, the Pit Crew must power off all their robots, remove them from the Competition Area, and empty all Projectiles inside the robots at the Projectile unloading area.

Penalty: The offending robot will be detained in the Projectile Unloading Area, until its Projectiles are cleared.

R9 After a match ends, Pit Crew must return all Projectiles used in the competition to the Projectile Unloading Area.

Penalty: Confiscation of Projectiles and disqualification of the offender from subsequent matches in the current division. The highest penalty that can be imposed on the offender is disqualification from the competition.

R10 Except for emergencies, teams must be present at the Inspection Area at least 55 minutes before the start of each match for Pre-Match Inspection. Teams must stand by at the Staging Area 10 minutes before each match.

Penalty: The highest penalty is an immediate Round Forfeiture.

R11 Without the permission of the Referee, a team cannot power on and commission or repair a robot in areas outside the Battlefield from the time the robot passes an Inspection to the end of a match.

Penalty: The highest penalty is an immediate Round Forfeiture.

R12 The identities and number of personnel of each team entering designated areas such as the Preparation, Inspection, Staging, and Competition Areas must meet the relevant requirements.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R13 Pit Crew members must wear corresponding labels which must not be covered. One member must wear the designated Captain label.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R14 Without permission from the Referee, Pit Crew entering the Competition Area must not communicate with anyone from the outside.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R15 Except for the Radar and wireless charging devices placed on the Battlefield, Pit Crew are not allowed to use official equipment power supply to supply power to personal devices within the Competition Area. However, they may carry their own power supply.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R16 Except for special circumstances, Pit Crew members are prohibited from wearing slippers into the Competition Area.

Penalty: The offender may receive a Red Card as the highest penalty.

R17 The team that makes an appeal must provide valid evidence or the specifications or rules in the official documents they are citing.

Penalty: The highest penalty that can be imposed on the offender is disqualification from the competition.

R18 The Captain is responsible for management and must ensure the following during preparation and participation:

- Participating robots must pass the review against the Rules of Usage for Fully Assembled Robots and Open-Source Robots.
- Team Coaches, Reserve Members, and other personnel who are ineligible for awards must not commit any violation during participation.
- Promptly stop any other violations.

Penalty: The Captain's management responsibility is determined based on the violation. The highest penalty that can be imposed on the offender is disqualification from the competition.

7.2.1.2 Battlefield Regulations

R19 Pit Crew must wear safety goggles when inside the Battlefield.

Penalty: A Verbal Warning will be issued, and the offender will be removed from the Battlefield. If the warning is ineffective, the offender will be issued a Red Card.

R20 During an Official Technical Timeout, Pit Crew members are not allowed to fix faults other than those in modules related to the Referee System.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R21 After the end of the Three-Minute Setup Period, Pit Crew must return to the designated area outside the Battlefield. During the competition, Pit Crew are not allowed to leave the area without the permission from the Referee.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R22 After the Three-Minute Setup Period, the remote controllers for the Sentry and Radar must be placed in the designated area at the Battlefield entrance or the Operation Room entrance. Remote controllers for other robots must be placed inside the Operation Room or at the designated location at the Battlefield entrance.

Penalty: If the incident occurs before the Seven-Minute Round, a Verbal Warning will be issued. If the warning is ineffective, a Red Card will be issued to the offending robot. If the incident occurs during the Seven-Minute Round, a Red Card will be issued immediately to the offending robot.

R23 After the Five-Second Countdown, the Pit Crew must not operate remote controllers located outside the Operation Room that correspond to fielded robots.

Penalty: The offending robot will be issued a Red Card, with the highest penalty being a Round Forfeiture.

R24 Pit crew members must ensure their robots are operating safely and will not cause harm to any person or equipment in the Competition Area.

Penalty: The offending team must bear the corresponding responsibility.

7.2.1.3 Operation Room Regulations

R25 During the Three-Minute Setup Period, only one person, in addition to the Operators and Gimbal Operators for the fielded robots, is allowed to enter the Operation Room. Personnel other than the Operators mentioned above must leave the Operation Room before the Three-Minute Setup Period ends.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R26 Operators must remain in their respective Operation Room to operate the relevant robot control equipment and wear the required headset during the 15-Second Referee System Initialization Period and the Seven-Minute Round, unless otherwise permitted by the Referee. After the Three-Minute Setup Period ends, Operators may not change their location.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R27 The Operator does not arrive at the Operation Room before the five-second countdown or leaves the Operation Room during the Seven-Minute Round.

Penalty: The offender will receive a Red Card.

R28 During the Seven-Minute Round, Gimbal Operators may use gimbal remote controllers or custom controllers for Drones, and remote controllers for Darts at the same time, while a Pilot may use only one remote controller. Besides, each Operator must configure a control device and custom client according to the requirements for the robot's control method.

Penalty: Verbal Warning. If the warning is ineffective, the offender will be issued a Red Card.

R29 The use of personal headphones or computers is prohibited in the Operation Room, except for Custom Clients.

Penalty: Verbal Warning. If the warning is ineffective, the team will be issued a Round Forfeiture.

R30 The Operation Room provides three USB Type-A receptacle ports, and expansion is prohibited. USB keyboards and mice that are produced by unqualified manufacturers, modified, homemade, or do not comply with the USB HID protocol are not allowed to be connected to the Operation Room computer.

Penalty: Verbal Warning and prohibition of using the violating device. If the warning is ineffective, the offender will be issued a Red Card.

R31 During the competition, communication with those outside the Operation Room is forbidden.

Penalty: Verbal Warning. If the warning is ineffective, the team will be issued a Round Forfeiture.

R32 During the match, Pilots operating Drones must obtain Pilot qualification and wear the corresponding labels.

Penalty: The offender will receive a Red Card. The highest penalty will be disqualification the offending team from the competition.

7.2.2 Robot

7.2.2.1 General Specifications

R33 Robots and related equipment to enter the field must pass the Pre-Match Inspection.

Penalty: Round Forfeiture.

R34 In the first round of a match, the robots must meet the minimum battle team roaster.

Penalty: Match Forfeiture.

R35 Robots must meet the requirements in the RoboMaster 2026 University Series Robot Building Specifications Manual.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

- The RMOC will conduct random checks on robots.
- Any report made against a robot for not complying with the robot building specifications manual must be supported by the relevant evidence.
- If the UWB Positioning Module or Speed Monitor Module is not installed according to requirements during the match, the corresponding module will be considered disconnected.

R36 In the event of a dispute, teams are obligated to show their robot's mechanisms, circuit design drawings and relevant code documents to the RMOC and answer relevant technical questions.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R37 Before the 15-Second Referee System Initialization Period, armor stickers that meet the requirements in the specifications documents must be attached to robots.

Penalty: Verbal Warning. If the warning is ineffective, the offending robot will be issued a Red Card.

R38 When waiting in the Staging Area, participants are not allowed to bring robots out of the Staging Area without permission.

Penalty: Verbal Warning. If the warning is ineffective, the offenders and robots will be issued a Red Card, with the highest penalty being disqualification.

R39 Robots must be free of safety issues including but not limited to short circuits, crashing, creating fumes or lighting flames, parts falling to the ground, and gas cylinder explosions. If a safety issue is identified or presents itself, participants must execute the relevant operations in accordance with the Referee's instructions.

Penalty: If it happens before the start of a match, the Pit Crew need to resolve the safety issue as required by the Referee. Otherwise, the offending robot will not be allowed to enter the field. If it is during the competition, a Verbal Warning will be issued. If the warning is ineffective, the offender or the offending robot will be issued a Red Card. Any incident involving serious safety hazards must be handled by the Head Referee in accordance with “8 Irregularities“.

R40 Robots are not allowed to fire Projectiles out of the Battlefield.

Penalty: Verbal Warning. If the warning is ineffective, the offending robot will be issued a Red Card.

R41 Dart Systems are not allowed to fire Darts out of the Battlefield.

Penalty: The offending robot will be issued a Red Card.

R42 During the Three-Minute Setup Period, the 15-Second Referee System Initialization Period, and the Five-Second Countdown, robots in the Battlefield are not allowed to leave their respective initial zones.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

R43 If any Projectile needs to be fired during the Three-Minute Setup Period, it must be launched into the Projectile clearance bag.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

R44 During the Three-Minute Setup Period, the replacement modules or parts used on robots must meet the requirements for Equivalent Parts.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

7.2.2.2 Ground Robot

R45 During the competition, no alive robot is allowed to block any of its Armor Modules with its body. No robot is allowed to block more than one Armor Module on any other alive robot in its team. When an Engineer is grabbing or carrying a Mobile Component, any one of its armors is allowed to be blocked by the carried Mobile Component and its relevant body structure, and the Armor Module obstructed can be different each time, but multiple Armor Modules are not allowed to be obstructed at the same time.

- Hero, Infantry, and Sentry robots are not allowed to obstruct their Armor Modules when carrying a Mobile Component.
- If the obstructed Armor Modules are disconnected, the robot will not be deemed in violation of rules.

Penalty: When one Armor Module is blocked, a warning is issued against the offending team based on their subjective intention. If the blocking was intentional, a Yellow Card will be issued along with a Verbal Warning. If the warning is ineffective, a Red Card will be issued. If the blocking was passive in nature, a Yellow Card will be issued. If more than one Armor Module is blocked, a Yellow Card and a Verbal Warning will be issued when a violation occurs. If the warning is ineffective, a Red Card will be issued.

7.2.2.3 Drone

R46 The remote controller used by a pilot is solely intended for functions related to the takeoff, landing, calibration, and other controls of the Drone. It must not be used for any other purposes.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

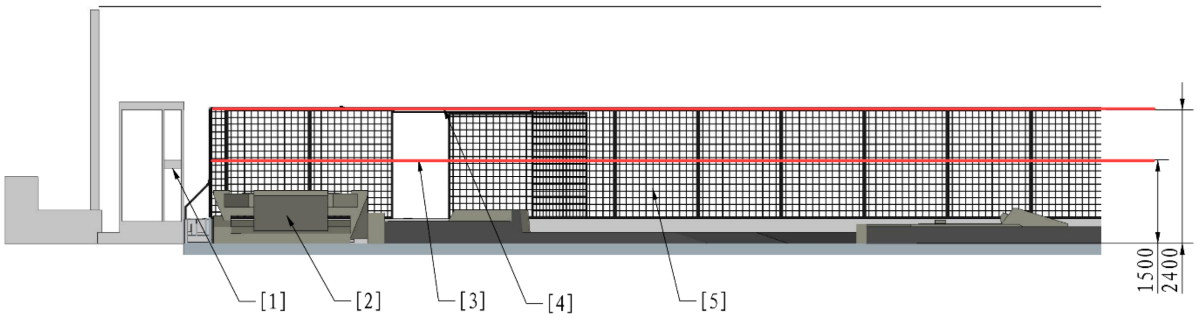
R47 During the Three-Minute Setup Period, Pit Crew members may debug the Drone near the Landing Pad, but must not start the propeller.

Penalty: The offender and the offending robot may be issued a Red Card as the highest penalty.

R48 The hook of the Aerial Safety Rope must be attached to the rigid ring of the Drone's vertical rigid safety rod.

Penalty: The offending robot will be issued a Red Card. Drones are not allowed to enter the other rounds in the same match and all matches after this competition stage.

R49 During the competition, the distance between the lowest point of a Drone and the Battlefield ground must not be less than 1.5 m, and no part of the Speed Monitor Module (17mm Projectile) carried by the Drone's Gimbal Launching Mechanism can exceed the highest point of the Perimeter Wall of the Flight Zone.



[1]	Pilot Operation Room	[2]	Landing Pad	[3]	Minimum Flight Altitude
[4]	Maximum Height of Perimeter Wall	[5]	Perimeter Wall		

Figure 7-1 Flight Altitude Restrictions

Penalty: A gesture or Verbal Warning is given to the Pilot, to remind him or her to adjust the flight altitude. If a warning is ineffective, the offending robot will be issued a Red Card and forbidden from entering any rounds in the same match.

R50 During the competition, Drones are forbidden from flying outside the Battlefield.

Penalty: The offending robot will be issued a Red Card, and the pilot will be disqualified. The Drone is not allowed to enter the other rounds in the same match.

R51 During the competition, a Drone is prohibited from launching Projectiles while in contact with the Landing Pad.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

R52 If a Drone experiences technical faults or is damaged due to the unreasonable design of the propulsion system or power supply system during the competition, it must be checked by the Referee and must be cleared by the Head Referee as hazard-free before being allowed to return to the match.

Penalty: The offending robot will be issued a Red Card. The offending robot is not allowed to enter the other rounds in the same match.

- If a Drone crashes in the Battlefield, the Referee may, depending on the situation, retrieve the robot to the Landing Pad and any consequences arising from the retrieval process must be borne by the participating team. If the robot has suffered severe structural damage, the Aerial Safety Rope hook has fallen off, or the competition is about to end, the Referee may not retrieve the robot.
- For safety reasons, if a Drone is flying erratically during the competition, the Head

Referee will eject the robot, and the Pilot must land the robot and stop its propeller, and is no longer allowed to operate the robot.

7.2.2.4 Other Robots

R53 The Dart System must not remain in a ready-to-launch state other than during the Seven-minute Round.

Ready-to-launch state: The energy storage component used for providing initial kinetic energy for Darts is in a tense, inflated, or rotating state. The energy storage component includes but is not limited to rubber band, pneumatic cylinder, friction wheel, etc.

Penalty: Verbal Warning. If the Verbal Warning is ineffective, the offending robot will be issued a Red Card.

7.2.3 Interaction

7.2.3.1 Interaction between Robots

R54 A robot may not use any of its body structures to strike an opponent robot in collision. If a defeated robot is blocking a key path, the robot can be slowly pushed away.

- In any collision between a Drone and a Ground Robot while the Drone is flying, the Drone will be deemed the offending robot.
- In any collision between a Drone and a Ground Robot after the Drone has landed on the Battlefield or while it is being retrieved, the Ground Robot will be deemed the offending robot.
- In any collision between two Ground Robots, the offending robot will be the one deemed by the Referee to be the initiator.

Penalty: Warnings will be issued against the offending robot based on their subjective intention and the degree of collision.

Table 7-3 Collision Violation Penalty Standards

Violation Level	Description
Yellow Card	● Actively causing frontal and high-speed collision ● Active pushing that causes the other team's robot to move noticeably ● Active pushing that impedes the normal movement of the other team's robot

Violation Level	Description
Red Card	● Actively causing high-speed, repeated, and intense frontal collision ● Active pushing that causes the other team's robot to move for a longer distance ● Active pushing that seriously impedes the normal movement of the other team's robot

If any collision has caused serious consequences to the other team's robot or rendered the competition unfair, in the next round of the same competition stage, the offending robot (identified by its number) will be barred from the Battlefield and cannot be substituted by another robot.

R55 A robot must not get entangled with any other robot (except for robots that cannot be respawned) due to active interference, blocking, or collision.

Penalty: Counting from when an entanglement is determined, warnings should be issued against the offending robot based on the length of the violation. If it exceeds 10 seconds, a first Yellow Card will be issued. Thereafter, each 20 seconds will incur a further Yellow Card. This should carry on until the offending robot is ejected. Regardless of whether the offending robot is alive, if the violation lasts longer than 90 seconds, the offending team may be issued a Round Forfeiture, which is the highest penalty, based on their subjective intention.

R56 No robot can attack the Drone, Dart Launcher, Radar, or wireless charging devices.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R57 No robot and its actions are allowed to block an opponent robot's entry into its Resupply Zone.

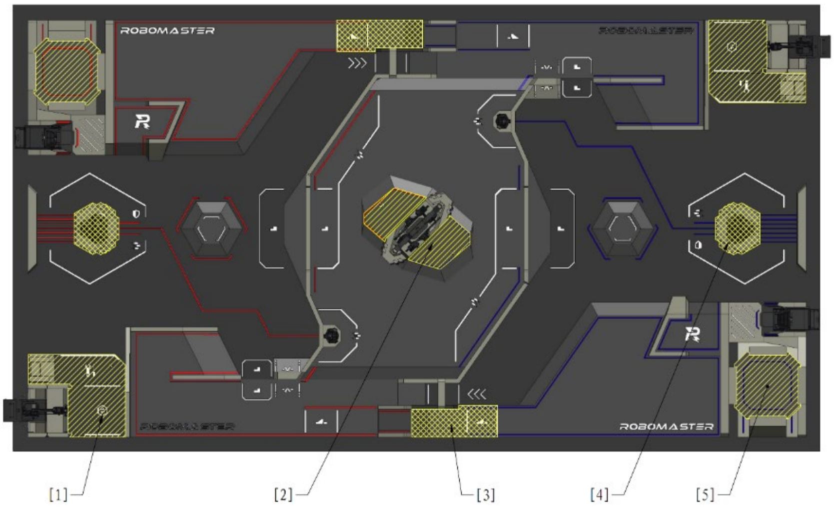
Penalty: The offending robot will be issued a Yellow Card. If the warning is ineffective, the offending robot will be issued a Red Card.

R58 Robots are not allowed to intentionally carry opponent Darts.

Penalty: Verbal Warning. If the Verbal Warning is ineffective, the offending robot will be issued a Red Card.

7.2.3.2 Interaction between Robots and Battlefield Components

Multiple Penalty Zones are set in the Battlefield. The Penalty Zone covers the ground and the air space directly above it. If robots themselves or any items they carry (such as Mobile Components) come into contact with or enter this space, it will be deemed that they have entered the Penalty Zone. The Penalty Zones are shown below.



[1]	Resupply Penalty Zone	[2]	Assembly Penalty Zone	[3]	Road Penalty Zone
[4]	Base Penalty Zone	[5]	Landing Pad Penalty Zone		

Figure 7-2 Battlefield Penalty Zones

Interactions between robots and Penalty Zones must comply with the following rules; otherwise, they will be deemed to have entered the Penalty Zone in violation of rules.

Single-side Penalty Zone: Only robots on the side can enter, and all opponent robots are prohibited from entering.

Both-side Penalty Zone: All robots from both sides are prohibited from entering.

Table: Battlefield Penalty Zone Access Rules

Penalty Zone Name	**Penalty Zone Attribute**
Resupply Penalty Zone	Penalty Zone for one side
Assembly Penalty Zone	Penalty Zone for one side
Road Penalty Zone	Penalty Zone for both sides
Base Penalty Zone	Penalty Zone for both sides
Landing Pad Penalty Zone	Penalty Zone for one side

R59 Ground Robots are forbidden from entering the Base Penalty Zone, Road Penalty Zone, or Landing Pad Penalty Zone in violation of rules.

- If a robot completely traverses the 17° slope from the bottom to the top of the Launch Ramp and then enters the Road Penalty Zone, losing mobility and being unable to leave the Penalty Zone, it will not be considered a violation. In all other cases, it will be regarded as a violation.

- A robot is not deemed in violation of rules if it is unable to leave the Road Penalty Zone from a Terrain Crossing Buff Point (Launch Ramp) after a failed attempt at the Launch Ramp, because the opponent's robot is also on the Buff Point.

Penalty: Warnings will be issued against the offending robot based on how long the robot remained in the Penalty Zone and the impact of the violation. If it exceeds 3 seconds, a Yellow Card is issued. Thereafter, every 10 seconds will incur a further Yellow Card. This carries on until the offending robot is ejected. An offending robot that causes serious damage to an opponent robot by remaining in a Penalty Zone will be issued a Red Card.

R60 Ground Robots are forbidden from entering the Resupply Penalty Zone or Assembly Penalty Zone in violation of rules.

If a robot is defeated or ejected in any Penalty Zone, the Referee may activate the robot temporarily as required and guide the robot's Operator in controlling the robot to leave the Penalty Zone.

Penalty: Warnings will be issued against the offending robot based on how long the robot remained in the Penalty Zone and the impact of the violation. If it exceeds 3 seconds, the first Yellow Card will be issued. Thereafter, every 5 seconds will incur a further Yellow Card. This will carry on until the offending robot is ejected. A defeated or abnormally disconnected robot that stays in a Penalty Zone longer than 20 seconds may be imposed a Round Forfeiture as the highest penalty.

R61 Robots are not allowed to bring Mobile Components into both teams' Road and Base Penalty Zones, and their team's Resupply Penalty Zone, and the other team's Dart Launching Station.

Penalty: A Yellow Card will be issued against the offending robot. If any subsequent Mobile Component has a decisive impact on the other team's Launch Ramp, Projectile supply, Dart launches, and target hits, or affects the normal operation of any Core Component, the offending robot will be issued a Red Card.

R62 Mobile Components used in the competition must not be brought onto the Battlefield. During the competition, robots can only use Projectiles and Mobile Components supplied by the RMOC within the Battlefield.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R63 Robots shall not obtain, carry, or use already assembled Energy Units in any way.

Penalty: The authorities imposed a loss by default.

R64 During the competition, robots are not allowed to damage the Battlefield, Battlefield Components, or Projectiles, nor affect the normal function of the Battlefield Components.

Penalty: The highest penalty that can be imposed on the offending team is disqualification from the competition.

R65 Teams are not allowed to hit a Power Rune with 42 mm Projectiles or Darts.

Penalty: The offending robot may be issued a Red Card as the highest penalty.

7.3 Serious Violations

The following actions are considered serious violations. The highest penalty the RMOC can impose on the offending team for serious violations is disqualification from the competition. In the event of any violation of local laws and regulations, the RMOC will fully cooperate with the relevant authorities to hold the offender legally responsible.

Table 7-4 Types of Serious Violations

Regulation	Type
1	Malicious destruction of the Battlefield, Battlefield Components, other official equipment, or the robots or equipment of other teams.
2	Falsification, assumption of a false identity, or any other behavior determined as cheating.
3	Tampering with or damaging the Referee System, or interfering with any detecting function of the Referee System by technical means.
4	Circumstances that violate the specifications documents and are determined by the Chief Referee as a serious violation.
5	Disobedience over penalties, refusal to cooperate, deliberate delay, disrupting the competition, forfeiting without valid reasons, boycotting, or other behavior that hinders the competition.
6	Match throwing or manipulation.
7	Providing property to others or illegally soliciting or accepting property from others for the purpose of obtaining an unjust competition outcome or improper benefits.
8	Uncivilized and immoral conduct involving defamation, verbal abuse, rude gestures, malicious heckling, or malicious throwing of objects.
9	Publishing, spreading, or disseminating to the media false or irresponsible

Regulation	Type
	remarks.
10	Deliberately attacking or colliding with others in a manner that endangers themselves or others.
11	Carrying hazardous items or contrabands.
12	Other behavior that violates the spirit of the competition and are deemed a serious violation.
13	Other conduct that violates core socialist values, sports ethics, public order and norms, the culture and discipline of the competition, laws and regulations, or that causes an adverse impact on society.

8 Irregularities

There will be a certain delay in the Referee's manual penalties and handling of abnormal situations. If it has a major impact on the result of the competition, the Chief Referee will determine the final result according to the actual situation.

If any of the following abnormalities occur, it shall be handled according to the corresponding process, to which both teams cannot object. The handling process is as follows:

- When a serious safety hazard or abnormality has occurred, such as: a battery explosion, stadium power outage, explosion of a compressed gas cylinder, or interpersonal conflict, the Head Referee will notify both teams' Operators after discovering and confirming the emergency, and eject all robots through the Referee System. The result of the round will be invalidated. The round will restart after the safety hazard or abnormality has been eliminated. The RMOC will give priority to addressing safety issues and will make every effort to protect Robots within reasonable limits. However, it cannot guarantee that Robots will be entirely unaffected, and any loss or impact resulting from safety measures shall be borne by the the Participating Teams.
- If non-key Battlefield Components are damaged during a match (damage to the PVC flooring, the light indicators on the competition area, or the light effects on the Base), which do not affect the fairness of the match, the match will proceed as usual.
- The competition will carry on despite any anomaly with a robot's armor light effects or light indicator effects or any damage to an Armor Module Sticker.
- If functional abnormalities or structural damage occur to key Official Equipment and Battlefield Components during the competition, thus rendering it unfair, for example, network issues causing robots to go disconnected, failure to trigger the Buff Effect after striking the Power Rune, or Battlefield Components malfunctioning, the Referee will manually address such issues using the Referee System. If the failure cannot be dealt with manually, the Referee will notify the Operators of both sides and eject all robots at the same time. The competition will end immediately, and the result of the competition will be invalid. When problems are solved, there will be a rematch.
- During a match, if key Official Equipment and Battlefield Components experience functional abnormalities or structural damage that affects the fairness of the competition, and the Head Referee did not confirm and end the game in time, leading to a situation where a game that should have ended continues and has a winner, the results for the round shall be invalidated once the Chief Referee has made a determination to that effect within 5 minutes after the end of the round, and a rematch shall be held.
- In the case of a serious violation that would clearly have triggered a penalty of forfeiture, and the Head Referee did not confirm and execute it in time, the results for the round shall be invalidated

once the Chief Referee has made a determination to that effect within five minutes after the end of the round, and the offending team will be issued a forfeiture.

- During the competition, if any situation has occurred that may affect the fairness of the competition, the Chief Referee will notify the Captains of both teams of the situation and suspend the results confirmation process within five minutes after the end of the match, and will make a determination within 60 minutes and notify both Captains of the final course of action. The handling outcome is final and cannot be challenged by both teams.

9 Appeal

Every team has one appeal opportunity during each of the Regional Competition, Wild Card Competition, and Final Tournaments. Appeal opportunities cannot be used cumulatively across competitions. If an appeal is successful, the team involved retains its right to appeal again in future matches. If it is unsuccessful, the team will have exhausted its one opportunity to appeal. When a team has exhausted its opportunity to appeal, the Arbitration Commission will no longer accept any appeal from the team. After the appeal is accepted, the Arbitration Commission will deliberate on the appeal materials and relevant evidence. On behalf of the Arbitration Commission, the Chief Referee will then communicate the appeal decision to both teams. The Arbitration Commission has the right to final interpretation of the appeal decision.

The following situations do not constitute a basis for appeal:

- Verbal Warnings and Yellow and Red Cards issued as penalties for violations
- Types and processes of Technical Timeouts initiated
- Regular Battle Damage occurring at the Referee System Onboard Module

No appeal is allowed five minutes after a Match Results Confirmation Form has been signed or a match has ended.

9.1 Appeal Process

The Participating Teams filing an appeal need to follow the procedures as shown below:

Lodging of an appeal	A team may lodge an appeal within five minutes after the end of a match (based on the time recorded on the Match Results Confirmation Form). Teams during the Group Stage may appeal on all rounds of a match, while in the Knockout Stage the losing team is only allowed to appeal on the rounds in which it has lost. A Captain who wishes to file an appeal should submit an appeal request at the results confirmation counter. The confirmation of an appeal request means the start of the appeal process and the consumption of one appeal opportunity. If a team has signed on the Results Confirmation Form or makes an appeal five minutes after a match has ended, the RMOC will not accept any of their appeal request. No appeals are allowed before and during the competition. Note: If the reason for the appeal is related to the robots of any team in the competition, the appealing party shall propose that the relevant robots be isolated and tested, which will be implemented after confirmation by the Arbitration Commission.
Filling in and signing the Appeal Form	Within 10 minutes after the team is provided with an appeal form, the Captain of the appealing team needs to fill in and sign the Appeal Form at the results confirmation counter. In addition to the Captain, no more than 2 pit crew members are allowed to assist the filling of the appeal form. The Arbitration Commission will arbitrate between the two parties only on the contents of the "Appeal Reason" column, which cannot be changed after it has been signed. If the appealing team does not sign and submit the appeal form within 10 minutes after they are provided with an appeal form, they will be deemed as "not providing any appeal reasons". Therefore, the subsequent processes will be skipped and the arbitration result will be "maintain the original match results".
Gathering of evidence or defense materials	If either team needs to collect appeal materials, they have to be submitted to the Chief Referee within 60 minutes after the appeal request has been accepted. The time and location of the submission must be agreed beforehand, and the team is required to explain the appeal materials during the submission. The Arbitration Commission will not accept any new appeal material that is provided beyond the submission period. If both teams do not need collect appeal materials, or the Arbitration Commission has identified sufficient evidence, or the appeal reason filled by the complainant is illegal, proceed directly to the next step.
Arbitration and communication	The Arbitration Commission will arbitrate between both parties according to related evidence and call the Captains of both teams to the Arbitration Room for further communication. Both teams must be at the Arbitration Room within 10 minutes after being notified by the Arbitration Commission. The arbitration can only be attended in person. No more than three members from each team may be present, and they must only consist of Regular Members or the Supervisor. The absence of any team shall be deemed as a forfeiture of its rights to the arbitration communication. If the appealing party's reasons for appeal are obviously inconsistent with the facts or use illegal basis, the Arbitration Commission may only explain the situation to the appealing party and maintain the original competition results. When the representatives of both parties agree with the arbitration result or the Arbitration Commission considers that the necessary communication has been completed, the arbitration communication ends, directly advancing to the nest step.
Confirmation of arbitration results	The Arbitration Commission will indicate its final decision on the Appeal Form. One member in the Arbitration Room from each team must sign on the Form on behalf of the team. The signing of the Form means that the arbitration decision is known by the team and no longer disputable. If the Appeal Form is not signed within five minutes after the decision has been announced, the decision is deemed to be known by the team. If the appealing party's reasons for appeal are obviously inconsistent with the facts or use illegal basis for the appeal, the Arbitration Commission will maintain the original competition results after explaining the situation to the appealing party, and the party under appeal does not need to sign and is deemed to be aware of the arbitration result.
Continued appeal	If a round is a rematch due to the arbitration result being "rematch between both teams", after the rematch is over, the team that still has the opportunity to appeal can file an appeal for the rematch. In this scenario, if the previous team that made the initial appeal wishes to appeal again (known as a "continued appeal"), this will exhaust the team's appeal chances regardless of the appeal results, because it will cause serious delays to the competition schedule. If a continued appeal is filed, the time for submission of evidence or defense materials is shortened to within 30 minutes of lodging the appeal.

Figure 9-1 Appeal Process

9.2 Appeal Materials

Each file submitted by the Participating Teams as appeal materials must not exceed 500 MB in size. No more than 10 files can be submitted.

9.3 Appeal Decision

Appeal Decisions include: maintaining the original match results, forfeiture against the respondent, and rematch between both teams. After the Arbitration Commission confirms the appeal decision, neither team is allowed to challenge it.

- Successful appeal: forfeiture against the respondent, rematch between both teams.
- Failed appeal: maintaining the original match results.

If the communicated appeal decision is a rematch, but neither team is willing to accept a rematch, the appeal will be deemed failed and the original match results will be maintained.

- Provided it does not affect the schedule of the entire competition, the rematch will in principle be held on the same day after all the other matches, depending on the actual situation.
- The flow of the rematch will be the same as regular matches. Both teams are required to compete according to the time stipulated by the RMOC and comply with relevant rules.

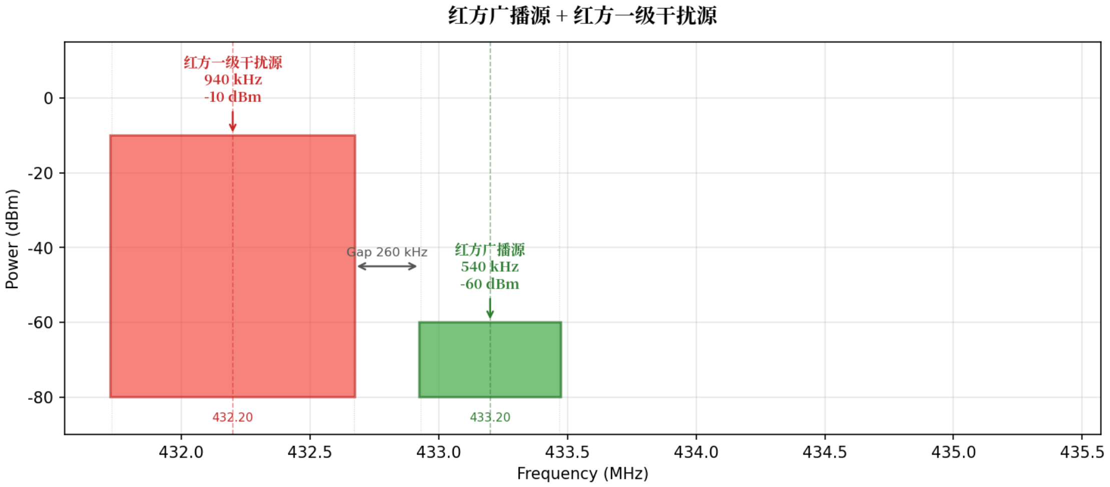
If the round involved in the appeal decides the final winner of a match, the match winner must be re-decided.

Example: In a BO3 match with a result of 1:2, the Red Team wins the first round, while the Blue Team wins the second and third rounds. After the end of round, the Red Team files an appeal regarding the second round, and the appeal decision is a rematch between both teams. If the Red Team wins the rematch round, the final match result becomes 2:0; if the Blue Team wins the rematch round, the final result remains 1:2.

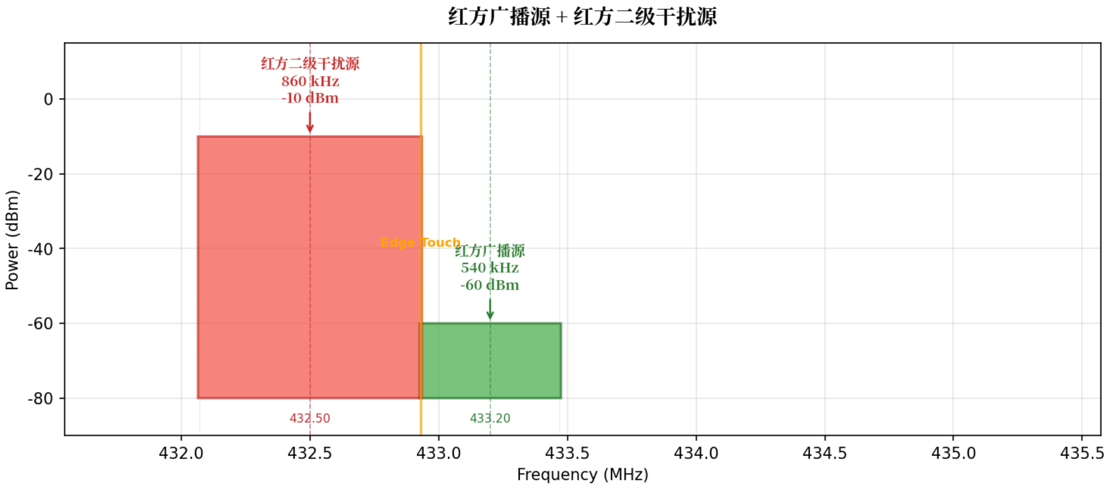
Appendix 1: Radar Information Wave Characteristics

Schematic diagram of spectral distribution

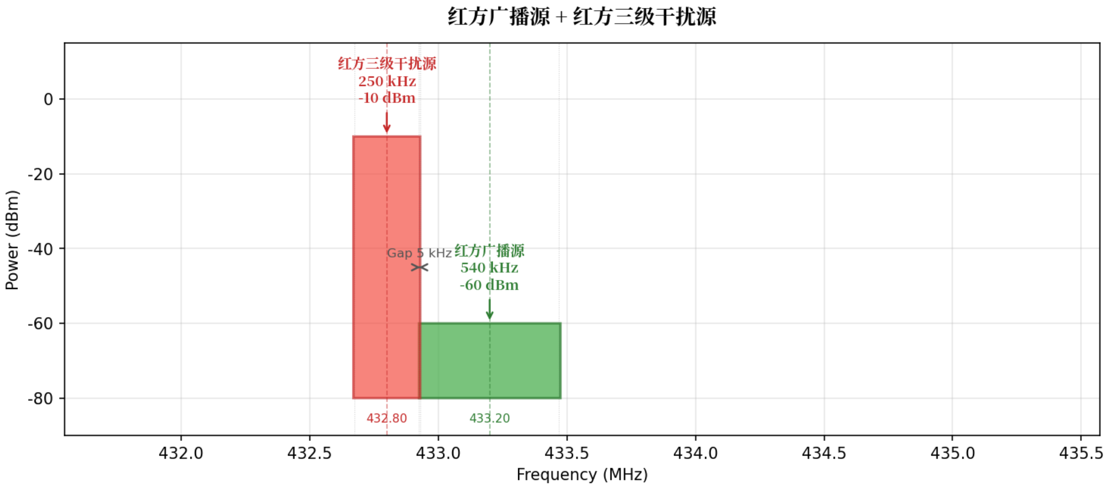
Red broadcast source + Red first-level interference source:



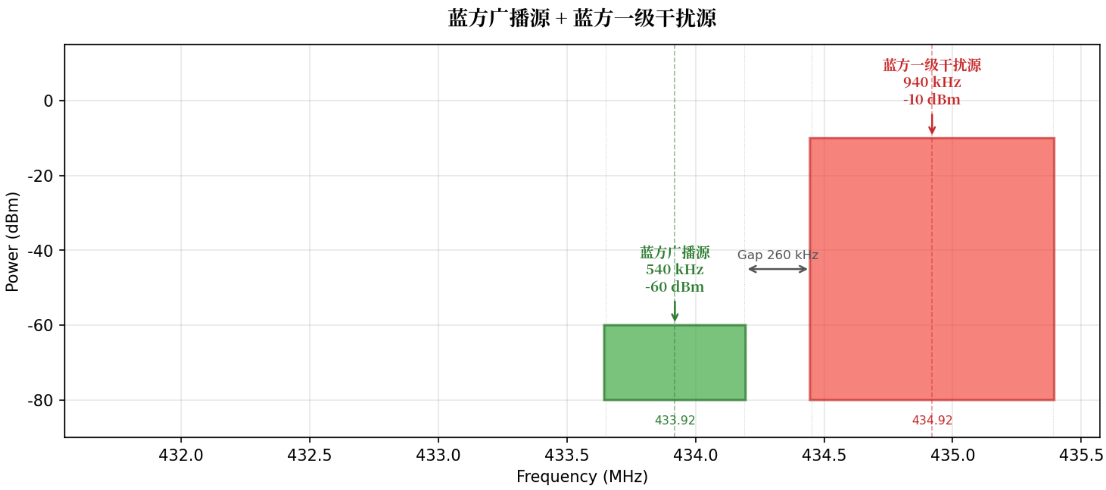
Red broadcast source + Red second-level interference source:



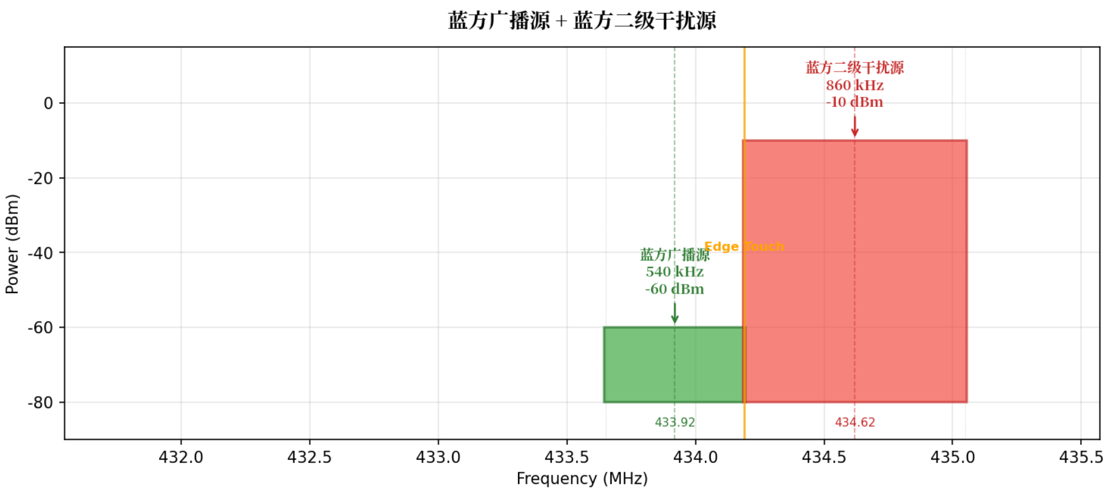
Red broadcast source + Red third-level interference source:



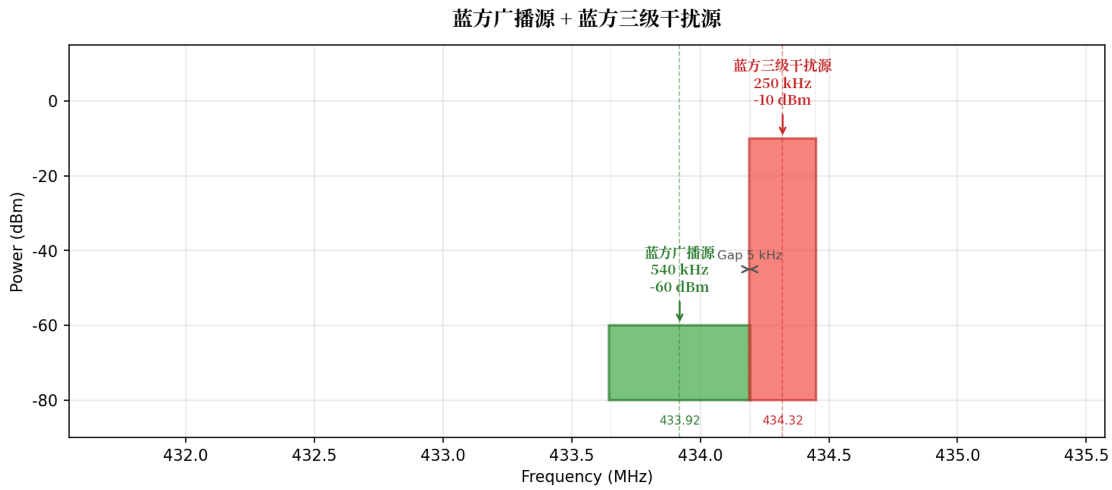
Blue broadcast source + Blue first-level interference source:



Blue broadcast source + Blue second-level interference source:

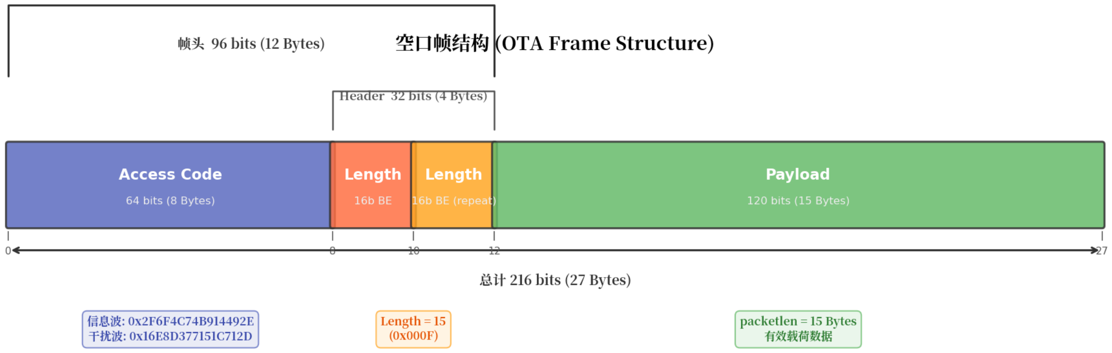


Blue broadcast source + Blue third-level interference source:



Air-interface frame structure

The structure of each transmitted data packet over the air interface is as follows:



Air-interface frame structure table:

Field	Length	Value/Description
Access Code	8 bytes(64bits)	Information Wave: 0x2F6F4C74B914492E, Interference Wave: 0x16E8D377151C712D (big-endian)
Header	4 bytes (32bits)	Contains two identical Payload Length fields (each 16 bits, big-endian, value is 15 i.e. 0x000F). The receiver checks if the two are consistent; if inconsistent, the packet is discarde
Payload	15 bytes	Payload data

Note: Both red and blue sides use the same Access Code. That is, both red and blue information waves use 0x2F6F4C74B914492E, and both red and blue interference waves use 0x16E8D377151C712D.

The data stream is sliced into groups of 15 bytes each, with each group serving as Payload of an air interface packet. After adding the Access Code and Header, it is transmitted. That is, the data frame (including all fields such as SOF and CRC) is continuously segmented into 15-byte fragments, which are sequentially assembled into air interface packets for transmission.

Supplementary notes for Modulation Link:

Project	**Description**
Preamble	There is no independent preamble, and the frame starts directly with the Access Code
Packet Interval	No fixed packet interval, data is transmitted upon arrival
Gaussian Filter Cutoff Length	Default 4 symbol periods (GNURadio digital_gfsk_mod default value)
Data Whitening (Whitening)	None
Forward Error Correction (FEC)	None

Time-domain characteristics after GFSK modulation (taking the Red broadcast source as an example):

Feature	**Value**
Duration per bit	52 μs
Duration per packet	11.23 ms (216 bits × 52 μs)
bit=1 instantaneous frequency offset	+250,769 Hz
bit=0 instantaneous frequency offset	-250,769 Hz
Gaussian filter 3dB bandwidth	BT/Tb = 0.35/52μs ≈ 6,731 Hz

Data frame protocol

Data push rate:

Source	Push Rate	Data Content
Information Wave	1350 byte/s	All valid data 0X0A01-0X0A05 (10Hz)
Interference Wave	1350 byte/s	Contains 0x0A06 data (10Hz), the remaining bytes are randomly filled

For the specific structure of the communication protocol, please refer to the RoboMaster 2026 University Series Communication Protocol.

Appendix 2: Pre-Match Procedures

This season does not have a separate Sentry Mapping section.

Mock Inspection

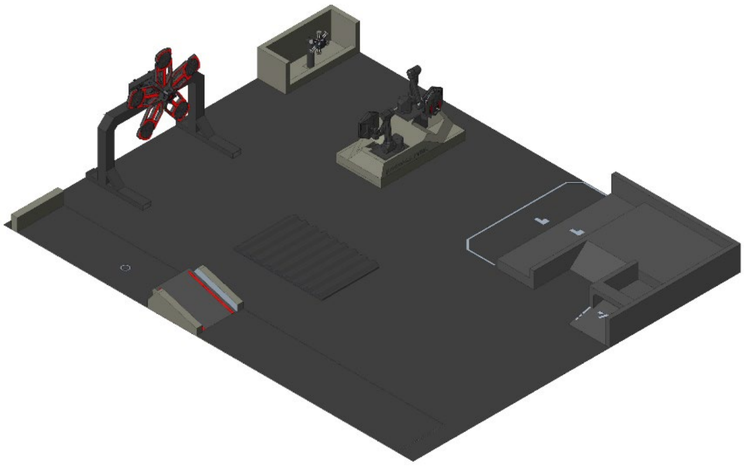
Mock Inspection is a check carried out before official inspection, intended to help the Participating Teams identify and resolve any issues with their robots in advance. During Mock Inspection and Practice Match, the RMOC will make every effort to assist the Participating Teams in identifying and resolving robot malfunctions. However, the inspection results are not equivalent to official inspection results. If a robot is found not to meet inspection standards during Pre-Match Inspection, it will not be allowed to enter the field.

Unique identification registration is only allowed during Mock Inspection. After Mock Inspection ends, only robots with existing unique identification can complete replacement registration. If a team needs to repair or replace a component with unique identification, they must apply to the RMOC for re-registration and re-labeling. Each team can submit up to six such applications within the same division. Unauthorized replacement of a uniquely identified component will disqualify the robot from entering the field.

- Non-emergency registration procedure: If more than two hours remain before a team's next match, or if it is outside RMOC working hours, the team can replace the relevant parts independently. They must submit either detailed before-and-after photos of the robot or a complete video of the replacement process through the Robot Unique Identifier Supplementary Registration portal. Teams must also book an inspection slot via the same link and report to the second Inspection Area at the scheduled time, where an RMOC official will carry out the review. Robots that pass the review will receive a new unique identifier.
- Emergency reporting procedure: If less than two hours remain before a team's next match and it falls within the RMOC's working hours, the team must contact the Referee in the second Inspection Area to request urgent repairs. After the RMOC staff confirm the application, they will accompany the team to the Preparation Area and supervise the entire repair process. In this case, robots can pass inspection and participate in the current match even if they do not have all unique identifiers. However, once the team has finished their next match, the robots must be brought to the second Inspection Area for a follow-up review and to assign any missing unique identifiers.

Battlefield Component Training

The battlefield component training area is equipped with components and wooden structures similar to those used in official matches, as shown below, providing the Participating Teams with a space for testing. Pre-Match Inspection is not required before battlefield component training. The Participating Teams must ensure that their robots' Referee System is functioning properly. If the Referee System fails on-site, any time spent resolving the issue will be counted against the team's training time.



Time and Booking:

Each participating team must arrive at the Staging Area at least 20 minutes in advance. If a team fails to arrive on time, its training session will be timed as scheduled and will not be extended. Each team has two training sessions, with 30 minutes for each session, including entering and exiting the field. The first training session will be scheduled by the RMOC, and the second must be booked by the Participating Teams themselves. Please enter and exit the field strictly according to the allocated time, and do not use any other team's training time.

Personnel and Equipment:

A maximum of 12 people from each team can enter the training area, and all personnel must wear goggles. The RMOC will provide only two Operator computers for use on the video transmission channel. If you require additional Player's Clients for more robots, you may bring your own computers, but they are not allowed to connect to the competition network.

Components, Projectiles, and Safety:

The RMOC will provide only Mobile Components for testing and will not supply 17 mm or 42 mm Projectiles. Teams must bring their own Projectiles if testing is required. When testing with 42 mm Projectiles, teams must not intentionally strike the Power Rune. Due to venue limitations, ambient lighting may affect the training area, and teams should make the necessary adjustments. If a team refuses to comply with the on-site Referee or commits any other violation, its training qualification will be revoked immediately, and the team will be required to immediately leave the field; in serious cases, its Practice Match qualification will also be canceled.

Practice Match

Practice Match offers the Participating Teams an opportunity for commissioning and adaptation before the official match, with the main objectives as follows:

- Familiarize yourself with the match environment and procedures.
- Test and adjust the status and strategies of robots.

Rules for Practice Match:

- Due to the tight schedule of Practice Match, which is primarily intended for commissioning, matches will generally not be suspended unless a critical issue occurs with official equipment.
- In the event of a malfunction in the Referee System Onboard Module, an official technical timeout of up to two minutes may be granted by the RMOC. This pause will be included in the practice match duration.
- Team technical timeouts are not permitted during Practice Match.

Rules for the Free Commissioning Phase:

- During Projectile launch testing, the Participating Teams can ask the Referee to borrow a clearing bag and can only launch Projectiles into the clearing bag.
- During the free commissioning phase, the match status will show as "Not Started", and the Participating Teams must adapt to and become familiar with the transitions between each match phase and the related requirements.

Eligibility and Stage Division:

- Robots that fail to pass inspection on time, or pose safety risks, are not allowed to join Practice Match.
- Robots that fail inspection due to serious non-compliance with inspection standards can only take part in the free debug phase of Practice Match.
- The number of robots participating in Practice Match must not exceed the maximum allowed in the Robot Line-up.

Appendix 3: Future Rule Planning

In future seasons, the RMOC will place greater emphasis on guiding and encouraging the robot capabilities, scenario adaptability, and technical directions outlined in the table below.

Core Robot System	Subsystem	Capability/Technology	Typical Robot
Perception System	External Sensor	Multi-sensor data fusion on a single platform	All
	Inter-robot Communication	Inter-robot information fusion	All
	Guidance System	Self-motion status acquisition	Dart System
Decision-making System	Computing Platform	Low-power operation	All
	Decision-making Algorithm	● Decision-making capability under minimal or no external influence (especially advanced semantic analysis and robust decision-making in extreme conditions) ● High-efficiency algorithm design with low time complexity under limited computational power	Sentry and Radar System
	Guidance System	Target localization and automated recognition	Dart System
Actuation System	Mobility System	● Cross narrow terrain by folding or transforming ● Stably traverse rugged, uneven, and harsh terrain	Ground Robot
	Projectile Launching Mechanism	● New type of launching mechanisms beyond friction wheel systems (such as pneumatic or electromagnetic launching) ● Irregular object recognition and high-precision launch ● Beyond-visual-range strike	Hero, Infantry, and Sentry

Core Robot System	Subsystem	Capability/Technology	Typical Robot
	Object Manipulation Mechanism	● Multi-degree-of-freedom motion mechanism ● Multi-degree-of-freedom terminal executive mechanism	Engineer
	Flight and Power System	● High payload-to-weight ratio ● Folding and transformation for compact transport	Drone
	Guidance Mechanism	● Equipped with propulsion, capable of autonomously altering its trajectory (non-aircraft), with particular emphasis on terminal maneuverability ● Non-parabolic trajectory design	Dart System
	Modular Mechanism	Designed as a functional module that can operate independently of the overall robot system to perform a specific task, while also allowing rapid integration into the system, for example, an independent Projectile launching mechanism	Ground Robot
	Complete Unit	Low-power operation	All
	Complete Unit	● The ability to make effective decisions in complex environments ● Collaboration between automatic robots and manual robots	Sentry and Radar System
Human-Robot Interaction	Complete Unit	● Robots efficiently and accurately convey the information necessary for humans to make decisions ● Make robot easier to operate, while highlighting the irreplaceable value of human decision-making	All

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